SRS Document

A Unified Management System for NIMC

Group 1

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Introduction

The National Institute of Mass Communication (NIMC) conducts training courses, seminars and workshops, mostly related to media. Alongside training, it runs a library for its employees, course participants and outside visitors, and a dormitory (Shibly Hostel) for participants who need accommodation. NIMC has teams of internal research, IT, finance and accounting teams. Its internal structure resembles most government offices and relies mainly on modules developed under A2I.

Despite this range of activities, NIMC has no classroom management system, its library has no management software, and its dormitory is not automated. Participants notably struggle to get information and seats in the dormitory, and there is no system to hold proper online courses. An online application system exists only for non-government individuals; applications from government offices still require a written letter sent to NIMC. The accounting and finance teams use government centralized systems such as dNothi.

“A Unified Management System for NIMC” aims to bring Classroom, Library and Dormitory management onto one platform that shares a single NIMC account. The platform also includes AI assistants for course resources, book recommendations and hostel help.

Purpose

This document describes the Software Requirements Specification and Analysis of the web application “A Unified Management System for NIMC”. It records the functional, non-functional and supporting requirements and sets a requirements baseline for development. The requirements are organized by topic so that the SRS can serve as an official means of communicating user requirements to the developers and as a common reference point for the development team and the stakeholder community. The SRS will evolve as users and developers work together to validate, clarify and expand its contents.

Intended Audience

Our intended audiences are:

1. Participants and other library and dormitory users (the people who use the services)
2. Course Coordinator, Course Coordination Team and Speakers (who run the classroom activities)
3. Librarian and Assistant Librarian (who manage the library)
4. Hostel Manager, Hostel Super/Warden and the approving authorities: Director General, Additional Director General and Director (who manage dormitory services)

5. NIMC Accounts and the NIMC IT team (who handle payments and operate the system)

This SRS will be used to verify whether the developer team has created a product that is acceptable to NIMC and its users. The developers will use it to plan milestones and delivery dates and as the basis for developing the system's functionality, linking each requirement to the software they create. The designers will use it as the basis for the system's design and refer back to it to make sure the design fulfills stakeholder needs. The testers will use it to derive test plans and test cases for each documented requirement.

Conclusion

This analysis of the audience helped us focus on the users who will rely on this document. The document as a whole will help everyone related to the project gain a clearer idea of what the NIMC management system must do.

1. Inception

1.1 Identifying the Stakeholders

Direct stakeholders:

1. Classroom Management
    - Participants (the people doing the course)
    - Speakers (invited to take the course sessions)
    - Course Coordination Team
    - Course Coordinator (lead)
2. Library Management
    - Librarian
    - Assistant Librarian
    - All office employees
    - Participants of courses
    - External visitors (including public users of the e-library)
3. Dormitory Management (Shibly Hostel)
    - Director General (DG)
    - Additional Director General (ADG)
    - Director
    - Hostel Manager (called Hostel Officer in the stakeholder slides)
    - Hostel Super / Warden
    - Residents: non-government participants, government participants and government external guests
4. NIMC Accounts (receives payment information and processes dormitory fees)

Indirect stakeholders:

1. NIMC IT team (operates and maintains the platform)
2. NIMC research, finance and accounting teams
3. Government offices that send their employees as participants (currently by written letter)
4. A2I and government centralized systems such as dNothi, with which NIMC's existing modules and teams work

1.2 Stakeholder Viewpoints

a. Participant Viewpoint

Participants are the largest user group and use all three modules through one shared NIMC account. The problems identified are:

1. Difficulty getting information about the dormitory and getting a seat.
2. No single place to find the session plan, speaker information and learning resources.
3. No proper way to attend or revisit online sessions.
4. No catalogue they can search to find books or check whether a book is available.

The requirements they need are:

1. One account (name, company, designation, phone, email, optional NID) with email verification, sign-in and password reset.
2. Joining a classroom using a classroom code.
3. Viewing the updated session plan, speaker details, learning resources and class recordings, with search, sort and filter.
4. Viewing their own attendance, and viewing assignment and examination notices.
5. Rating speakers after each session, anonymously or with their name, and giving feedback on the overall course.
6. Receiving a certificate with a QR code after the course.
7. A soft library membership card, book search, borrowing status, reviews and book requests.
8. Dormitory application, fee information, room details, stay extension or cancellation requests, issue reporting and progress tracking.
9. Notifications by email, and by SMS in special cases.

b. Course Coordinator and Course Coordination Team Viewpoint

The Course Coordinator leads course management and is supported by the Course Coordination Team. They need to:

1. Create a course classroom, share a unique classroom code, and remove ineligible participants.
2. Assign speakers and prepare and change the session plan (topic, date, time, speaker).
3. Upload resources before or after a session and store recordings of online sessions.
4. Record attendance for each session and see each participant's overall attendance.
5. See an engagement indicator generated from attendance, assessment and other information, to identify participants who need attention.
6. Evaluate assignments and examinations, and review AI-generated plagiarism and AI-content flags before taking any academic decision.
7. See the speaker's overall rating; the speaker's rating history across courses is visible only to the Course Coordinator.
8. Manage course evaluation and issue certificates according to configurable eligibility conditions.

c. Speaker Viewpoint

Speakers are invited to conduct sessions. They need to know their assigned topic, date and time, to be told when the schedule changes, and to share the materials used in their session through the Course Coordination Team. Their performance is judged through participant ratings, which are summarized for the Course Coordinator.

d. Librarian and Assistant Librarian Viewpoint

The library currently has no management software. They need to:

1. Assign memberships (course participant, office employee, government employee, external visitor) with validity, and issue membership cards with a QR code.
2. Check membership and the limit of three borrowed books, then issue and return books, set return deadlines and extend them in special circumstances.
3. See the catalogue and all circulation records: borrowed books, borrowers, issue dates, due dates and overdue status.
4. Apply a configured overdue penalty, and override it in special cases with the decision recorded.
5. Manage the catalogue and the e-library, including whether a digital resource may be downloaded.
6. Moderate reviews, review the book-request list to judge demand, and notify users when a requested book is added.
7. Generate a monthly library report in a downloadable or printable format.

e. Library User Viewpoint (Office Employees and External Visitors)

Office employees and external visitors need a valid library membership (external visitors also provide identification), search by title, author, catalogue information, ISBN or category, see only general availability, view their own borrowed books and deadlines (not other users'), submit reviews and book requests, and use the e-library. Public e-library resources should be readable without a physical membership.

f. Approving Authorities Viewpoint (DG, ADG, Director)

They review dormitory applications in sequence: the DG first, then the ADG, then the Director. They want key information in front of them (seat availability, participant company, designation, stay time, reason for accommodation), a direct approval link by email as well as through the portal, and the DG's authority to approve fee waivers for eligible government employees.

g. Hostel Manager and Hostel Super / Warden Viewpoint

The Hostel Manager reviews the application after the approvals, sees the list of available rooms and seats, allocates a room and seat, and records check-in and check-out (with designated staff). The Hostel Manager also wants live availability (total and empty rooms, total, occupied and available seats), a dashboard and a monthly report. The Hostel Super/Warden receives issue reports such as a broken air conditioner, takes action and updates the issue status until it is resolved.

h. NIMC Accounts Viewpoint

NIMC Accounts needs the dormitory fee and any approved waiver sent by the system, and confirms payment before room allocation proceeds.

i. Resident Viewpoint

Residents want to see their allocated room and seat, ask for extensions or cancellations, report problems and follow their progress, rate dormitory services and rooms, read a digital noticeboard and receive email alerts. They also want a Hostel AI that answers questions (for example the Wi-Fi password for their own room), gives directions on a 3D hostel map and, where supported devices are connected, controls lights or a geyser.

1.3 Working towards Collaboration

Conflicts:

1. Government applications are manual (written letters) while non-government applications are online.

2. Participants may rate speakers anonymously, but the Course Coordination Team manages the evaluation and only the Course Coordinator may see a speaker's rating history.
3. The AI analysis flags possible plagiarism or AI-generated work, but academic decisions belong to the Course Coordination Team.
4. The e-library should be publicly accessible, but physical membership and download permission are controlled by the library.
5. Staff may already have entered a person's details, while that person needs an account of their own.
6. The stakeholder slides name the Hostel Officer and Hostel Super for check-in and check-out, while the user story names the Hostel Manager or designated dormitory staff.
7. Dormitory approval passes through several officers in sequence (DG, ADG, Director, Hostel Manager) while participants want a quick answer on seats.
8. Hostel AI should answer room-specific questions and control devices, but information such as the Wi-Fi password must not reach other residents, and not every room has supported IoT devices.

Negotiations:

1. Online application for government participants is a requirement of the new system, but letters stay valid until government offices adopt it. (NIMC management and government offices)
2. Ratings may be anonymous or named, and the rating history is shown only to the Course Coordinator. (Participants and Course Coordinator)
3. AI results are shown only as flags and indicators; the Course Coordination Team reviews them and decides. (Course Coordination Team and developers)
4. Public resources are open to all; other resources need membership, and the uploading librarian decides whether a file can be downloaded. (Librarian and library users)
5. A person whose identity was entered by staff activates the existing identity instead of opening a duplicate account. (NIMC staff and participants)
6. The system supports check-in and check-out by the Hostel Manager or any designated dormitory staff, so both role names are covered. (Hostel Manager, Hostel Super and developers)
7. Approval links are sent by email and shown in the portal, and an application that is approved but has no free seat stays pending. (DG, ADG, Director and Hostel Manager)
8. Hostel AI answers only with information tied to the asker's assigned room, and device commands are limited to supported, integrated devices. 3D map and IoT are

treated as exciting features, not core requirements. (Residents, IT team and developers)

These conflicts and negotiations are drawn from the stakeholder slides and the user story. Please confirm them with NIMC.

1.4 Manual System / Existing System

NIMC currently has an online application system only for non-government individuals. Applications for government participants are fully manual and need a written letter sent to NIMC. The accounting and finance teams use government centralized systems such as dNothi. NIMC has no classroom management system and no library software, and its dormitory is not automated. The activities below therefore happen without a dedicated platform (details to be confirmed with NIMC).

Course admission:

1. A non-government individual applies online.
2. A government participant's office sends a written letter to NIMC.
3. NIMC staff process the application manually.

Classroom activities:

1. The Course Coordinator contacts speakers and prepares the session plan.
2. Schedule changes, resources and notices are passed to participants without a single platform.
3. Attendance is taken session by session by the Coordinator.
4. Speaker evaluation, assignment checking and course feedback are handled manually, and online courses cannot be held properly.
5. Certificates are prepared after the course.

Library activities:

1. A user asks for a book at the library desk.
2. The librarian checks the user's eligibility without software and keeps borrowing and return records manually.
3. Return deadlines (course end date for participants), overdue penalties, book requests and monthly reports are tracked by hand.

Dormitory activities:

1. A participant asks for accommodation and often struggles to get information or a seat.
2. The application goes through the DG, ADG, Director and Hostel Manager.

3. The fee is collected through NIMC Accounts, with government employees paying a lower fee or receiving a DG-approved waiver.
4. The Hostel Manager allocates a room and seat, and check-in, check-out, extension and cancellation are recorded manually.
5. Problems in rooms are reported to the Hostel Super/Warden and followed up informally.

1.5 Proposed System

We need to note that this system is sub-automated. The software aids the manual decisions of NIMC officers and does not replace them.

1.5.1 Shared NIMC Account

1. Participants and NIMC employees open an account with name, company, designation, phone number, email address and optionally NID number.
2. They verify the email address through a verification email and then sign in with email and password.
3. Account holders can update their information, and can reset a forgotten password through the registered email.
4. The same account works for Classroom, Library and Dormitory services. If an authorized employee has already entered a person's details, the person activates that identity instead of creating a duplicate.
5. A module may ask for extra information; for example, a dormitory application requires the participant's NID number.

1.5.2 Classroom Management

Course Coordinator and Course Coordination Team

1. Create a classroom for a course and share a unique classroom code; remove an ineligible participant.
2. Assign speakers and prepare the session plan (topic, date, time, speaker), and change it when schedule, topic or speaker changes.
3. Upload resources (documents, presentations, videos, notes); online class recordings are stored for authorized participants.
4. Record attendance for each session; the system calculates overall attendance and generates an engagement indicator from attendance, assessment and other information.
5. Record assignment and examination results; AI-based analysis flags similarities, possible plagiarism and AI-generated content (text and other supported media), and the team reviews each flag.

6. View the speaker's overall rating (rating history is visible only to the Course Coordinator), manage course evaluation, and issue certificates with a QR code when the configured eligibility conditions are met.

Participant

1. Join a classroom using the code, view the updated session plan and speaker information, and search, sort and filter resources.
2. View own attendance and recorded sessions.
3. Rate the speaker after each session with a numerical rating and optional comment, anonymously or with name, and give overall course feedback.
4. Ask the course LLM assistant questions answered from the stored course resources.
5. Receive email notifications (and SMS in special cases) about schedule changes, speaker changes, online session information and assignment or examination notices.

1.5.3 Library Management

Librarian and Assistant Librarian

1. Assign membership to a shared account, record the category (course participant, office employee, government employee or external visitor) and set validity; participant memberships follow the course duration. The member receives a soft membership card with a QR code.
2. Before issuing a book, check that the membership is active and that the user holds fewer than three books. Record the issue date and due date; the book status changes from Available to Borrowed. Participants' default return date is the course end date, and the assistant librarian can extend it in special circumstances.
3. Record returns, mark the book Available again, and apply the configured penalty if the book is overdue. The librarian may override a penalty, and the adjustment is recorded.
4. View all circulation records (borrowed books, borrowers, issue and due dates, overdue status).
5. Manage the catalogue (title, author, catalogue information, ISBN, category, publisher and publication details, copy status) and the e-library, including whether a resource can be downloaded.
6. Moderate reviews, review book requests as a demand list, and generate the monthly report (catalogue totals, new books, borrowed and returned books, overdue books, active and new members, most borrowed books, requests, reviews and ratings, e-library usage) in a downloadable or printable form.

Library users

1. Search the catalogue and see whether a book is a physical copy, an online resource or both; ordinary users see only general availability.
2. View their own borrowed books and deadlines only.
3. Read or download e-library resources according to permission; public resources do not require membership.
4. Submit book reviews and ratings, and request books that are not in the library.
5. Receive reminders before due dates and notifications when a waiting or requested book becomes available or is approved.
6. Ask LibraryAssistantAI for recommendations and follow-up details (reviews, availability, related resources); it refers the user to the librarian when a question is outside the library data.

1.5.4 Dormitory Management

1. **Application:** a participant submits an application with the preferred accommodation type, expected period of stay and single-seat or double-seat requirement. It is forwarded to the Hostel Manager.
2. **Approval:** the approval view shows seat availability, participant company, designation, stay time and reason. Approval links are emailed and also available in the portal. The order is DG, then ADG, then Director, then the Hostel Manager, who checks room and seat availability. If no suitable room is free, the approved application stays pending.
3. **Payment:** the system calculates the fee from participant category and seat type and sends the payment information to NIMC Accounts. Eligible government employees may receive a waiver approved by the DG. Room allocation proceeds after payment is confirmed or a waiver applies.
4. **Allocation:** the Hostel Manager allocates a room and seat, the room and seat database is updated, and the participant can view the allocation and stay details.
5. **Stay:** the Hostel Manager or designated staff record check-in and the stay becomes active. Participants can request stay extensions or cancellations, and the system updates the stay once a request is approved.
6. **Issues:** a participant reports a problem; the Hostel Super/Warden takes action, updates the status until resolved, and the participant follows the progress.
7. **Check-out:** staff record check-out; the occupancy becomes inactive, the seat is released and the stay history is kept.
8. **Availability, dashboard and reports:** total and empty rooms, total, occupied and available seats, room type and capacity, plus a dashboard and a downloadable monthly report (applications, approvals, allocations, check-ins and check-outs, extensions, cancellations, issues).

9. **Notifications and evaluation:** email notifications for approval, payment, allocation, extension or cancellation decisions, check-in or check-out and issue updates; residents may rate services and rooms.
10. **Hostel AI:** answers room-specific questions (for example the Wi-Fi password for the resident's own room), guides to locations on a 3D hostel map when available, and sends supported IoT commands such as turning off lights or starting the geyser.

The current dormitory fee structure is:

Participant type	Single seat fee	Double seat fee
Government employee	200	100
Non-government participant	400	200

2. Elicitation

2.1 Quality Function Deployment (QFD)

This maps user needs to system functions across NIMC's core operational domains, helping prioritize features for the digital management system.

2.1.1 Normal Requirements

- **Classroom Management:**
  - The system will display all learning resources in a single, centralized platform for participants.
  - Participants can submit speaker evaluations after classes.
  - Lesson plans and speaker information will be clearly displayed to participants.
  - Course coordinators can create lessons, assign speakers, and upload course resources.
  - Course coordinators can maintain individual digital attendance sheets for participants.
  - The platform will provide online class support for participants and handle certificate issuance.
- **Dormitory Management (Shibly Hostel):**
  - Potential occupants can submit online dormitory applications.
  - NIMC officers can oversee and execute the digital application approval workflow.
  - Approved occupants can securely pay fees online.
  - Hostel Supers can perform check-in and check-out operations and access real-time occupancy information.
  - Occupants can submit requests for stay extensions or cancellations.
- **Library Management:**
  - Assistant librarians can utilize the existing user database to let participants borrow books and calculate return deadlines based on course durations.
  - Assistant librarians can view the catalog, withdrawn books, and borrower records.
  - Users can search for books by catalogue number, author, or name.
  - The system will provide soft membership cards for users.

2.1.2 Expected Requirements

- **Classroom Management:**
  - Speaker performance scores derived from evaluations will be displayed to course coordinators.

    - Participants will have access to advanced resource searching and sorting features.
    - The system will maintain authorized user access and a password reset mechanism for participants.
    - Course coordinators can perform student performance analysis based on attendance, exams, and assignments.
    - The system will send automated SMS/email notifications to participants.
- **Dormitory Management:**
    - The system will generate monthly reports for hostel managers and NIMC officers.
    - Occupants can rate services and rooms.
    - A digital noticeboard will be accessible for residents.
    - The system will dispatch automated email alerts to residents.
- **Library Management:**
    - Newly added books will automatically update the master book list on the main NIMC website.
    - The system will generate monthly library activity reports.
    - Users can submit wishlists to request new book acquisitions.
    - An internal book review system will be available for members.
    - Book details can be accessed internally via computers in the library or online.

2.1.3 Exciting Requirements

- **Classroom Management:**
    - Integration of a Large Language Model (LLM) to answer participant queries dynamically based on uploaded course resources.
- **Dormitory Management:**
    - Provision of an interactive 3D hostel map for easy navigation.
- **Library Management:**
    - Deployment of an LLM-powered chatbot to deliver personalized book recommendations.

A Unified Management System for NIMC

Shared NIMC Account

Participants and NIMC employees open a NIMC account using their name, company, designation, phone number, email address, and optionally their NID number. They verify their email address by responding to a verification email sent by NIMC. They sign in using their registered email address and password.

Account holders update their account information when any details change. If they forget their password, they request a password reset through their registered email address.

The same NIMC account is used for authorized Classroom, Library, and Dormitory services. A module may still request additional information when required; for example, a dormitory application requires the participant's NID number.

Classroom Management

The National Institute of Mass Communication (NIMC) conducts different training courses, seminars and workshops. The Classroom Management System manages the activities of these courses through a single platform. It involves the participants doing the course, the speakers who are invited to conduct the sessions, the Course Coordination Team and the Course Coordinator who leads the course management activities.

For a particular course, the Course Coordinator enters the course name into the system and creates a classroom. Once the classroom is created, the Course Coordinator shares a unique classroom code with the participants. Participants use the classroom code to enter the appropriate classroom. The Course Coordinator can remove a participant from the classroom if the participant is found to be ineligible.

The Course Coordinator then contacts and assigns speakers for the course and prepares the initial session plan. The session plan contains information of the session - topic, date, time and assigned speaker. The session plan can be changed during the course when there is a change in schedule, topic or speaker. Participants can view the updated session plan and the assigned speaker information. A sample session plan is given below:

Session	Topic	Date	Time	Speaker
Session 1	Introduction to Mass Communication	10 Sept.	10:00 AM	Speaker A
Session 2	Media Production	12 Sept.	10:00 AM	Speaker B

Session	Topic	Date	Time	Speaker
Session 3	Digital Journalism	15 Sept.	11:00 AM	Speaker C

The Course Coordination Team or the Course Coordinator uploads learning resources before or after a session according to the materials used or delivered by the speaker. Resources include documents, presentations, videos, notes and other materials. Since a course may contain a large number of resources, participants can search, sort and filter the resources to find the required material. For online sessions, the system also stores the class recording so that authorized participants can access the recorded session after the class.

An LLM-based assistant that can answer participant questions using the course resources stored in the system. The assistant helps participants find and understand information from the available course materials without manually searching through individual documents and recordings.

During each session, the Course Coordinator or authorized member of the Course Coordination Team records the attendance of participants. Attendance is maintained for each session and is used to calculate the participant's overall attendance. Participants can also view their own attendance records through the system. If a participant repeatedly misses sessions or shows low participation, the system uses attendance, assessment and other available information to generate an engagement indicator for the Course Coordination Team. This allows the team to identify participants who require attention and also provides an overall indication of how actively participants are engaging with the course.

The system notifies participants about important classroom activities. Normal notifications are sent through email, while SMS are used in special cases. Notifications may include changes in session schedules, speaker changes, online session information, assignment or examination notices and other important course updates.

After each session is completed, participants can evaluate the speaker who conducted that session. Participants provide a numerical rating and may also provide comments. A participant may choose to submit a rating anonymously or with their name. The system collects the ratings and calculates the speaker's overall rating. The Course Coordinator can view the speaker's rating and rating history across different courses, while the Course Coordination Team manages the evaluation process. The rating history of a speaker is available only to the Course Coordinator.

At the end of the course, participants may be required to complete assignments or examinations. The Course Coordination Team evaluates the submitted work and records the results in the system. The system stores the assessment results together

with attendance information so that the Course Coordination Team can review participant performance.

The system also provides an AI-based analysis service for participant submissions. The service can analyze different types of submitted work, including text and other supported media, to identify similarities between participants' work and to provide indicators of possible plagiarism or AI-generated content. The resulting findings are presented as flags or indicators for the Course Coordination Team. The system does not make the final academic decision automatically; the Course Coordination Team reviews the flagged work and determines the appropriate action.

At the completion of the course, the Course Coordination Team manages the overall course evaluation process. Participants provide feedback about the overall course, while the Course Coordination Team reviews the collected evaluation information to understand the effectiveness of the course and identify areas for improvement.

After the course and its required assessments are completed, eligible participants receive certificates through the system. The exact eligibility conditions for receiving a certificate can be configured according to the requirements of the particular course. The certificate contains a QR code which can be scanned to view details.

Library Management

Assigning Library Membership to the Shared NIMC Account

Participants use their shared NIMC account for library services.

The assistant librarian verifies the information and assigns a library membership. The user’s category (course participant, office employee, government employee, or external visitor) is recorded, and membership validity is set accordingly. The assistant librarian may adjust this, with course participant memberships tied to their course duration.

Upon membership assignment, the participant receives a membership card featuring their details, membership ID, validity, category, company, designation, and a QR code.

The librarian uses this QR code to quickly identify the user’s account when issuing or returning books.

Checking Library Membership and Borrowing Eligibility

When a user wants to borrow a physical book, the librarian or assistant librarian checks the user’s membership through the system. The system verifies that the membership is active and that the user has not reached the maximum borrowing limit.

A user can borrow a maximum of three books at a time. The librarian or assistant librarian views the user’s current borrowed books and their due dates.

Course participants are allowed to borrow books according to their course membership. Their normal return deadline is based on the end date of their course. If there is a special circumstance, the assistant librarian can extend the borrowing period and update the return date in the system.

Office employees can borrow books according to the borrowing period selected or permitted by the library. External visitors can also borrow physical books after obtaining a valid library membership and providing the required identification.

Searching and Selecting a Physical Book

A library user searches the library catalogue to find a book. The user can search using information such as book name, author, catalogue information, ISBN, category, or other available book information.

The system displays the relevant book information and indicates whether the book is currently available as a physical copy or as an online resource.

Users can see general book availability, while detailed circulation information, such as which member currently has a withdrawn/borrowed book, is available to the librarian and assistant librarian.

Borrowing a Physical Book

When a user asks for a selected book to the library desk, the librarian or assistant librarian identifies the user through their membership ID, account, or QR code.

The librarian checks the user’s borrowing eligibility and selects an appropriate return period. The system records the user’s information, book information, issue date, and selected due date.

The book’s status changes from Available to Borrowed, and the book appears in the user’s borrowed-book list.

The librarian and assistant librarian can view the library’s circulation records, including which books are currently borrowed, who has borrowed them, issue dates, and due dates.

Before the due date, the system sends a notification to the user reminding them to return the book.

Returning a Book and Handling Overdue Books

When a user returns a book, the librarian or assistant librarian records the return in the system. The system stores the return date and changes the book’s status back to Available.

If the book is returned after the due date, the system identifies it as overdue and applies the library’s configured penalty. The penalty amount or rule is determined by the librarian.

In special cases, the librarian can override the penalty when appropriate. The system records the adjustment so that the library retains a history of the decision.

After the book is returned, it becomes available for the next user. If another user has requested or is waiting for the book, the system can notify them when the book becomes available.

Viewing Borrowed and Withdrawn Books

The librarian and assistant librarian can view the library's circulation information through the system. They can see the books currently borrowed, their borrowers, issue dates, due dates, overdue status, and return information.

They can also view withdrawn/issued book records and identify which user currently has each book.

Users can view their own borrowed books and deadlines but cannot access other users' borrowing information.

Using the E-Library

The NIMC Library also provides an e-library containing digital resources such as magazines and online copies of open-source books added by NIMC.

A user searches the catalogue and can see whether a resource is available as a **physical book**, an **online resource**, or both.

The user opens an online resource through the e-library. Depending on the permission assigned to that resource, the user can either read it online or download it. The librarian who uploads the resource determines whether downloading is permitted.

The e-library can also be accessed publicly. Physical library membership is not necessarily required for resources that have been made publicly accessible.

The system keeps usage statistics for digital resources so that the librarian can see information about e-library usage.

Getting Book Recommendations from LibraryAssistantAI

A user can ask **LibraryAssistantAI** for book recommendations based on their interests, course, or learning needs.

For example, a participant asks:

> **Participant:** "I am doing the Cybersecurity for Bankers course. What book should I read?"

LibraryAssistantAI searches the books and resources available through the NIMC library and recommends relevant resources.

> **LibraryAssistantAI:** “I recommend checking the following books available in the library. You may ask me for reviews of any.”

The user can ask follow-up questions about recommended books, including their reviews, availability, and related resources. The assistant uses information available from the library catalogue and e-library when providing recommendations.

If a recommended physical book is currently unavailable, the assistant can inform the user and suggest other relevant available resources. The assistant can also direct the user to consult the librarian when a question requires information or judgment outside the available library data.

Reviewing Books

A user with an NIMC library account can submit a review for a book. The user can provide a written review and a rating.

Reviews are associated with the relevant book and can help other users decide whether a resource may be useful to them.

The librarian can view and moderate submitted reviews and remove inappropriate reviews when necessary.

Users can also ask LibraryAssistantAI about available book reviews when looking for recommendations.

Requesting a New Book

A registered user submits a request for a book that is not currently available in the NIMC library.

The system records the requested book and adds it to the library’s book-request list. The librarian can review the collection of requests and use the overall demand to identify books that may be useful additions to the library.

Book requests are therefore not necessarily treated as individual purchase orders. Multiple users requesting the same or similar resources can help the librarian identify resources that should be considered for acquisition.

When a requested book is approved for addition, the relevant user can receive a notification.

Managing Books and the Catalogue

The librarian or assistant librarian manages the library catalogue through the system. They can add new books and maintain their information, including title, author, catalogue information, ISBN, category, publisher, publication details, availability, and other relevant information.

The system keeps track of the status of each physical copy, including whether it is available, borrowed, overdue, or withdrawn.

The librarian or assistant librarian can also manage digital resources in the e-library, including uploading permitted magazines and open-source digital books and selecting whether a digital resource can be downloaded.

Detailed catalogue and circulation information is available to library staff, while ordinary users receive only the information relevant to finding and using library resources.

Library Notifications

The system sends notifications to users when important library events occur.

A user receives a reminder before a borrowed book reaches its due date. Users can also be notified when a previously unavailable book they requested or were waiting for becomes available.

The system can also send relevant notifications when a requested book is approved or when other library actions require the user's attention.

Dormitory Management

NIMC provides dormitory facilities for participants who require accommodation during their courses, seminars and workshops. The Dormitory Management System is designed to manage the dormitory application, approval, payment, room allocation, check-in, stay management, issue reporting, check-out and reporting activities through a single platform.

Applying for Dormitory

A participant who requires accommodation submits a dormitory application through the platform.

Accommodation Information

- Preferred type of accommodation
- Expected period of stay
- Accommodation requirement: Single-bed or Double-bed

The participant submits the application for either a single-seat or double-seat accommodation. After submission, the recorded application is forwarded to the Hostel Manager for review.

Dormitory Application Approval

After the application is submitted by the participant, the Hostel Manager starts the formal dormitory approval process.

The approval process shows important information about the participant and accommodation requirement -

- Seat availability

- Participant company
- Participant designation
- Participant stay time
- Reason for accommodation

Email notifications containing a direct approval link are sent to the responsible persons. The same approval link is also available through the portal.

The Director General (DG) first reviews the application. After the DG's review, the application goes to the Additional Director General (ADG) for review then to the Director and then the hostel manager.

After the required approvals are completed, the Hostel Manager reviews the application again and checks whether a suitable room and seat are available.

A list of available rooms and seats is shown to the Hostel Manager. The Hostel Manager uses this information to determine whether suitable accommodation is available for the participant.

If the application is approved but suitable accommodation is not available, the application remains pending until an appropriate room or seat becomes available.

Payment

After an application is approved and suitable accommodation is available, the system determines the applicable dormitory fee based on the participant's category and the selected accommodation type.

For example, the current fee structure is:

Participant Type	Single Seat Fee	Double Seat Fee
Government Employee	200	100
Non-Government Participant	400	200

Government employees may also receive a waiver when they are eligible and the waiver is approved by NIMC Director General.

The system applies the appropriate fee or approved waiver to the participant's application and sends the required payment information to NIMC Accounts for processing.

After the applicable dormitory fee is determined, the payment is processed through NIMC's Accounts. When payment is confirmed, or when an approved waiver applies, the system proceeds with room allocation.

Room Allocation

The Hostel Manager allocates an available room and seat according to the approved application and available dormitory capacity. The system updates the room and seat database so that the allocated space is no longer shown as available.

The participant can view the allocated room, seat and relevant stay information through the system.

Checking In and Managing the Stay

When the participant arrives at the dormitory, the Hostel Manager or designated dormitory staff records the participant's check-in through the system. The participant's stay status then changes to active.

The system maintains occupancy information throughout the participant's stay. The participant can view their accommodation information and stay details.

If the participant needs to remain in the dormitory beyond the originally approved period, they can submit a stay-extension request through the system. Similarly, a participant can submit a cancellation request when they no longer require the accommodation. Special cancellation cases can also be handled by the responsible dormitory authority.

The system updates the participant's stay information after the request is approved.

Reporting Dormitory Issues

During their stay, a participant can report problems related to the dormitory through the system. The issue report may contain a description of the problem and any other relevant information.

The issue is forwarded to the Hostel Super or Warden for action. The responsible staff member can update the status of the issue as it progresses.

For example:

> **Participant:** "The room's air conditioner is not working."

The Hostel Super/Warden receives the report, assigns or takes the necessary action, and updates the issue status until it is resolved.

The participant can view the progress of their reported issue through the system.

Checking Out

When the participant's approved stay ends, the participant checks out of the dormitory. The Hostel Manager or authorized dormitory staff records the check-out through the system.

After check-out, the participant’s occupancy status becomes inactive and the previously occupied room or seat is marked as available. The system retains the participant’s stay and check-out information for future reference.

If the participant leaves before the original end date, the same check-out process can be used after any necessary cancellation or adjustment has been approved.

If an approved cancellation, adjustment, or early departure requires a refund, the Hostel Manager and NIMC Accounts handle the applicable refund through the payment service.

Viewing Dormitory Availability

The Hostel Manager can view the current availability of dormitory rooms and seats through the system.

The system can display information such as:

- Total rooms
- Total seats
- Occupied seats
- Available seats
- Empty rooms
- Room type and capacity

The availability information is updated whenever a room or seat is allocated, cancelled, or released after check-out.

This allows the Hostel Manager to determine the current dormitory capacity while processing new applications and room allocations.

Dormitory Dashboard and Reports

The system provides a dashboard for dormitory management. The dashboard provides staff with an overview of current occupancy and availability.

At the end of each month, the system can generate a monthly dormitory report containing information such as:

- Total rooms and seats
- Occupied and available seats
- Empty rooms
- Number of applications
- Approved and pending applications
- Room allocations
- Check-ins and check-outs
- Stay extensions and cancellations
- Reported issues and their status

The Hostel Manager can view the report and generate a suitable downloadable or printable version for record keeping.

Dormitory Notifications and Service Evaluation

The system can send email notifications to participants when important dormitory events occur. These may include application approval, payment information, room allocation, stay-extension decisions, cancellation decisions, check-in or check-out information and issue-status updates.

In special cases, additional notification methods may be used when necessary.

The system may also allow participants to provide ratings or feedback about dormitory services and rooms. The collected ratings can help NIMC evaluate the quality of its dormitory facilities and identify areas that require improvement.

Navigation and AI Assistance

A participant can ask Hostel AI questions about relevant dormitory information. For example, the participant may ask, “What is the Wi-Fi password?” Hostel AI responds with the information available for the participant’s assigned room, such as, “The Wi-Fi password for Room 301 is nimco-wifi-301.”

The participant can also ask questions about the hostel layout, such as, “Where is the bathroom switch?” Hostel AI provides appropriate location guidance and may show the location on a 3D hostel map when that information is available.

Where supported IoT devices are integrated with the hostel, the participant can ask Hostel AI to perform actions such as “Turn off the lights” or “Start the geyser.” Hostel AI sends the supported command to the relevant device and confirms the result to the participant.

Level 0

Name: NIMC Management System

Primary Actor: Participant, Course Coordinator, Hostel Manager, Hostel Super, Librarian, Assistant Librarian

Secondary Actor: Speaker, Director General (DG), Additional Director General (ADG), Director, External Certificate Viewer

External System: Notification Service, Payment Service, AISystem

Description:

1. **NIMC Management System:** Provides shared access to account, classroom, library, and dormitory management functions for authorized NIMC users.

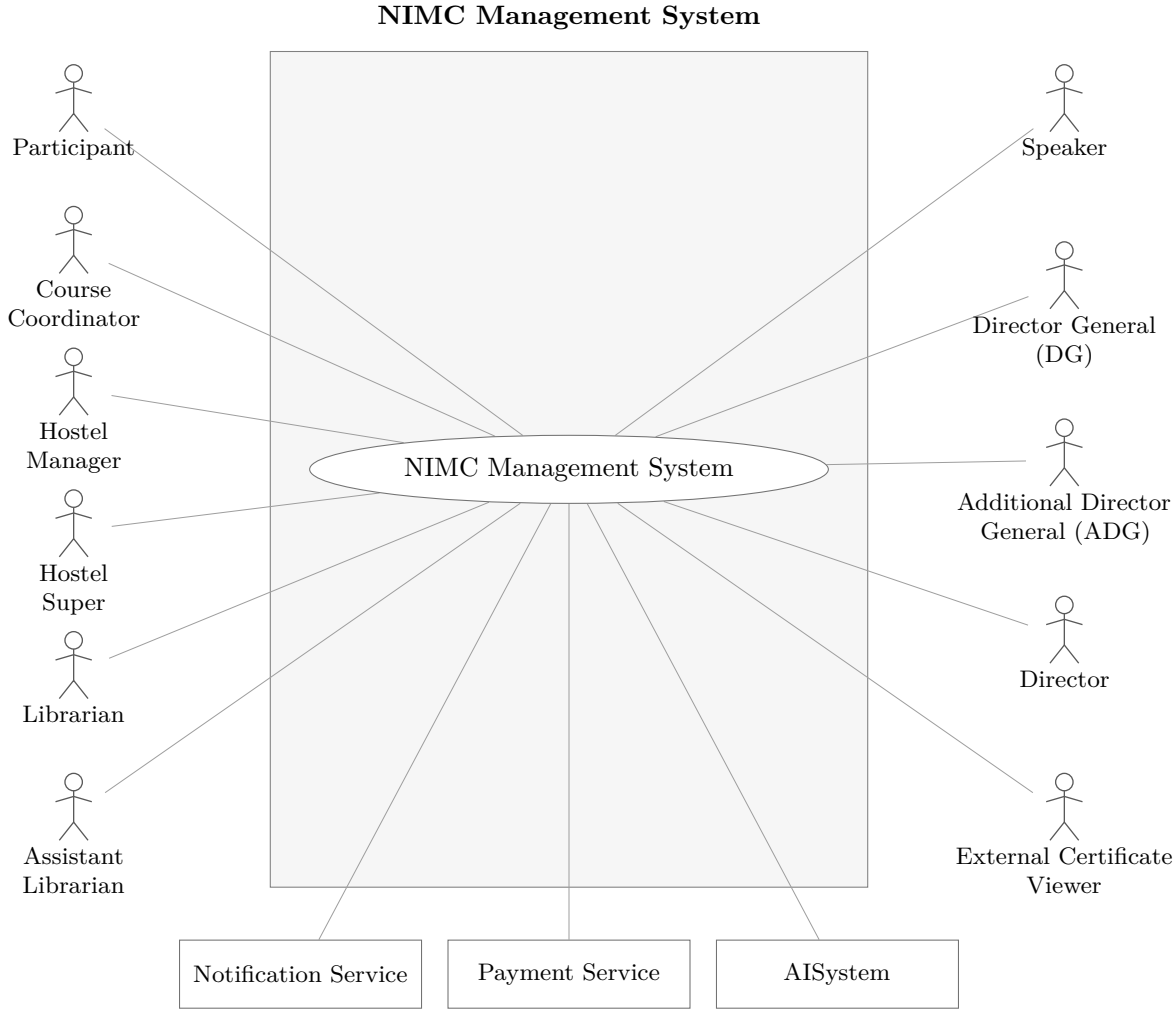


Figure 1: Level 0 — NIMC Management System

Level 1

Name: NIMC Management System

Primary Actor: Participant, Course Coordinator, Hostel Manager, Hostel Super, Librarian, Assistant Librarian

Secondary Actor: Speaker, Director General (DG), Additional Director General (ADG), Director, External Certificate Viewer

External System: Notification Service, Payment Service, AISystem

Description:

1. **Unified Account Management:** Maintains one NIMC identity and shared authentication for users across NIMC services.
2. **Classroom Management:** Manages training-course classroom activities.
3. **Library Management:** Manages library membership, circulation, catalogue, and digital-library activities.
4. **Dormitory Management:** Manages accommodation applications, approvals, payment and allocation, stays, issues, reporting, and hostel assistance.

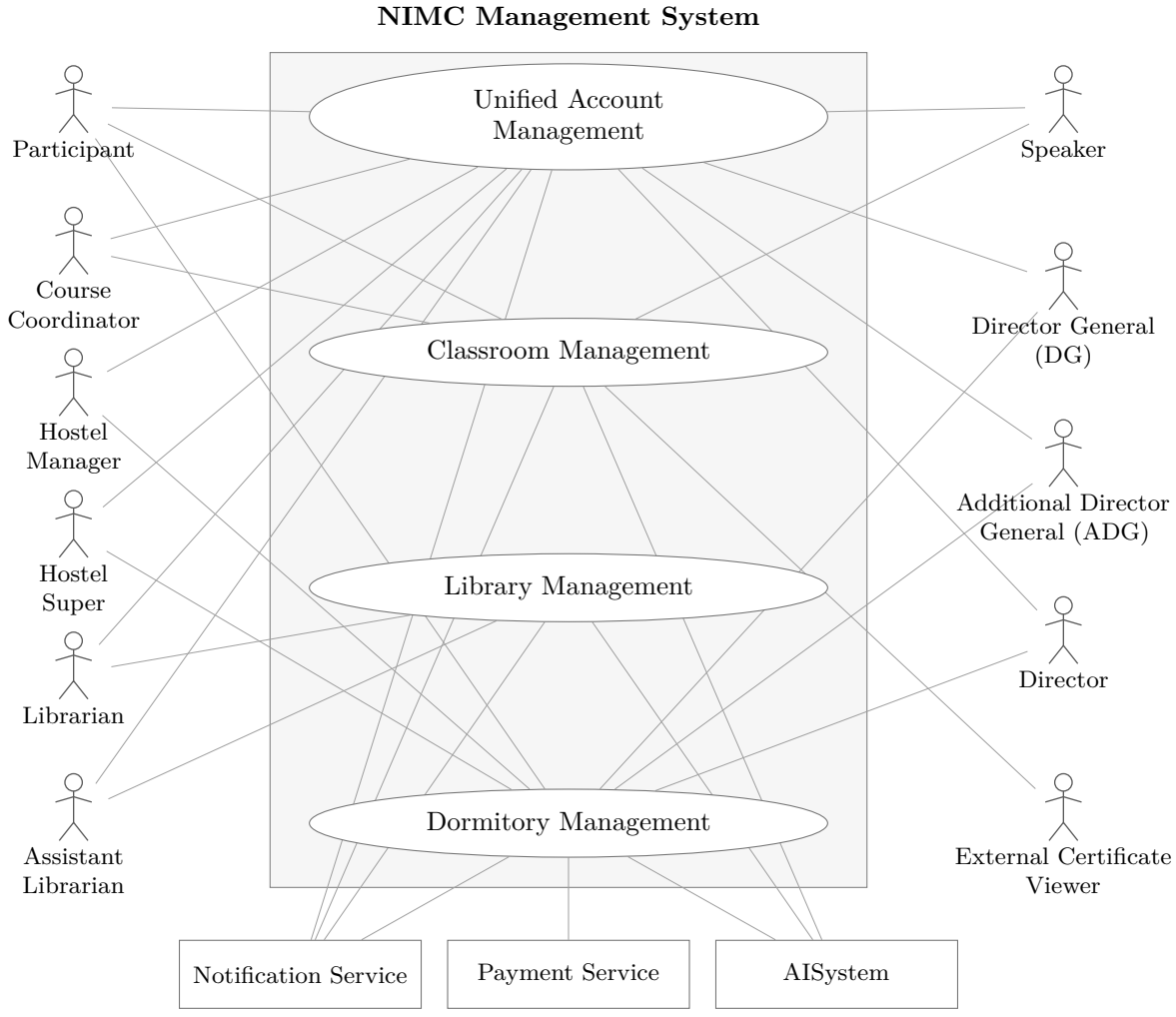


Figure 2: Level 1 — NIMC Management System

Level 1.1

Name: Unified Account Management

Primary Actor: Participant, Course Coordinator, Speaker, Hostel Manager, Hostel Super, Librarian, Assistant Librarian

Secondary Actor: Director General (DG), Additional Director General (ADG), Director

External System: Notification Service

Description:

1. **Accounts System:** A person registers a new NIMC account.
2. **Role and Access Management:** Account holders verify their registered email address, sign in and sign out, and recover their password through the registered email address. The Director and designated Course Coordinator, Hostel Manager, and Librarian assign or update roles and permissions for Classroom, Library, or Dormitory functions.
3. **Profile and Module Management:** Common identity information such as name, phone, email, company, and designation is maintained in one place.

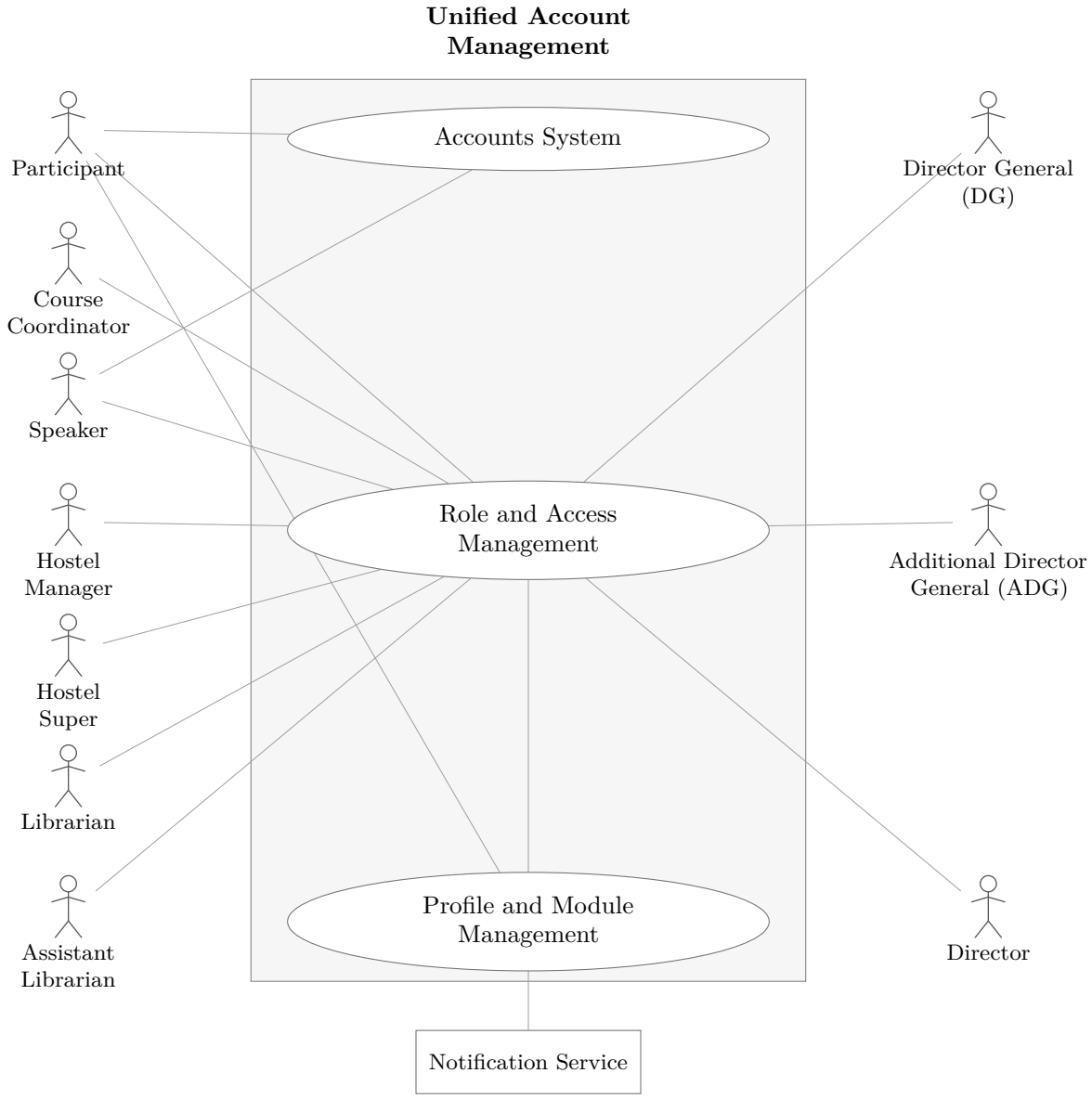


Figure 3: Level 1.1 — Unified Account Management

Level 1.2

Name: Classroom Management

Primary Actor: Participant, Course Coordinator

Secondary Actor: Speaker, External Certificate Viewer

External System: Notification Service, AISystem

Description:

1. **Course Management:** Handles course creation, enrollment, participant information, participant review when required, removal of ineligible participants, speaker management, and certificate management.
2. **Session Management:** Combines planning, scheduling and rescheduling, session and speaker information, notifications, attendance recording, and AI-supported engagement review.
3. **Course Resource Management:** Combines learning-resource upload, organization, access, search, sorting and filtering, resource notifications, and course-resource AI assistance.
4. **Academic Assessment Management:** Handles exams and assignments, participant submissions, AI-supported work review and evaluation, and assessment-result recording.

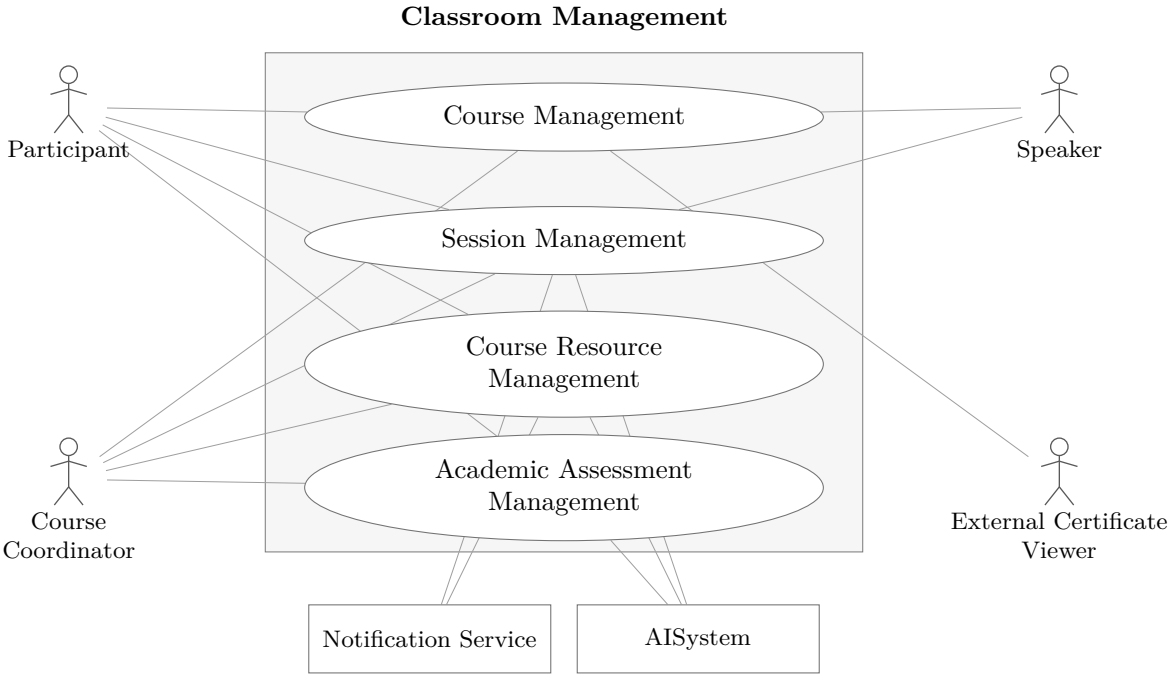


Figure 4: Level 1.2 — Classroom Management

Level 1.2.1

Name: Course Management

Primary Actor: Participant, Course Coordinator

Secondary Actor: Speaker, External Certificate Viewer

External System: None

Description:

1. **Course Creation and Participant Management:** The Course Coordinator creates a classroom for a particular course. The Course Coordinator provides the classroom code to eligible participants. A participant uses the code to request or complete enrollment in the appropriate classroom. The Course Coordinator records participant information, reviews an incoming participant before enrollment when required, accepts or rejects the participant as appropriate, and removes a participant later found to be ineligible.
2. **Speaker Management:** The Course Coordinator records and maintains information about invited speakers. Participants can view the profile of the speaker assigned to a session. After a session, a participant can submit a numerical rating and optional comment for the speaker and may choose a named or anonymous submission. The Course Coordinator reviews the speaker's overall rating and rating history across courses. Cross-course rating history remains restricted to the Course Coordinator.
3. **Certificate Management:** After required course and assessment conditions are satisfied, the Course Coordinator issues the certificate to an eligible participant. The participant accesses the issued certificate together with its QR information. An External Certificate Viewer uses the certificate QR information to check the issued certificate and the participant information associated with it.

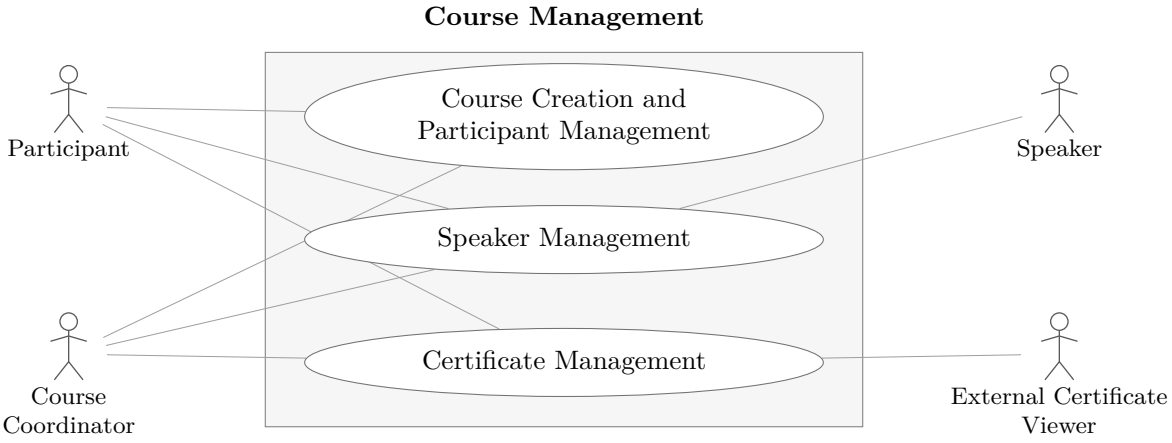


Figure 5: Level 1.2.1 — Course Management

Level 1.2.2

Name: Session Management

Primary Actor: Participant, Course Coordinator

Secondary Actor: Speaker

External System: Notification Service, AISystem

Description:

1. **Session Scheduling:** The Course Coordinator prepares the initial session plan and maintains or reschedules its topic, date, time, and assigned speaker.
2. **Session Information and Notification:** Participants and speakers access the session plan and assigned speaker information. The Course Coordinator sends important schedule or speaker-change notices by email, or by SMS for a special case. Participants and speakers receive the notices.
3. **Attendance System:** The Course Coordinator records attendance and reviews attendance, assessments, participation, and AI-supported engagement indicators to identify participants requiring attention. Participants access their own attendance records.

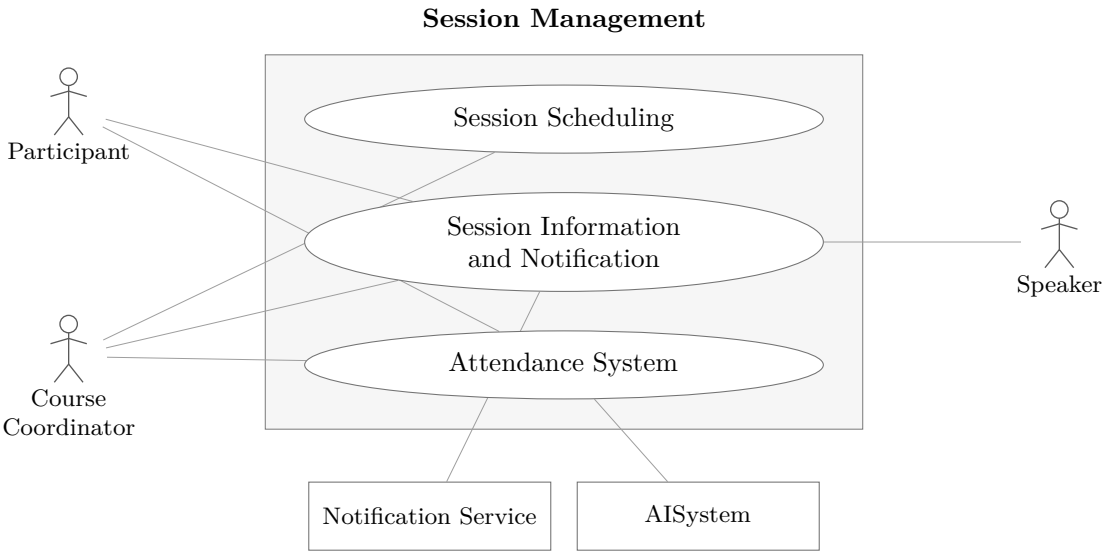


Figure 6: Level 1.2.2 — Session Management

Level 1.2.3

Name: Course Resource Management

Primary Actor: Participant, Course Coordinator

Secondary Actor: None

External System: Notification Service, AISystem

Description:

1. **Learning Resource Management:** The Course Coordinator uploads and organizes documents, presentations, videos, notes, and authorized session recordings. Participants access resources and use search, sort, and filter options to find material.
2. **Resource Notification:** The Course Coordinator emails participants when an important new resource or recording is available.
3. **Classroom AI:** The planned course assistant lets participants ask course questions, receive answers grounded in available resources, find relevant material, and request explanations.

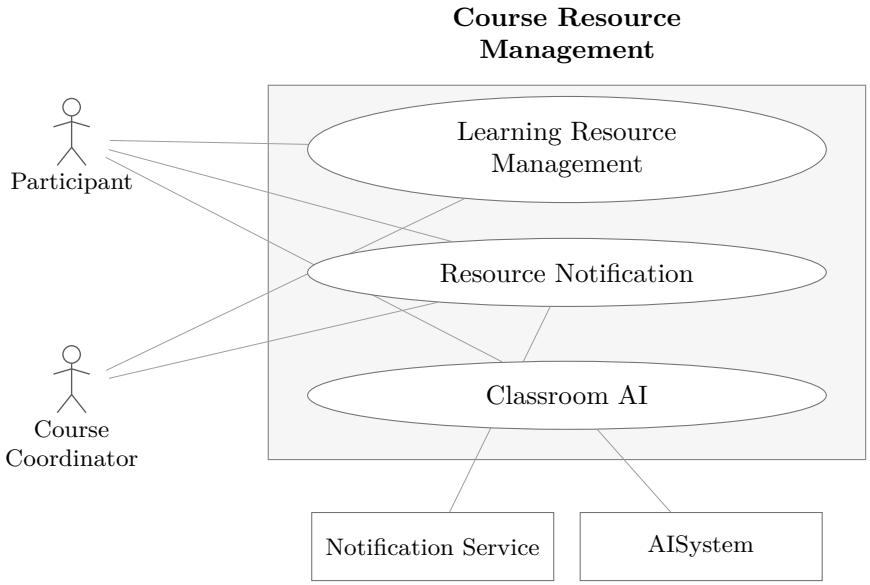


Figure 7: Level 1.2.3 — Course Resource Management

Level 1.2.4

Name: Academic Assessment Management

Primary Actor: Participant, Course Coordinator

Secondary Actor: None

External System: AISystem

Description:

1. **Exam or Assignment Management:** The Course Coordinator publishes examination or assignment requirements. Participants access the requirements and submit their work.
2. **Work Review and Evaluation:** The Course Coordinator requests and reviews AI indicators for similarity, possible plagiarism, and possible AI-generated content. These indicators support review; the Course Coordinator makes the final academic decision. The Course Coordinator evaluates the submitted work and records assessment results after review. Participants access their results.

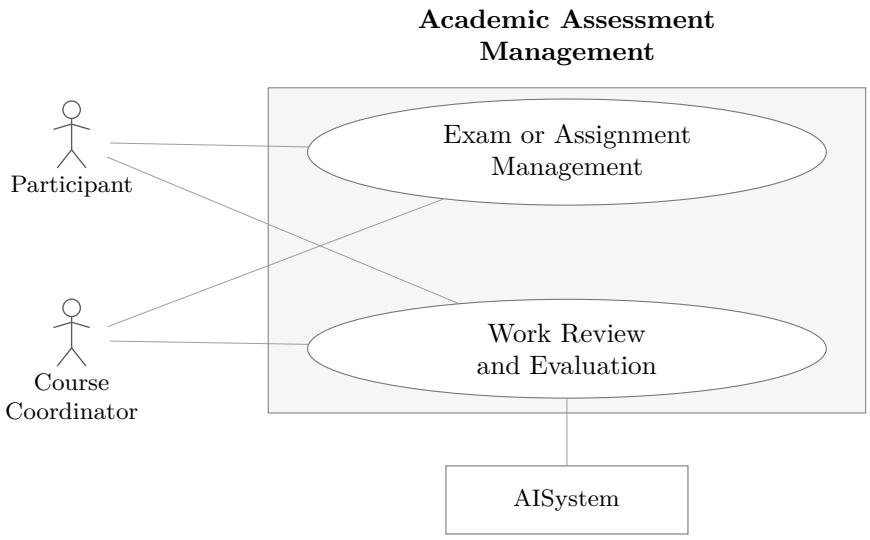


Figure 8: Level 1.2.4 — Academic Assessment Management

Level 1.3

Name: NIMC Library Management System

Primary Actor: User, Librarian, Assistant Librarian

Secondary Actor: None

External System: Notification Service, AISystem

Description:

1. **Membership and Access:** Verifies membership and issues digital membership cards.
2. **Library Resource Management:** Combines catalogue search, availability, staff circulation records, recommendations, reviews, and library AI assistance.
3. **Circulation Management:** Handles book issue and return, overdue penalties, waitlist notifications, and wishlists.
4. **Digital Library:** Supports digital-resource discovery, permitted reading and downloading, usage analytics, and collection management.

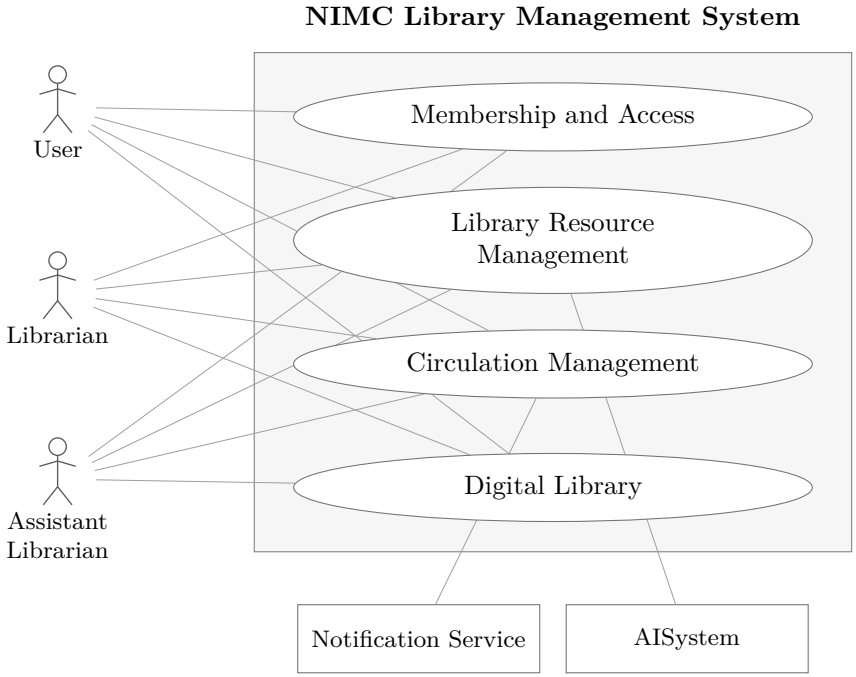


Figure 9: Level 1.3 — NIMC Library Management System

Level 1.3.1

Name: Membership and Access

Primary Actor: User (Course Participant, Office Employee, Government Employee, External Visitor), Assistant Librarian

Secondary Actor: None

External System: None

Description:

1. **Membership Verification and Assignment:** The Assistant Librarian verifies submitted information and assigns a membership category and appropriate validity period, adjustable for cases such as course duration.
2. **Digital Membership Card Issuance:** Upon approval, the system generates a digital card showing member details, membership ID, validity, category, company, designation, and a QR code.

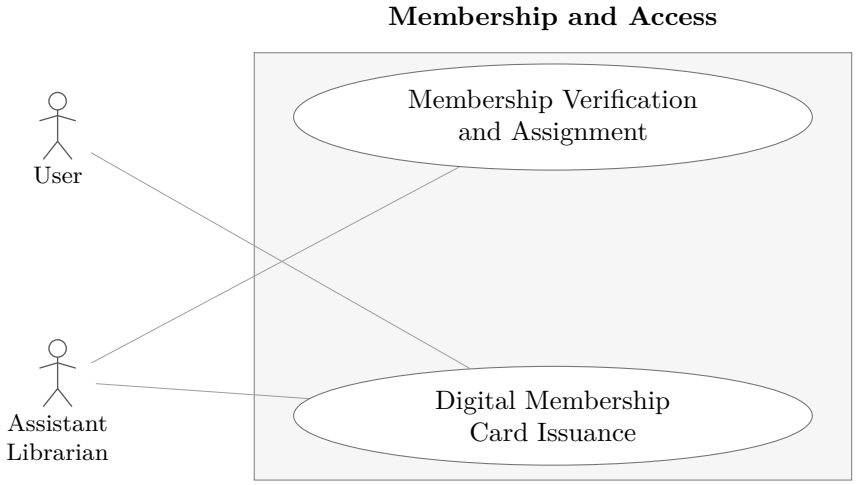


Figure 10: Level 1.3.1 — Membership and Access

Level 1.3.2

Name: Library Resource Management

Primary Actor: User, Librarian, Assistant Librarian

Secondary Actor: None

External System: AISystem

Description:

1. **Library Catalogue System:** Users search by title, author, ISBN, category, or other book information and check physical or online availability. Librarians and assistant librarians additionally access detailed circulation records, including borrower, issue date, and due date.
2. **Recommendations and Reviews System:** Users receive recommendations and submit book ratings and reviews; the Librarian moderates reviews.
3. **LibraryAI System:** Users ask for resource recommendations and follow-up information about reviews, availability, and related resources. Questions requiring staff judgment are referred to the Librarian.

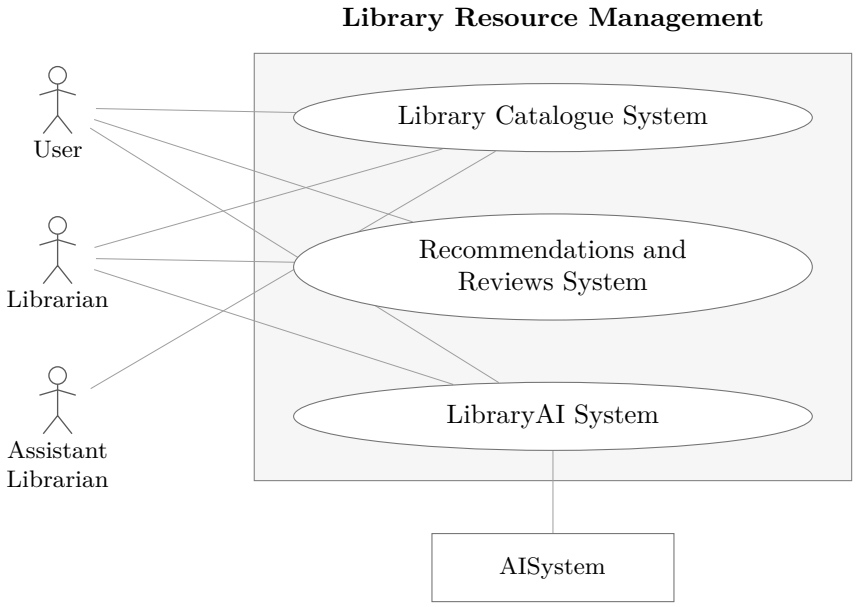


Figure 11: Level 1.3.2 — Library Resource Management

Level 1.3.3

Name: Circulation Management

Primary Actor: User, Librarian, Assistant Librarian

Secondary Actor: None

External System: Notification Service

Description:

1. **Book Issue and Return:** Staff checks borrowing eligibility, records issue and due dates, and updates availability when a physical book is issued or returned.
2. **Overdue Penalty Processing:** The system flags overdue returns and applies the configured penalty. The Librarian may override the penalty, with the adjustment logged.
3. **Waitlisted User Notification:** Users receive due-date reminders and availability notifications for requested or held books.
4. **Wishlist System:** Users maintain a personal list of books or resources they want to access.

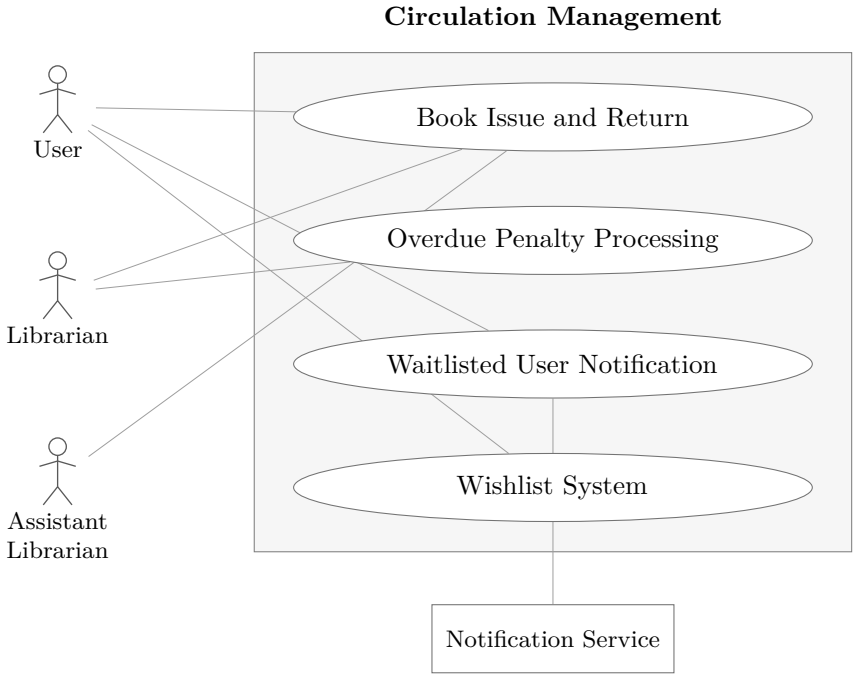


Figure 12: Level 1.3.3 — Circulation Management

Level 1.3.4

Name: Digital Library

Primary Actor: User, External Visitor

Secondary Actor: Librarian, Assistant Librarian

External System: None

Description:

1. **Resource Discovery:** Users and external visitors browse and search digital resources and check availability within their access permissions.
2. **Digital Resource Access:** Users read online or download digital resources according to the permissions set for each resource.
3. **Usage Analytics:** The Librarian reviews digital-resource usage statistics.
4. **Digital Collection Management:** The Librarian and Assistant Librarian upload and maintain permitted magazines and open-source books, including download permissions.

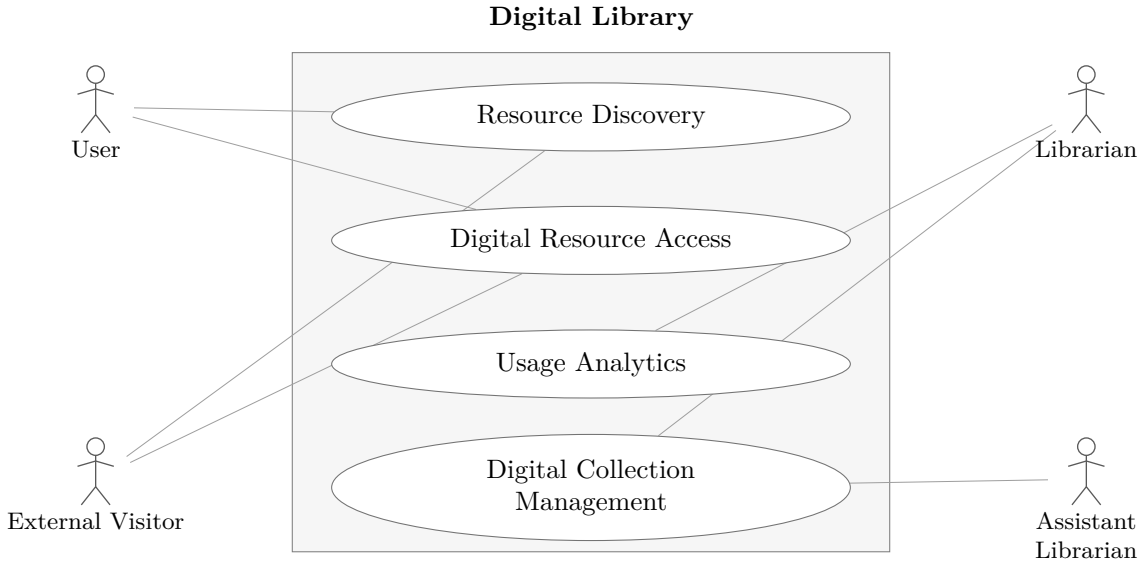


Figure 13: Level 1.3.4 — Digital Library

Level 1.4

Name: Dormitory Management

Primary Actor: Participant, Hostel Manager, Hostel Super

Secondary Actor: Director General (DG), Additional Director General (ADG), Director, NIMC Accounts

External System: Payment Service, Notification Service, AI Service

Description:

1. **Dormitory Application:** Combines accommodation applications, initial review, and the formal approval chain.
2. **Accommodation Allocation:** Handles fee waivers, dormitory payments, and room and seat allocation.
3. **Stay Management:** Combines check-in and check-out, ratings, refunds, extension or cancellation requests, and issue reporting.
4. **Dormitory Information System:** Combines occupancy information, monthly reports, application and stay statistics, and hostel-information assistance.

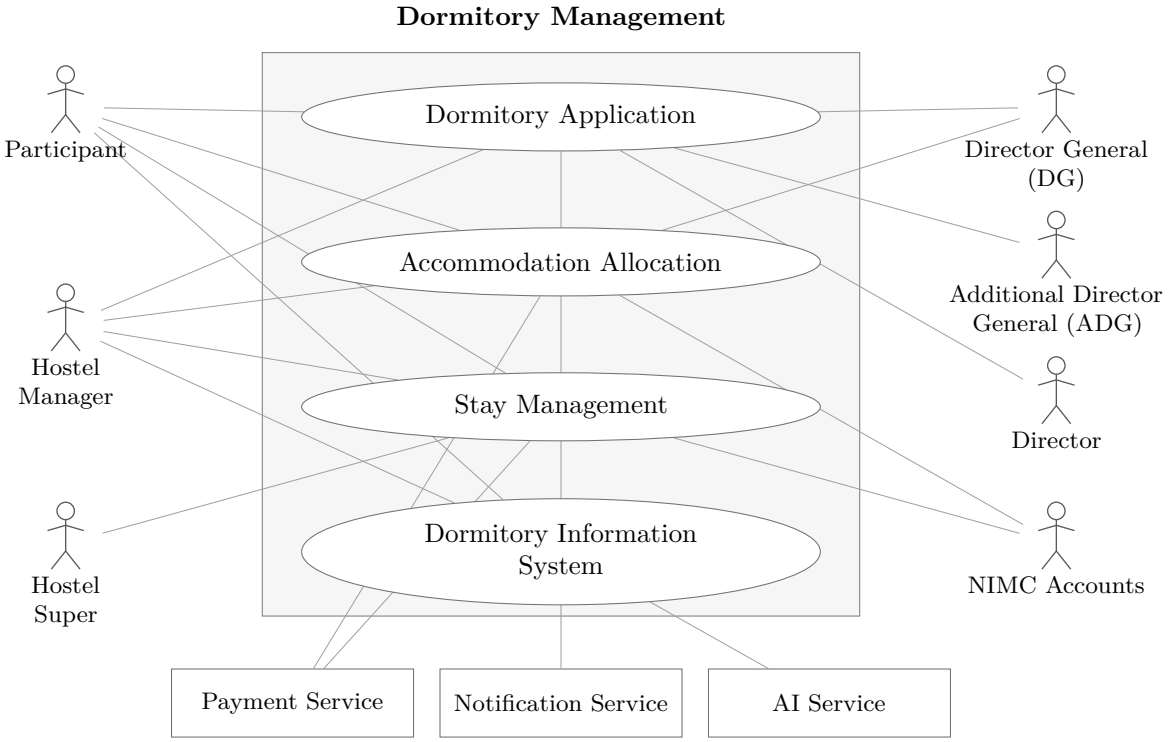


Figure 14: Level 1.4 — Dormitory Management

Level 1.4.1

Name: Dormitory Application

Primary Actor: Participant, Hostel Manager

Secondary Actor: Director General (DG), Additional Director General (ADG), Director

External System: Notification Service

Description:

1. **Accommodation Application System:** The participant creates and submits an application using the unified NIMC account, specifying accommodation type and stay dates. The Hostel Manager checks completeness and availability and requests formal approval.
2. **Application Approval System:** The application is reviewed in sequence by the Director General, Additional Director General, Director, and Hostel Manager. Notification Service sends approval links by email to the responsible reviewers.

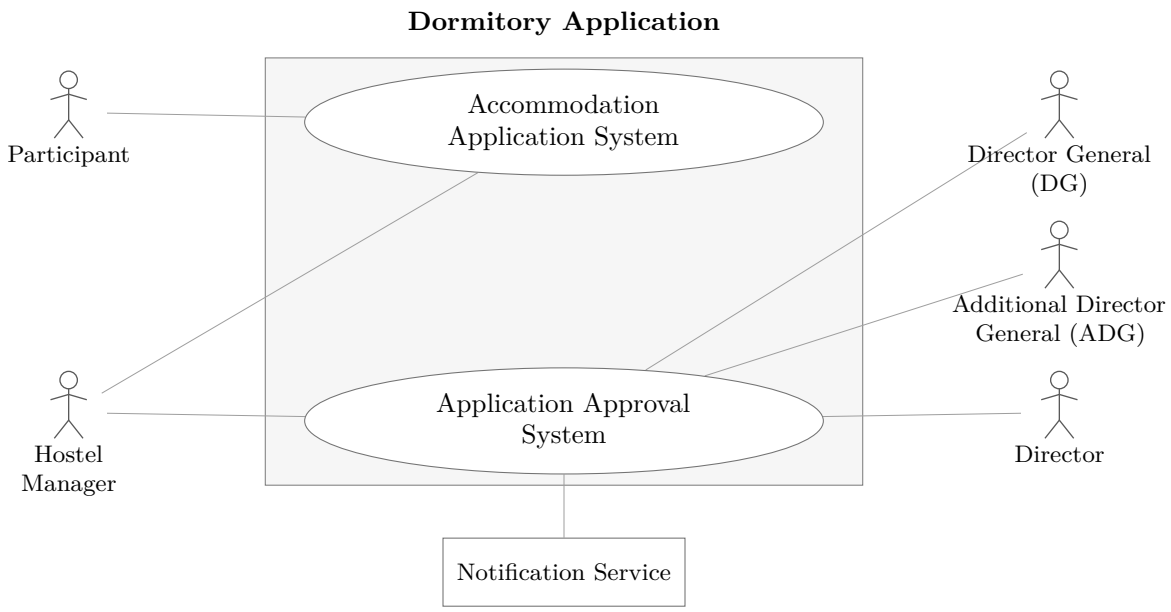


Figure 15: Level 1.4.1 — Dormitory Application

Level 1.4.2

Name: Accommodation Allocation

Primary Actor: Participant, Hostel Manager

Secondary Actor: Director General (DG), NIMC Accounts

External System: Payment Service

Description:

1. **Waiver System:** Eligible government employees request a fee waiver. The Director General approves eligible requests; the participant and Hostel Manager access the waiver status.
2. **Dormitory Payment System:** The participant and Hostel Manager access the applicable fee information. NIMC Accounts processes and confirms the dormitory payment through Payment Service.
3. **Room Allocation System:** After confirmed payment or an approved waiver, the Hostel Manager allocates a suitable room and seat. The participant accesses the allocation details.

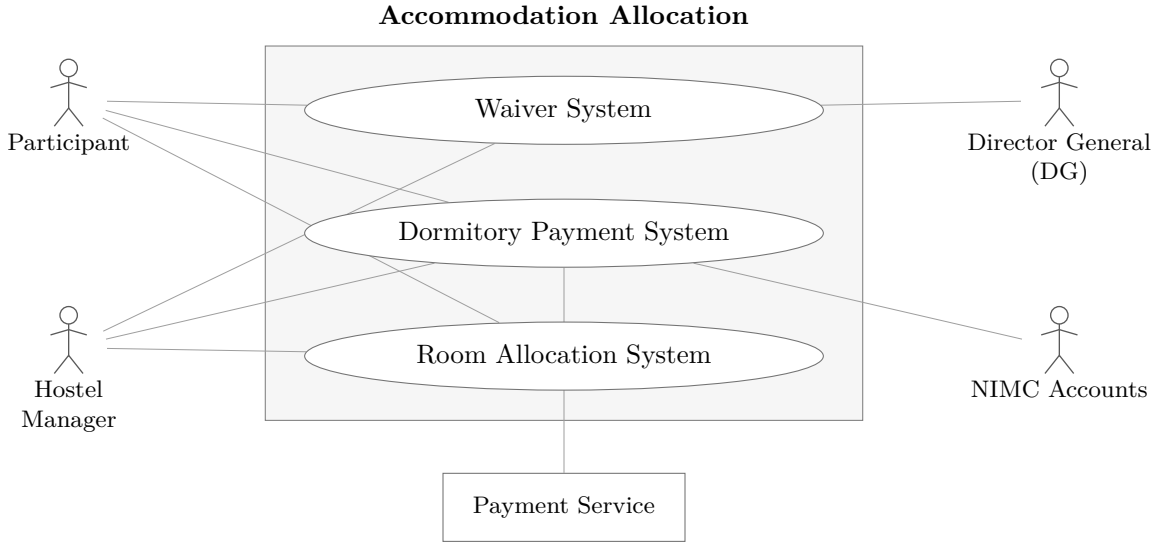


Figure 16: Level 1.4.2 — Accommodation Allocation

Level 1.4.3

Name: Stay Management

Primary Actor: Participant, Hostel Manager

Secondary Actor: Hostel Super, NIMC Accounts

External System: Payment Service

Description:

1. **Check-In and Out System:** The Hostel Manager records participant arrival and departure; participants access their accommodation and stay details. After the stay, the participant submits a rating and feedback about the dormitory room and services.
2. **Extension/Cancellation Application System:** The participant submits a request to extend or cancel a stay; the Hostel Manager handles the request. The Hostel Manager and NIMC Accounts handle an applicable refund after an approved adjustment or early departure through Payment Service.
3. **Issue Report System:** The participant reports a problem and tracks its progress. The Hostel Super takes or assigns action; the Hostel Super or Hostel Manager updates the issue status.

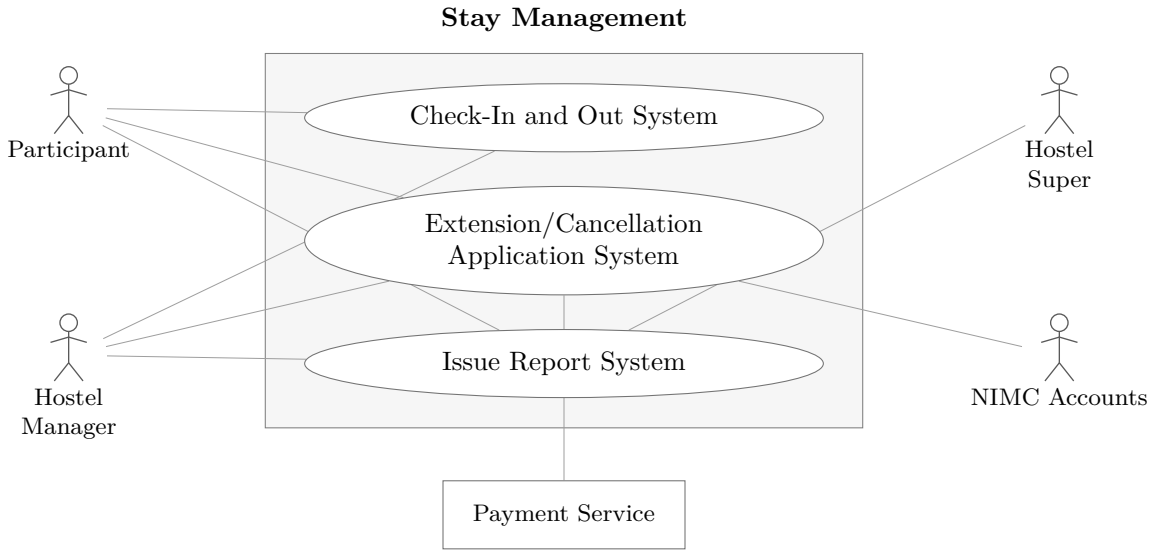


Figure 17: Level 1.4.3 — Stay Management

Level 1.4.4

Name: Dormitory Information System

Primary Actor: Hostel Manager

Secondary Actor: Participant

External System: AI Service

Description:

1. **Dormitory Information Access:** The Hostel Manager accesses room and seat totals, occupancy, availability, room types, and capacity.
2. **Monthly Dormitory Reporting:** The Hostel Manager generates a monthly report for digital access and record keeping.
3. **Application and Stay Statistics:** The Hostel Manager accesses application totals, approvals, pending applications, allocations, check-ins, check-outs, extensions, cancellations, and issue statistics.
4. **HostelAI:** Participants obtain hostel information and layout guidance and request supported IoT actions. The Hostel Manager supports hostel-information queries.

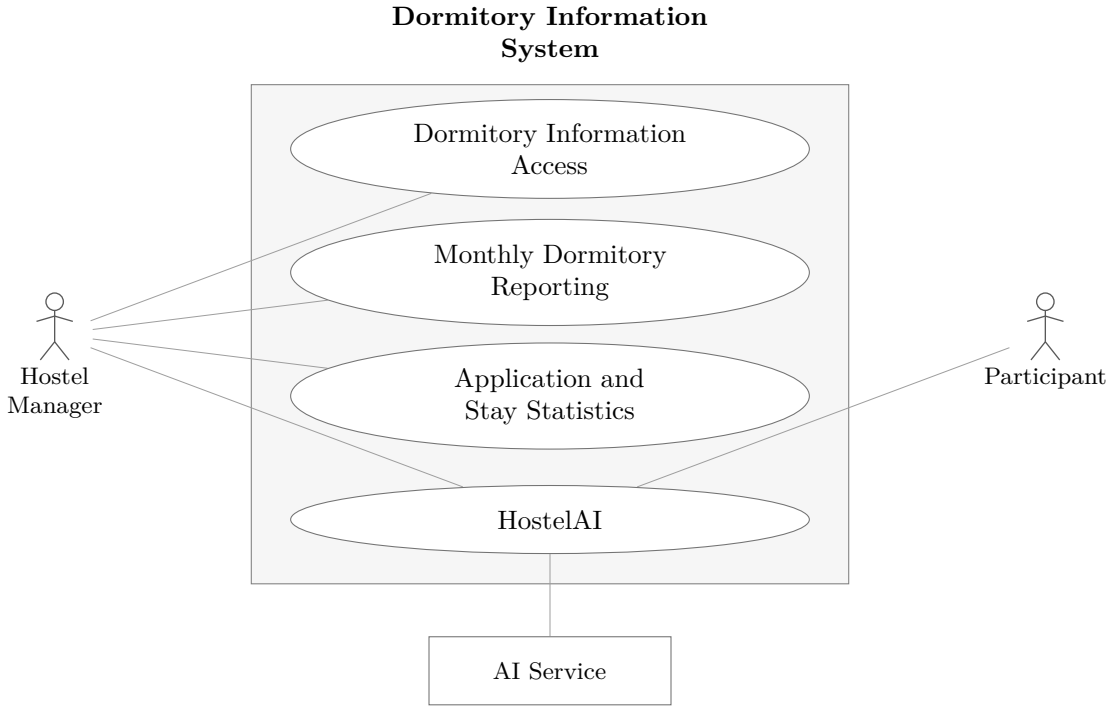


Figure 18: Level 1.4.4 — Dormitory Information System

Activity Diagram

Level 1.1:

Name: Unified Account Management

Reference: Use Case Diagram Level 1.1

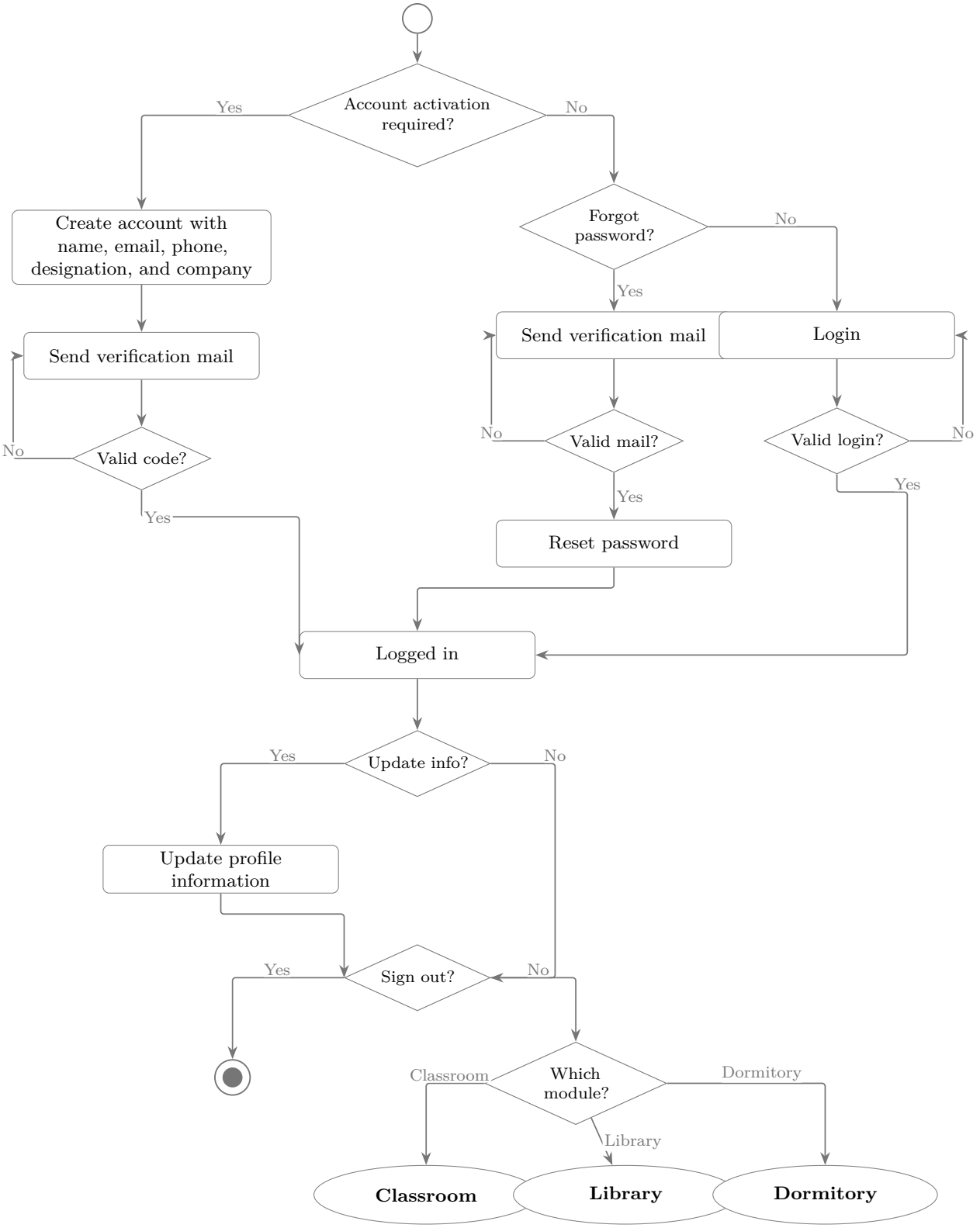


Figure 1: **Level 1.1: Unified Account Management**

Activities:

Accounts System:

A1: A person creates a NIMC account with name, email, phone, designation, and company. NID can be included.

R1: The account record is stored and is ready for email verification.

Role and Access Management:

A1: Notification System sends a verification mail. The user submits the code. An invalid code returns to the mail, and a valid code logs the user in.

R1: The verified account is open.

A2: If the password is forgotten, Notification System sends a verification mail, the user confirms the mail, and then resets the password. Otherwise the user logs in. The user can sign out.

R2: A valid login or a completed reset logs the user in. Signing out ends the session.

Profile and Module Management:

A1: When the user chooses to update info, name, phone, email, company, or designation is changed. Otherwise the flow continues.

R1: The shared profile is updated.

A2: If the user does not sign out, the user opens Classroom, Library, or Dormitory.

R2: Control passes to the selected module.

Level 1.2:

Name: Classroom Management

Reference: Use Case Diagram Level 1.2

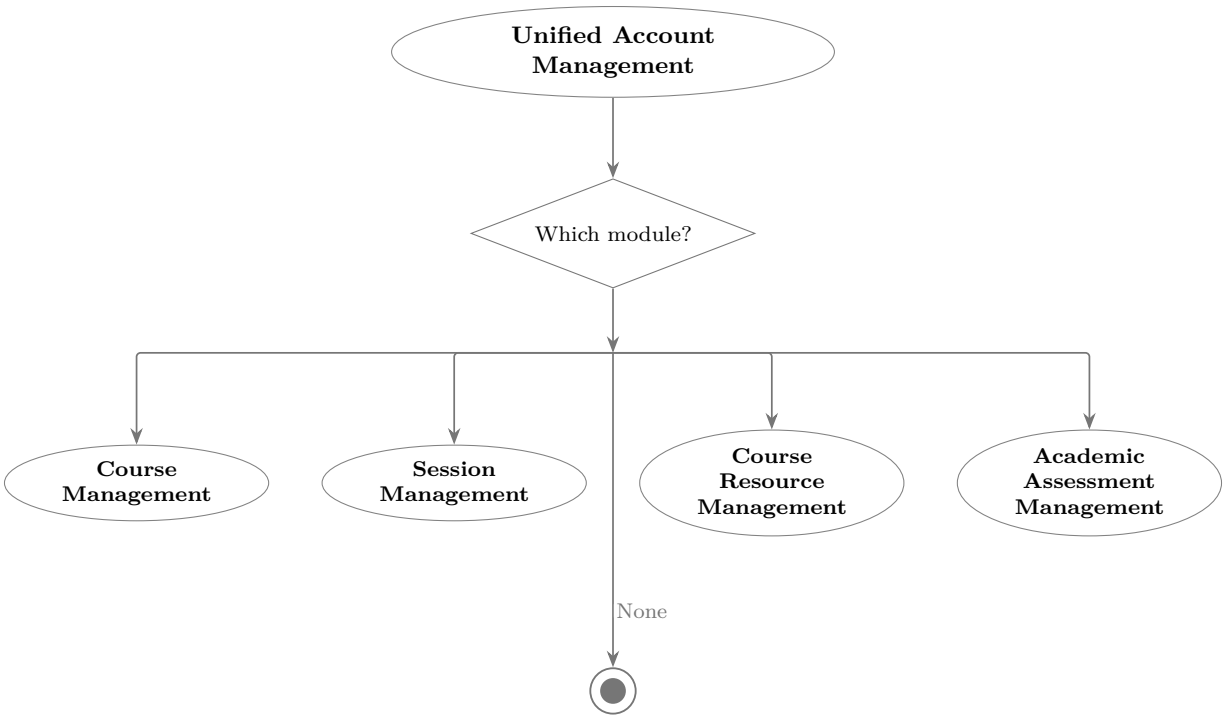


Figure 1: **Level 1.2: Classroom Management**

Activities:

Course Management:

A1: The actor asks for course creation, enrollment, participant review, or certificate functions.

R1: The request is routed to Course Management.

Session Management:

A1: The actor asks for session planning, session information, notification, or attendance functions.

R1: The request is routed to Session Management.

Course Resource Management:

A1: The actor asks for learning resources, a resource notice, or the course assistant.

R1: The request is routed to Course Resource Management.

Academic Assessment Management:

A1: The actor asks for an assignment, an examination, or result review.

R1: The request is routed to Academic Assessment Management.

Level 1.2.1:

Name: Course Management

Reference: Use Case Diagram Level 1.2.1

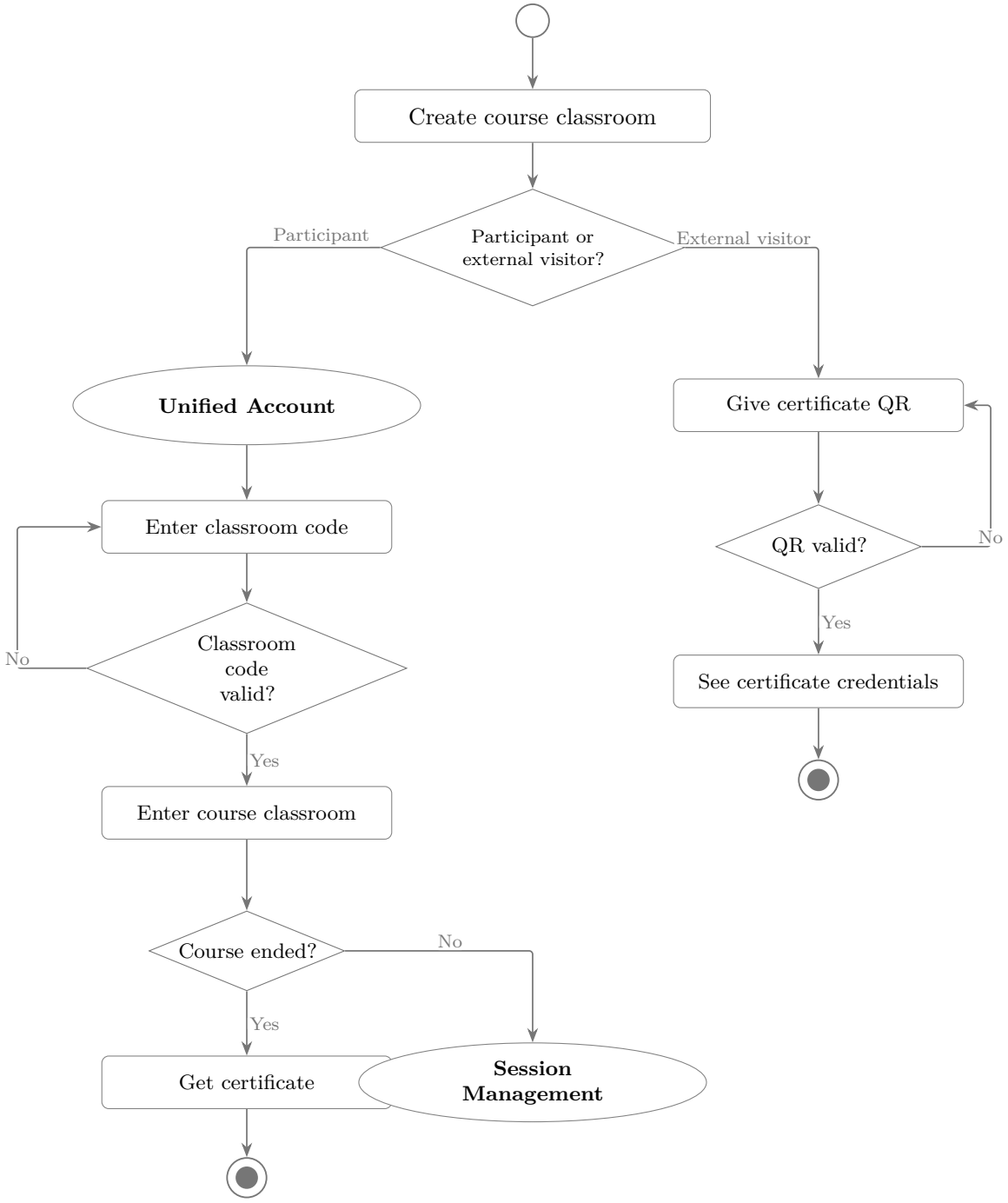


Figure 2: **Level 1.2.1: Course Management**

Activities:

Course Creation and Participant Management:

A1: Course Coordinator enters the course name, creates the classroom, and shares the classroom code with eligible participants.

R1: The classroom exists, and participants can use the code to enter it.

A2: Participant comes through Unified Account and enters the classroom code. A valid code opens the course classroom. The Course Coordinator can remove a participant later found to be ineligible.

R2: An invalid code returns the participant to enter the code again. A removed participant no longer has classroom access. When the course has not ended, control passes to Session Management.

Speaker Management:

A1: Course Coordinator keeps the invited speaker's information, and a participant can view the speaker assigned to a session. After a session, the participant submits a numerical rating and an optional comment, named or anonymous.

R1: The Course Coordinator can view the speaker's overall rating and the rating history across courses.

Certificate Management:

A1: When the course has ended, an eligible participant receives the issued certificate.

R1: The certificate and its QR information are available to the participant.

A2: External Certificate Viewer gives the certificate QR.

R2: A valid QR shows the certificate and the associated participant information. An invalid QR is given again.

Level 1.2.2:

Name: Session Management

Reference: Use Case Diagram Level 1.2.2

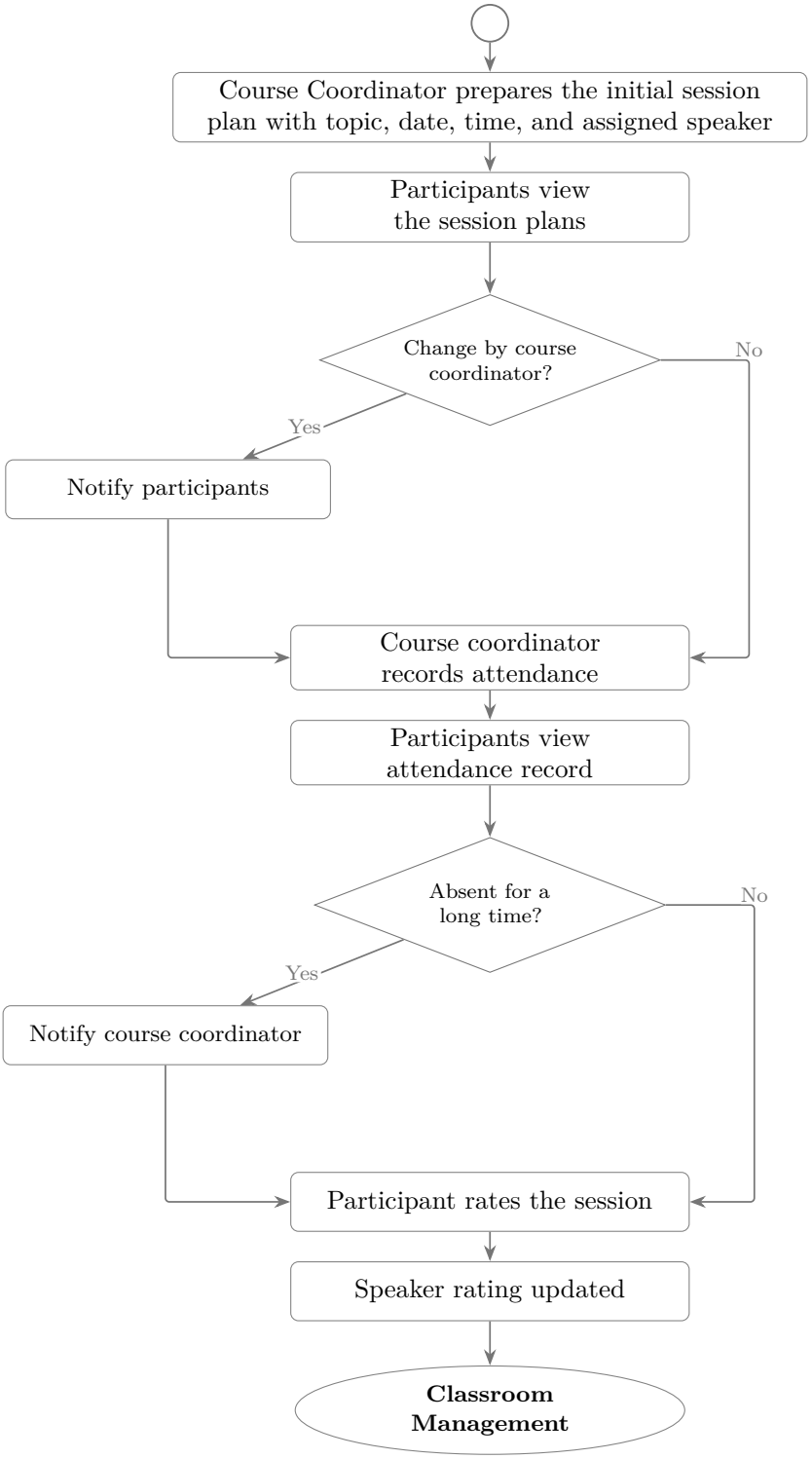


Figure 3: **Level 1.2.2: Session Management**

Activities:

Session Scheduling:

A1: Course Coordinator prepares the initial session plan with the topic, date, time, and assigned speaker.

R1: The session plan is available for the course.

A2: When the Course Coordinator changes the plan, participants are notified. When there is no change, the flow continues.

R2: Notification service delivers a change notice by email, or by SMS when the case requires it.

Session Information and Notification:

A1: Participants view the session plans, including the topic, date, time, and assigned speaker.

R1: The current session plan is shown to the participants and the assigned speaker.

A2: After the session, the participant rates the session and the speaker rating is updated.

R2: The rating and optional comment are stored, named or anonymous. The Course Coordinator can view the rating history. The flow then returns to Classroom Management.

Attendance System:

A1: Course Coordinator records attendance, and the participant views their own attendance record.

R1: The session attendance record is stored and shown to the participant.

A2: If a participant has been absent for a long time, the Course Coordinator is notified. Otherwise the flow continues.

R2: Attendance and the engagement indicator identify who needs attention.

Level 1.2.3:

Name: Course Resource Management

Reference: Use Case Diagram Level 1.2.3

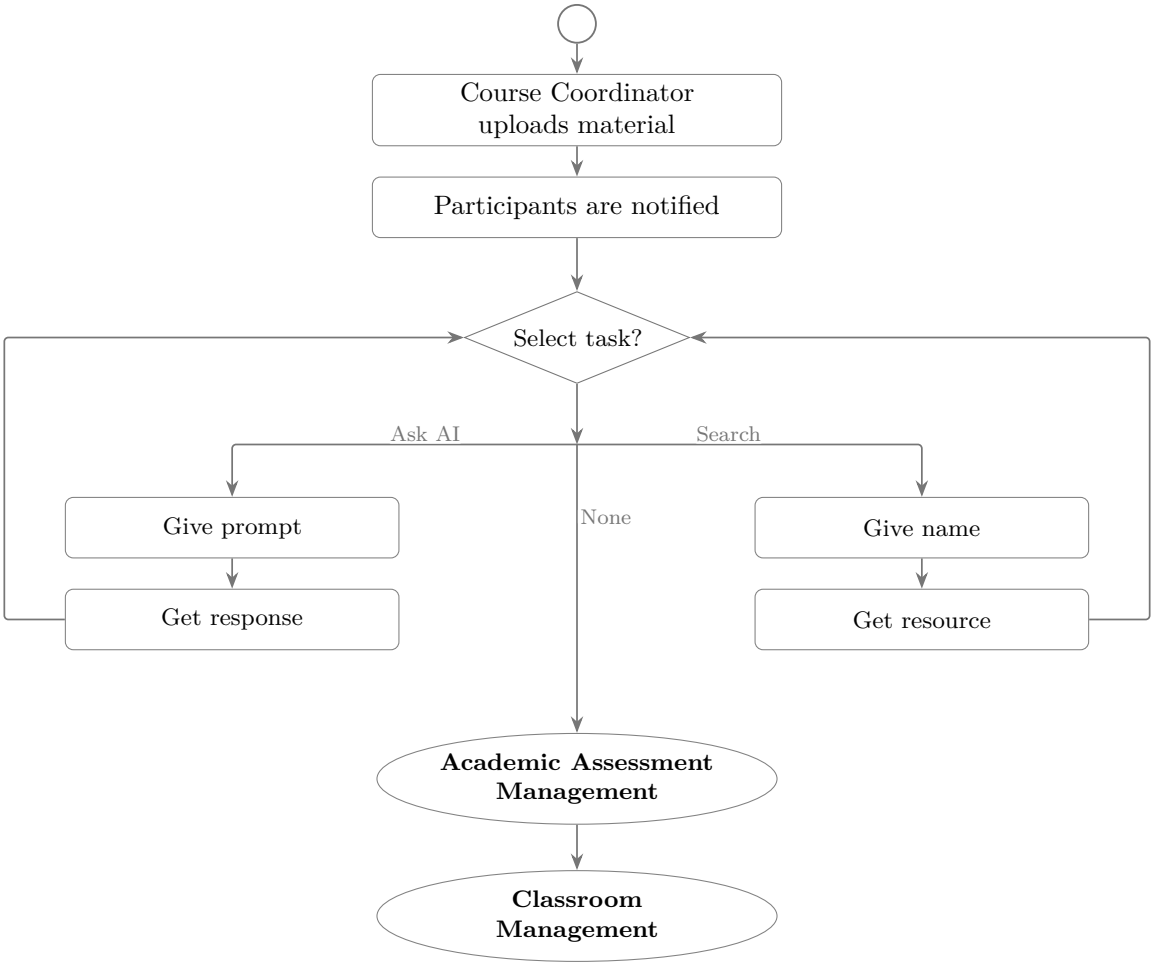


Figure 4: **Level 1.2.3: Course Resource Management**

Activities:

Learning Resource Management:

A1: Course Coordinator uploads documents, presentations, videos, notes, or an authorized session recording.

R1: The material is stored with the course resources.

A2: Participant searches by giving a resource name and receives the matching resource, then may choose another task.

R2: The resource is shown. Sort and filter remain available while looking through the material.

Resource Notification:

A1: Participants are notified when new material is available.

R1: Notification service delivers the resource notice.

Classroom AI:

A1: Participant gives a prompt and receives a response, then may choose another task.

R1: AISystem returns a response grounded in the stored course resources.

A2: When the participant selects no further task, the flow continues to Academic Assessment Management and then returns to Classroom Management.

R2: Control leaves Course Resource Management.

Level 1.2.4:

Name: Academic Assessment Management

Reference: Use Case Diagram Level 1.2.4

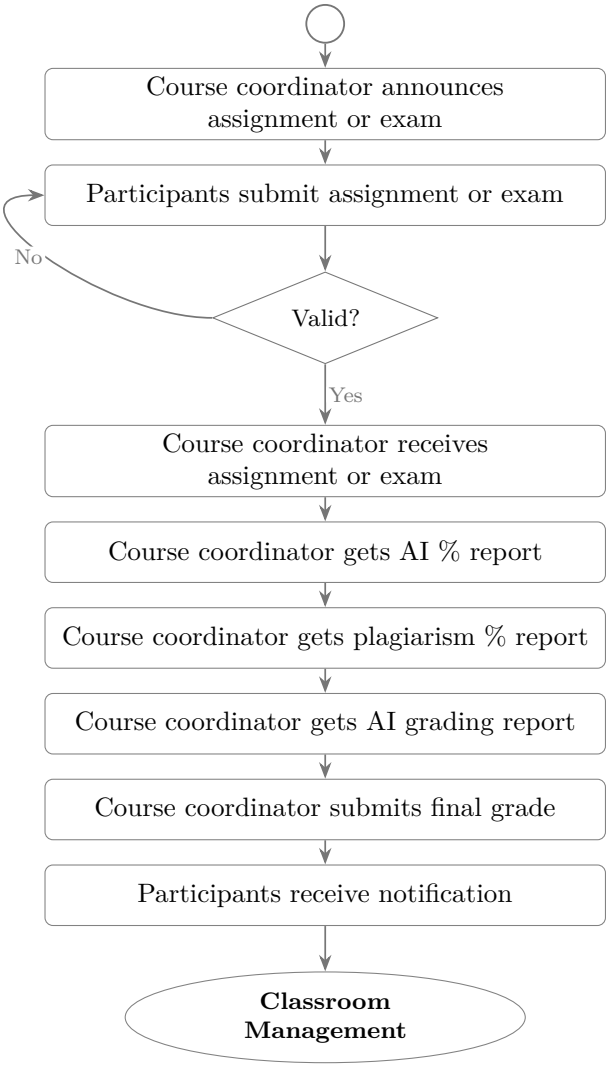


Figure 5: **Level 1.2.4: Academic Assessment Management**

Activities:

Exam or Assignment Management:

A1: Course Coordinator announces the assignment or examination paper.

R1: Participants can see the announced paper.

A2: Participants submit the assignment or examination. An invalid submission is submitted again.

R2: A valid submission is received by the Course Coordinator and is ready for review.

Work review and evaluation:

A1: Course Coordinator gets the AI-content percentage report, the plagiarism percentage report, and the AI grading report.

R1: AISystem returns the reports for review. The reports do not decide the grade.

A2: Course Coordinator submits the final grade, participants receive a notification, and the flow returns to Classroom Management.

R2: The final grade is recorded and Notification service delivers the result notice.

Level 1.3:

Name: NIMC Library Management System

Reference: Use Case Diagram Level 1.3

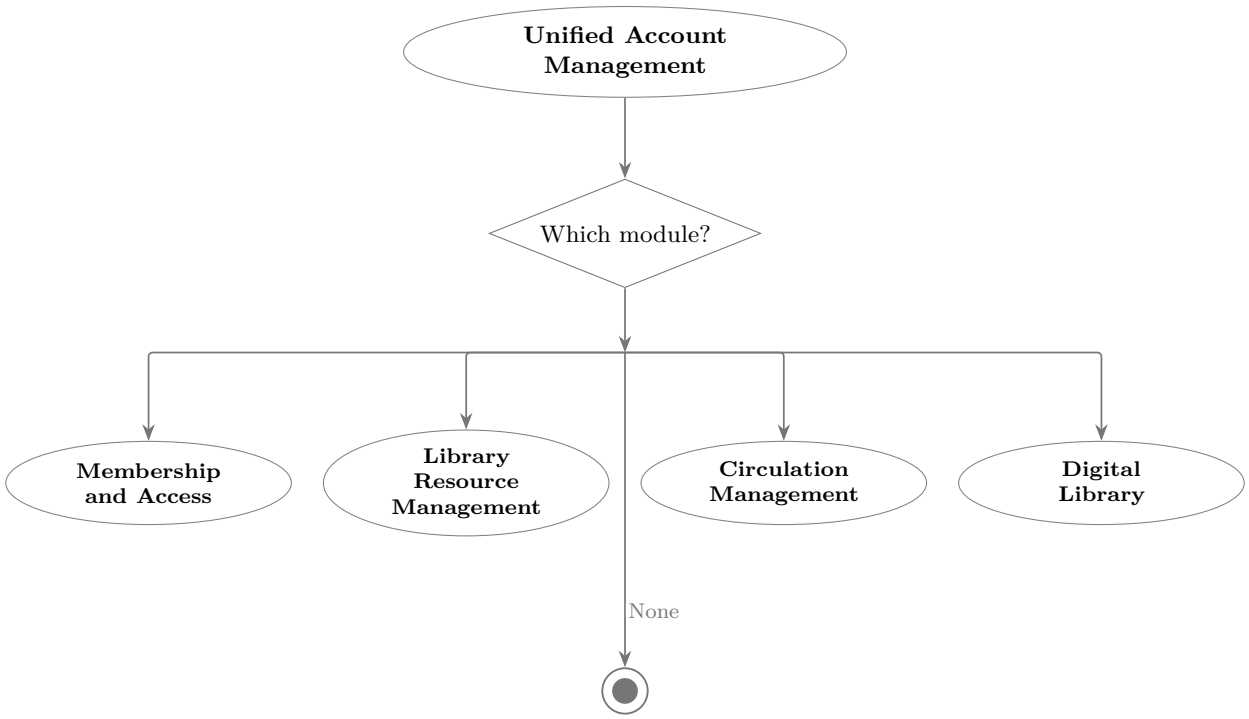


Figure 1: **Level 1.3: NIMC Library Management System**

Activities:

Membership and Access:

A1: The actor asks to verify a member or issue a membership card.

R1: The request is routed to Membership and Access.

Library Resource Management:

A1: The actor asks to search the catalogue, read a review, get a recommendation, or ask the library assistant.

R1: The request is routed to Library Resource Management.

Circulation Management:

A1: The actor asks to issue a book, return a book, or use a wishlist.

R1: The request is routed to Circulation Management.

Digital Library:

A1: The actor asks to find or open a digital resource.

R1: The request is routed to Digital Library. None ends the choice.

Level 1.3.1:

Name: Membership and Access

Reference: Use Case Diagram Level 1.3.1

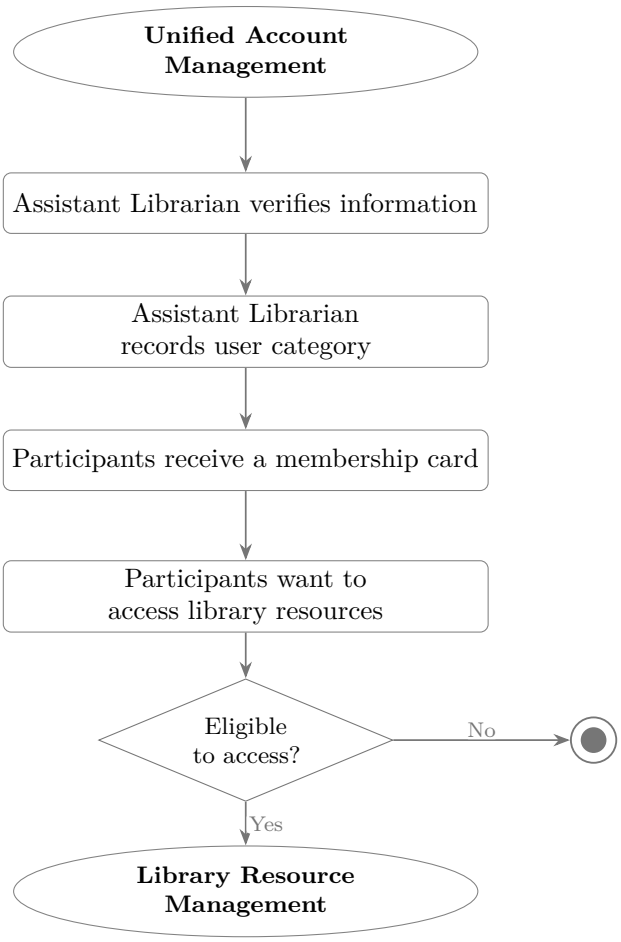


Figure 2: **Level 1.3.1: Membership and Access**

Activities:

Membership Verification and Assignment:

A1: The Assistant Librarian verifies the information submitted through the unified account.

R1: The member information is checked.

A2: The Assistant Librarian records the user category and the validity that fits that category.

R2: The membership category is stored.

Digital Membership Card Issuance:

A1: Participants receive a membership card.

R1: The card shows the member details, membership ID, validity, category, and QR code.

A2: When a participant wants to access library resources, eligibility is checked. An eligible participant continues to Library Resource Management. Otherwise the request ends.

R2: Access is opened only for an eligible member.

Level 1.3.2:

Name: Library Resource Management

Reference: Use Case Diagram Level 1.3.2

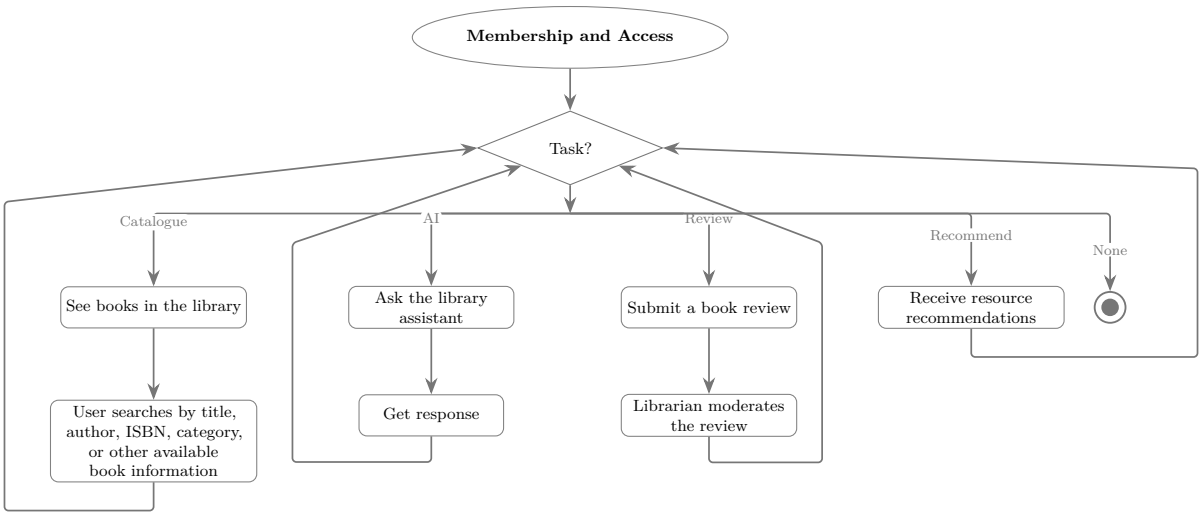


Figure 3: **Level 1.3.2: Library Resource Management**

Activities:

Library Catalogue System:

A1: The user chooses the catalogue and sees the books in the library.

R1: The catalogue is open.

A2: The user searches by title, author, ISBN, category, or other available book information, then may choose another task.

R2: Matching books and their availability are shown.

LibraryAI System:

A1: The user asks the library assistant and receives a response, then may choose another task.

R1: AISystem returns a response about resources, reviews, or availability.

A2: A question that needs staff judgment is referred to the Librarian.

R2: The Librarian receives the referred question.

Recommendations and Reviews System:

A1: The user submits a book review and the Librarian moderates it.

R1: An accepted review is stored.

A2: The user receives resource recommendations, then may choose another task. None ends the choice.

R2: Recommendations are shown from the stored catalogue and reviews.

Level 1.3.3:

Name: Circulation Management

Reference: Use Case Diagram Level 1.3.3

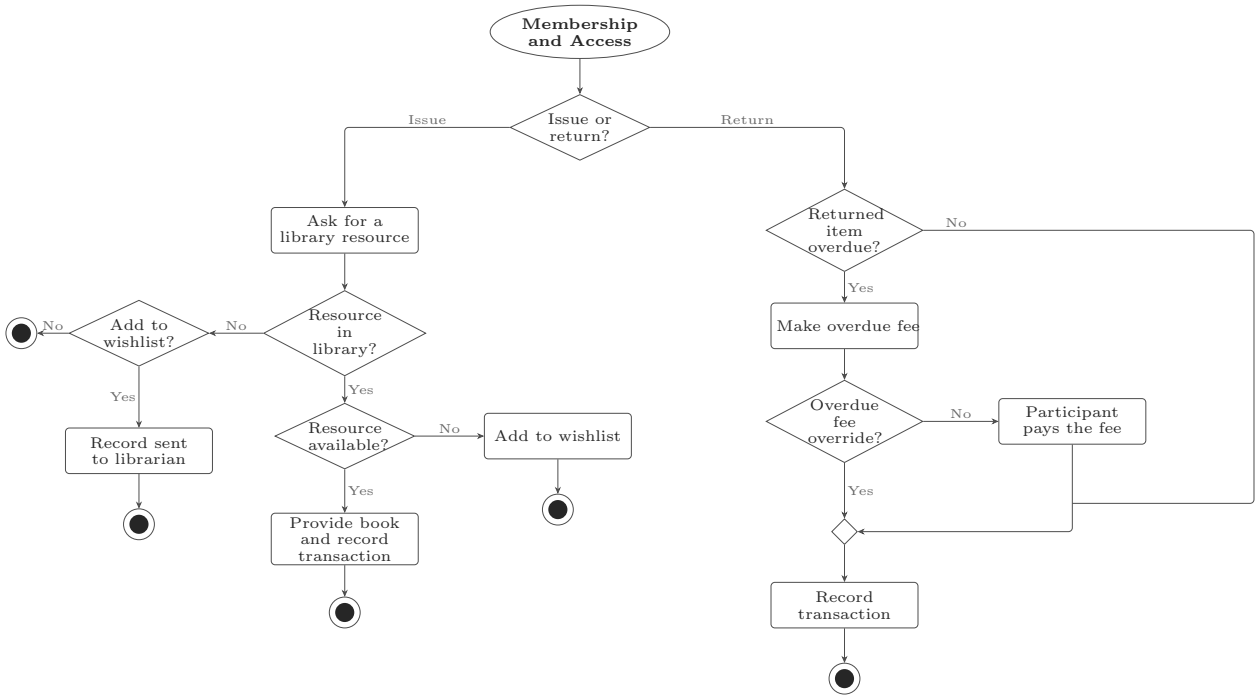


Figure 4: **Level 1.3.3: Circulation Management**

Activities:

Book Issue and Return:

A1: The user asks for a library resource. The librarian checks that the resource is in the library and that a copy is available, then provides the book and records the transaction.

R1: The issue, due date, and availability are recorded.

A2: On return, a transaction is recorded when the item is not overdue, when the librarian overrides the fee, or after the participant pays.

R2: The return is recorded and the copy becomes available.

Overdue Penalty Processing:

A1: An overdue return creates the overdue fee. The librarian may override it. Otherwise the participant pays the fee before the return is recorded.

R1: The fee or the override is stored with the transaction.

Wishlist System:

A1: If the resource is not in the library, the user may add it to a wishlist. The record is then sent to the librarian. Declining ends the request.

R1: The wishlist entry is stored.

A2: If the resource is in the library but not available, it is added to the wishlist and the issue ends.

R2: The user is waiting for that resource.

Waitlisted User Notification:

A1: The wishlist record is sent to the librarian, who can notify the user when the resource can be provided.

R1: Notification service can deliver the availability notice.

Level 1.3.4:

Name: Digital Library

Reference: Use Case Diagram Level 1.3.4

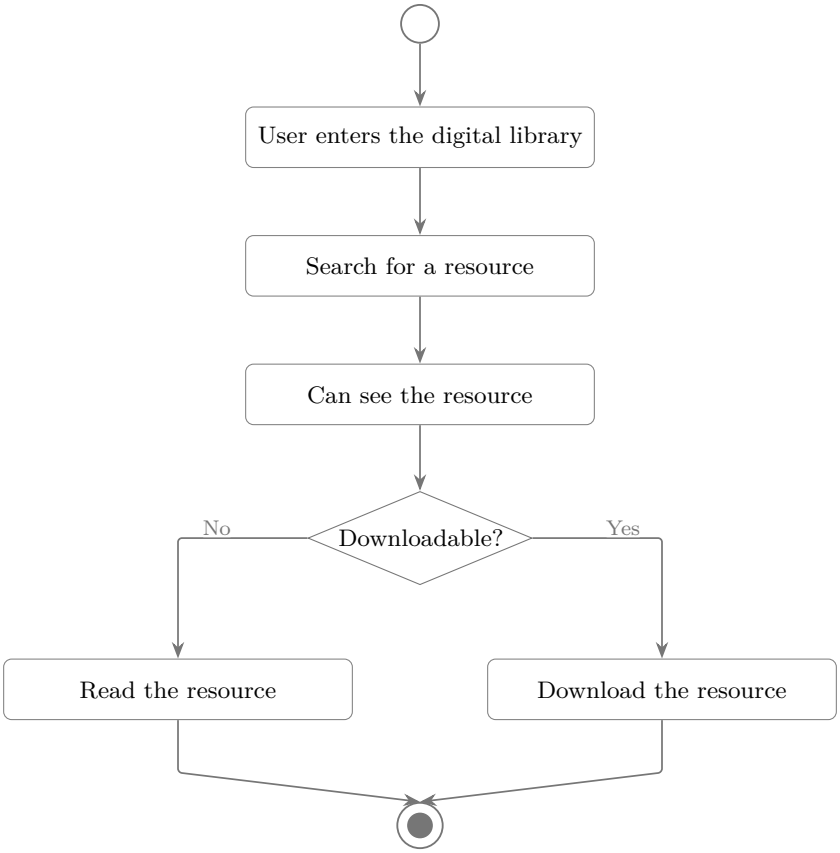


Figure 5: **Level 1.3.4: Digital Library**

Activities:

Resource Discovery:

A1: The user enters the digital library.

R1: The digital collection is open within the user's permissions.

A2: The librarian searches for the requested resource.

R2: Matching digital resources are found.

Digital Resource Access:

A1: The user can see the resource. If it is downloadable, the user downloads it. Otherwise the user reads it online.

R1: The resource is shown or saved within its permission.

Usage Analytics:

A1: The librarian reviews how often digital resources are opened or downloaded.

R1: Usage statistics are available for the collection.

Digital Collection Management:

A1: The librarian or assistant librarian uploads and maintains permitted magazines and open-source books, including download permission.

R1: The digital collection is updated.

Level 1.4:

Name: Dormitory Management

Reference: Use Case Diagram Level 1.4

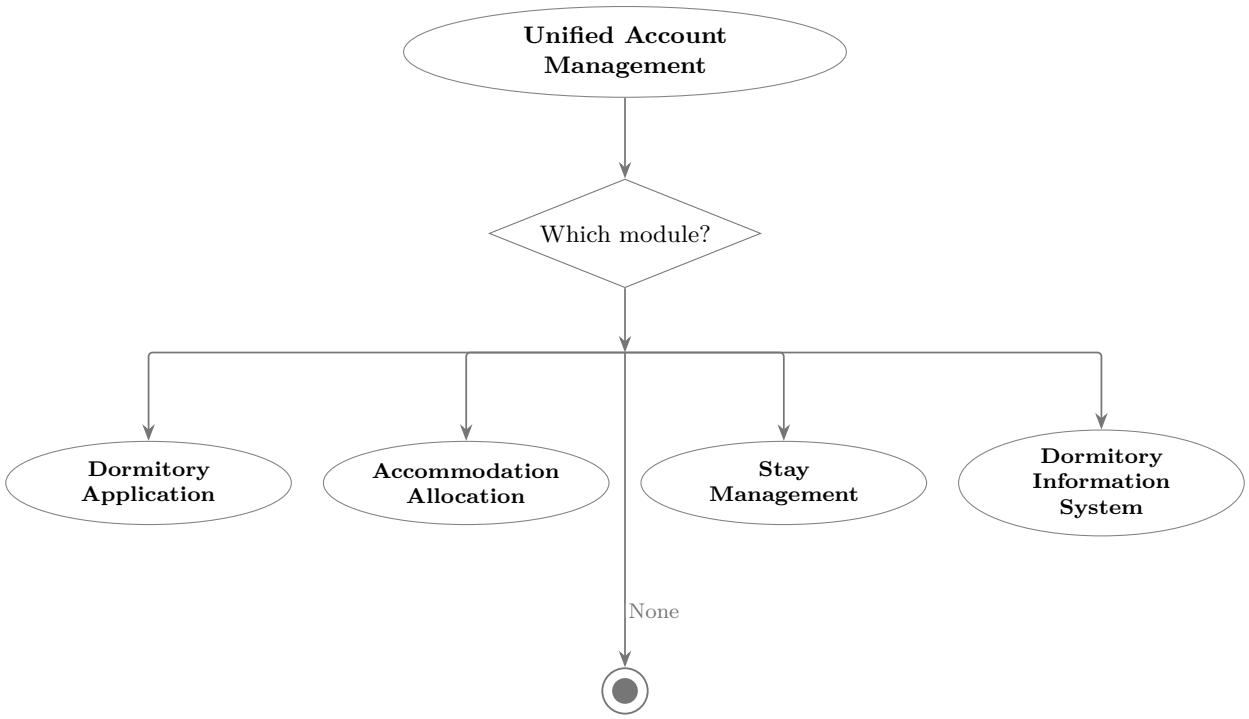


Figure 1: **Level 1.4: Dormitory Management**

Activities:

Dormitory Application:

A1: The actor asks to apply for accommodation or to review an application.

R1: The request is routed to Dormitory Application.

Accommodation Allocation:

A1: The actor asks for a fee waiver, dormitory payment, or room allocation.

R1: The request is routed to Accommodation Allocation.

Stay Management:

A1: The actor asks for check-in, a rating, an issue report, or a change to the stay.

R1: The request is routed to Stay Management.

Dormitory Information System:

A1: The actor asks for occupancy information, a monthly report, stay statistics, or hostel assistance.

R1: The request is routed to Dormitory Information System.

Level 1.4.1:

Name: Dormitory Application

Reference: Use Case Diagram Level 1.4.1

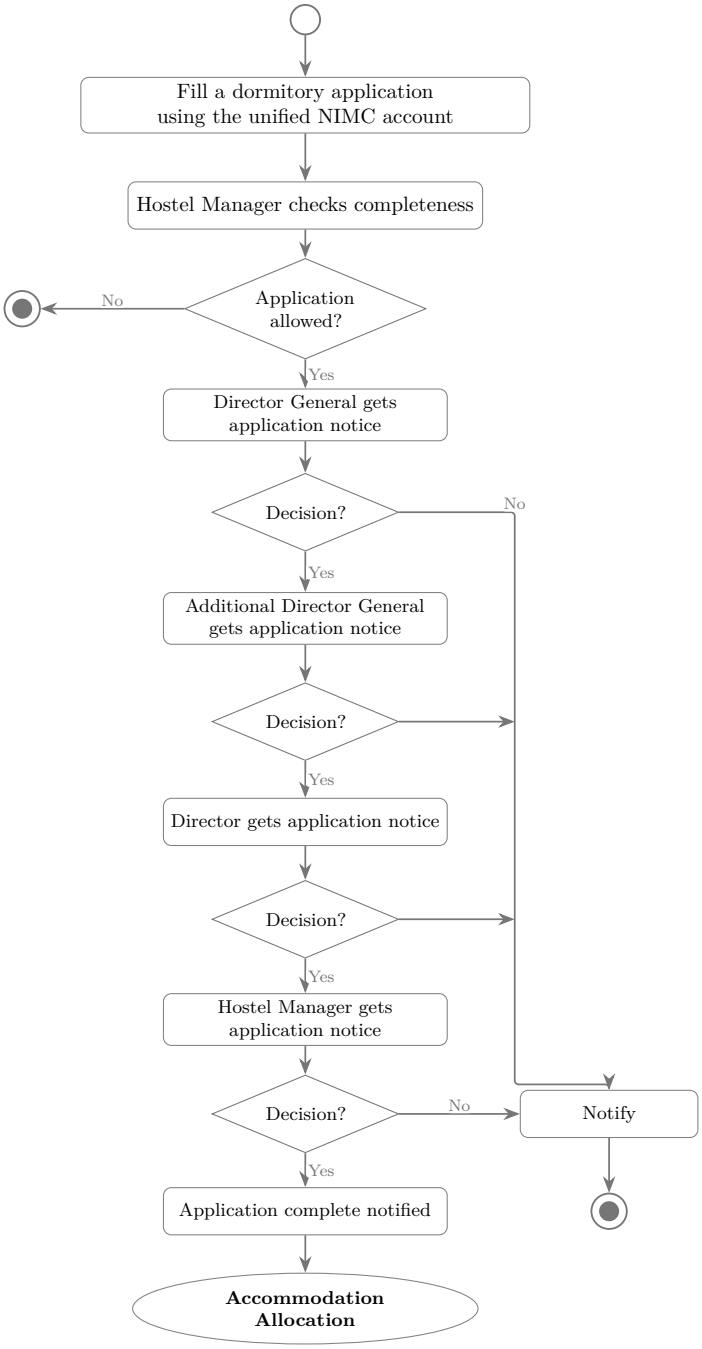


Figure 2: **Level 1.4.1: Dormitory Application**

Activities:

Accommodation Application System:

A1: The participant fills a dormitory application with the unified NIMC account.

R1: The application is stored for review.

A2: The Hostel Manager checks completeness. An application that is not allowed ends. An allowed application goes on for approval.

R2: A complete application is ready for the approval chain.

Application Approval System:

A1: The Director General, Additional Director General, Director, and Hostel Manager each receive the application notice and give a decision. A refusal is notified and the application ends.

R1: Notification System delivers the refusal.

A2: When every decision is yes, completion is notified and the flow continues to Accommodation Allocation.

R2: The approved application is ready for allocation.

Level 1.4.2:

Name: Accommodation Allocation

Reference: Use Case Diagram Level 1.4.2

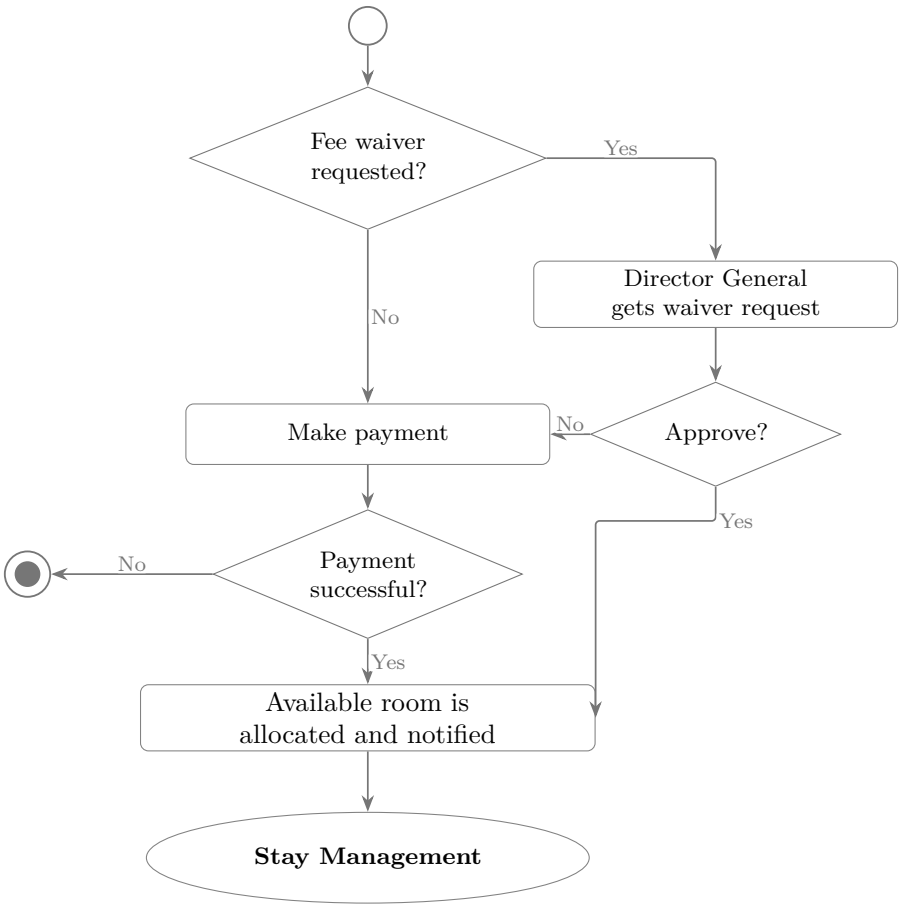


Figure 3: **Level 1.4.2: Accommodation Allocation**

Activities:

Waiver System:

A1: When a fee waiver is requested, the Director General receives the waiver request and approves or refuses it. An approved waiver continues to room allocation. A refusal continues to payment. When no waiver is requested, the flow continues to payment.

R1: The waiver decision is recorded.

Dormitory Payment System:

A1: The participant makes the dormitory payment. Payment System checks the result. An unsuccessful payment ends the allocation.

R1: A successful payment is confirmed.

Room Allocation System:

A1: An available room is allocated and the participant is notified. The flow continues to Stay Management.

R1: The allocated room is recorded and Notification System delivers the notice.

Level 1.4.3:

Name: Stay Management

Reference: Use Case Diagram Level 1.4.3

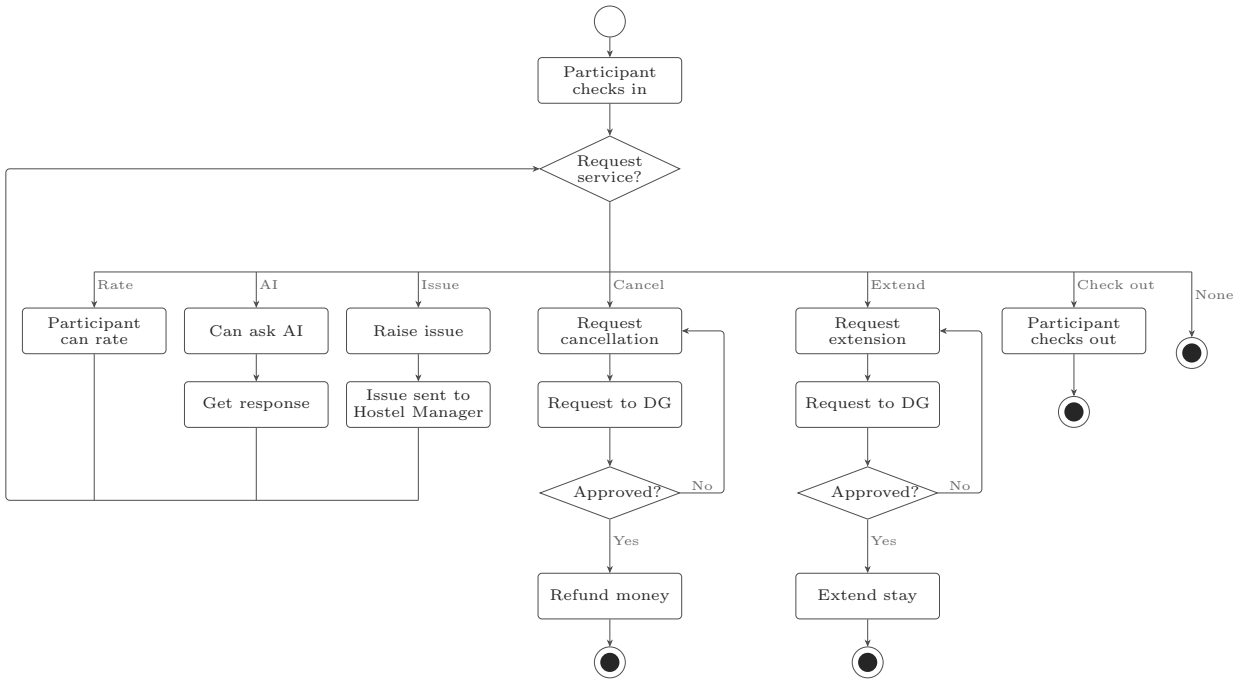


Figure 4: **Level 1.4.3: Stay Management**

Activities:

Check-In and Out System:

A1: The participant checks in. When the stay ends, the participant checks out.

R1: Check-in opens the stay. Check-out closes it and the room or seat is available again.

A2: The participant can rate the dormitory.

R2: The rating is stored. Check-out, rating, and the other completed requests end together.

Extension/Cancellation Application System:

A1: A cancellation is sent to the Director General. Approval refunds the money and ends the request. A refusal returns to the cancellation request.

R1: An approved cancellation closes the stay and Payment System refunds the money.

A2: An extension is sent to the Director General. Approval extends the stay and ends the request. A refusal returns to the extension request.

R2: An approved extension updates the stay.

Issue Report System:

A1: The participant raises an issue and it is sent to the Hostel Manager. The request then ends.

R1: Notification System delivers the issue to the Hostel Manager.

Level 1.4.4:

Name: Dormitory Information System

Reference: Use Case Diagram Level 1.4.4

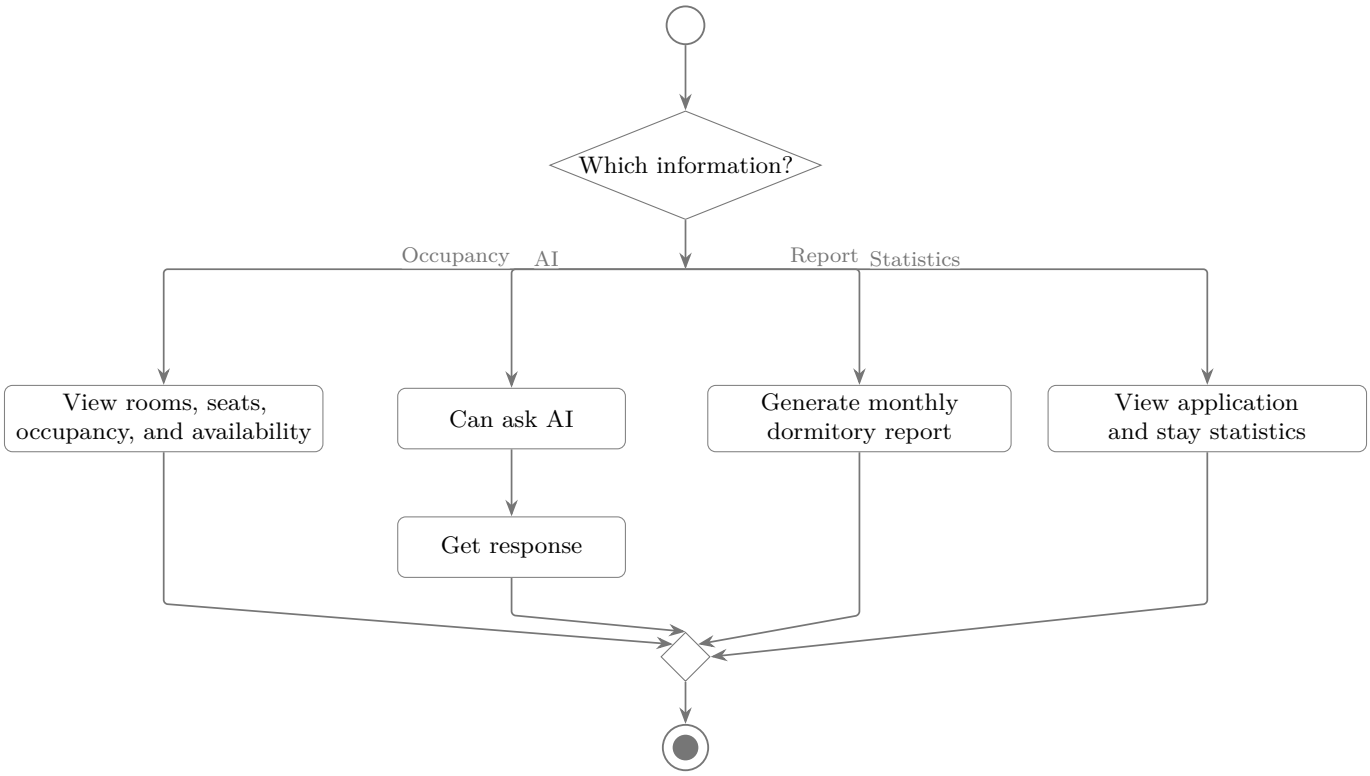


Figure 5: **Level 1.4.4: Dormitory Information System**

Activities:

Dormitory Information Access:

A1: The Hostel Manager views room and seat totals, occupancy, availability, room type, and capacity.

R1: The current dormitory availability is shown.

HostelAI:

A1: The participant asks the hostel assistant and receives a response.

R1: AI Service returns the response.

Monthly Dormitory Reporting:

A1: The Hostel Manager generates the monthly dormitory report.

R1: The report is available for digital access and record keeping.

Application and Stay Statistics:

A1: The Hostel Manager views application totals, approvals, pending applications, allocations, check-ins, check-outs, extensions, cancellations, and issue statistics.

R1: The stay statistics are shown.

Swimlane Diagram

Level 1.1:

Name: Unified Account Management
Reference: Activity Diagram Level 1.1

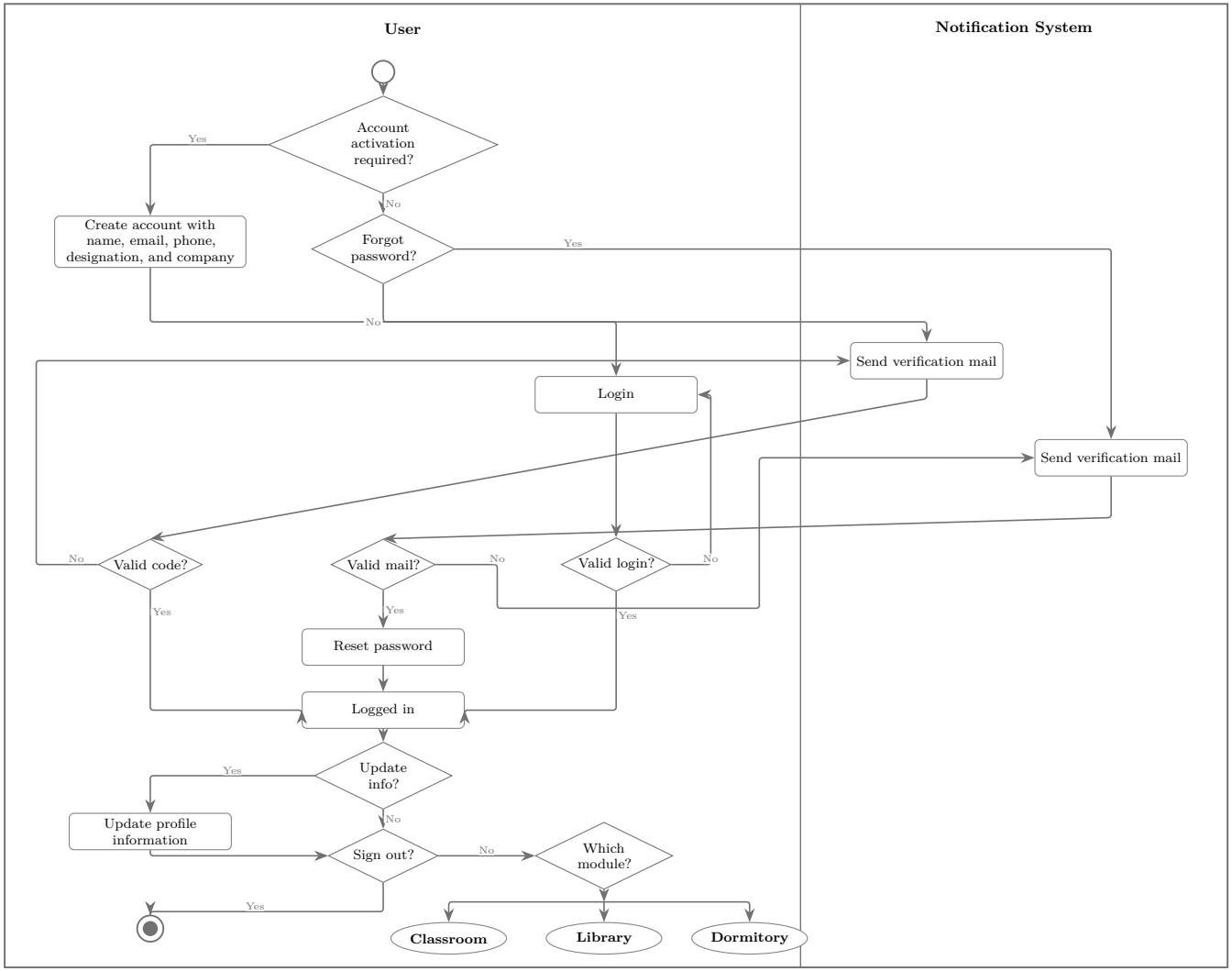


Figure 1: **Level 1.1: Unified Account Management**

Swimlane Diagram

Level 1.2.1:

Name: Course Management
Reference: Activity Diagram Level 1.2.1

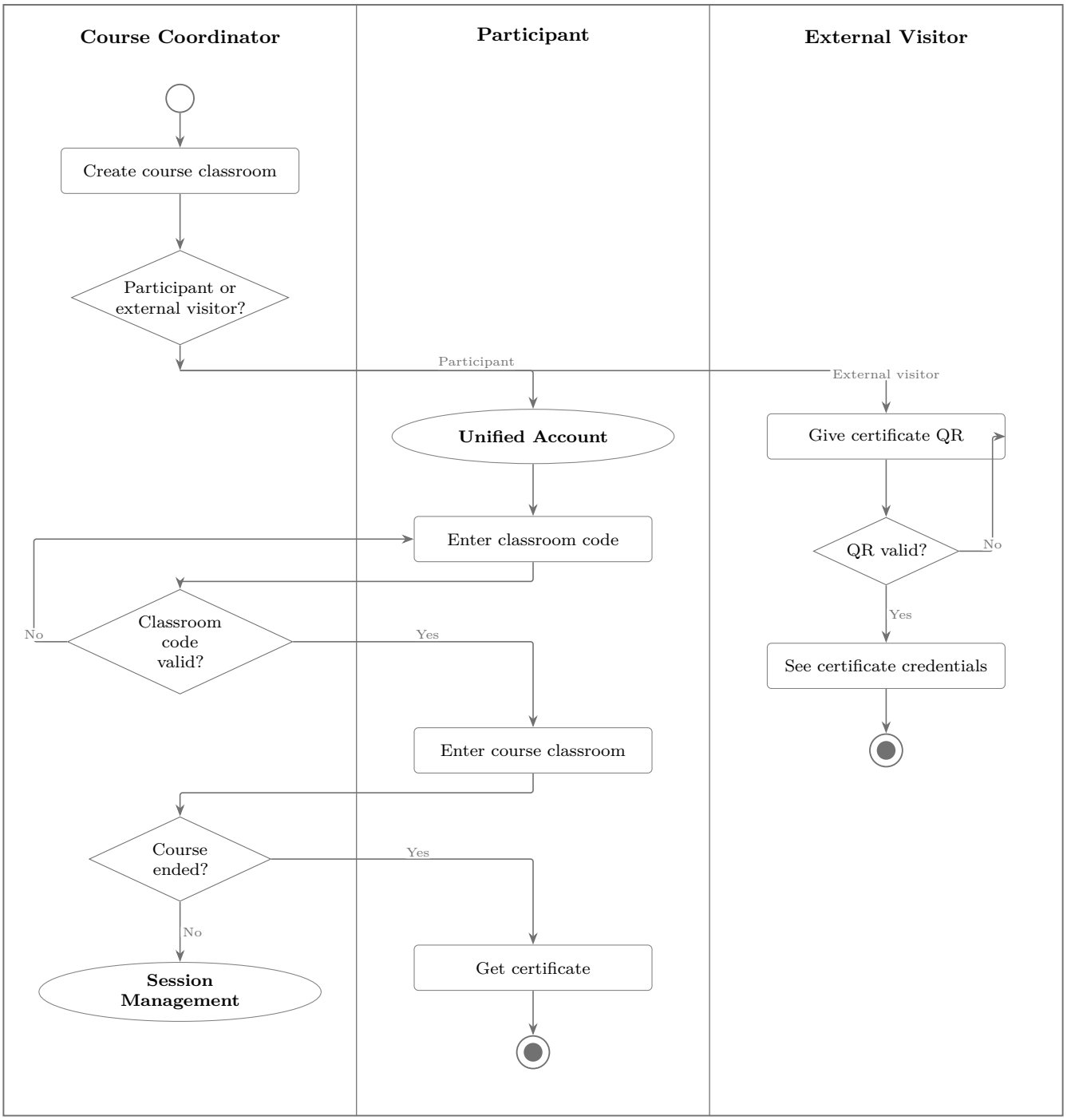


Figure 1: **Level 1.2.1: Course Management**

Level 1.2.2:

Name: Session Management
Reference: Activity Diagram Level 1.2.2

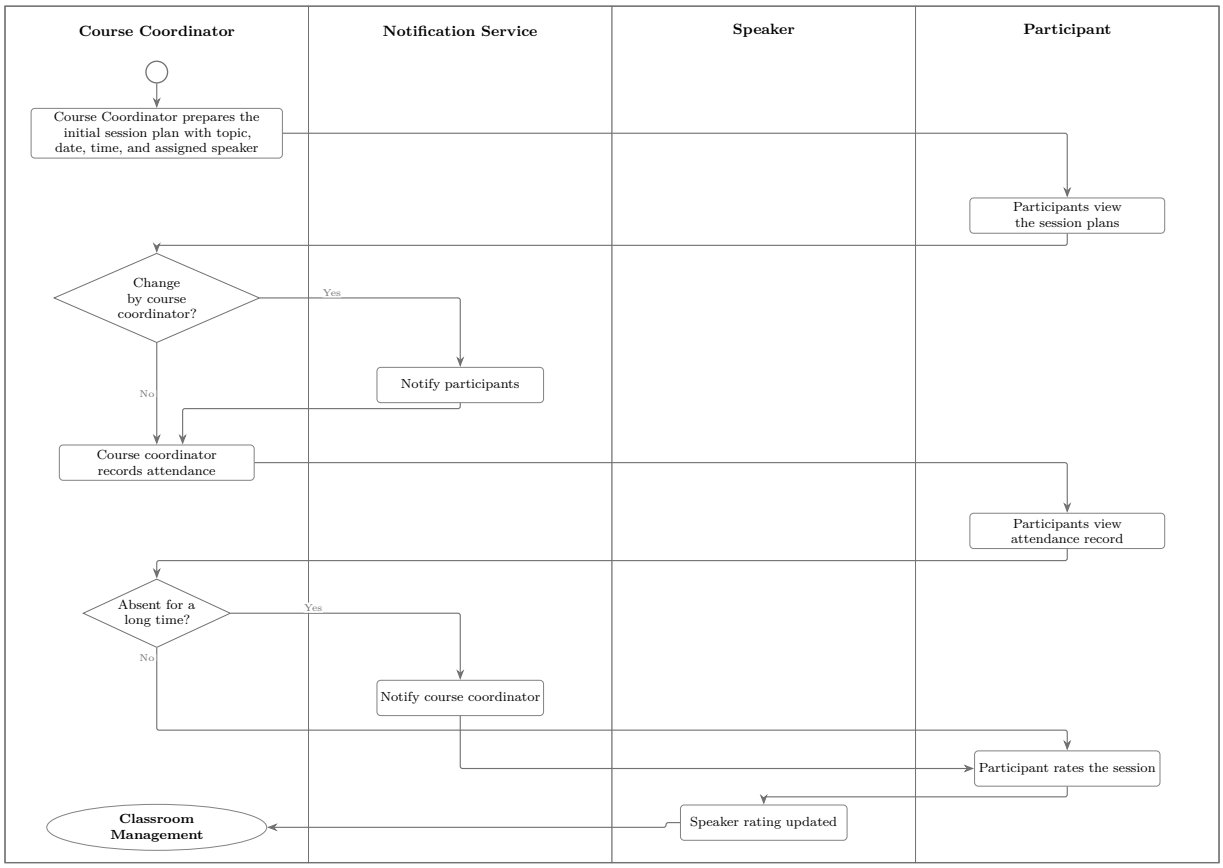


Figure 2: **Level 1.2.2: Session Management**

Level 1.2.3:
Name: Course Resource Management
Reference: Activity Diagram Level 1.2.3

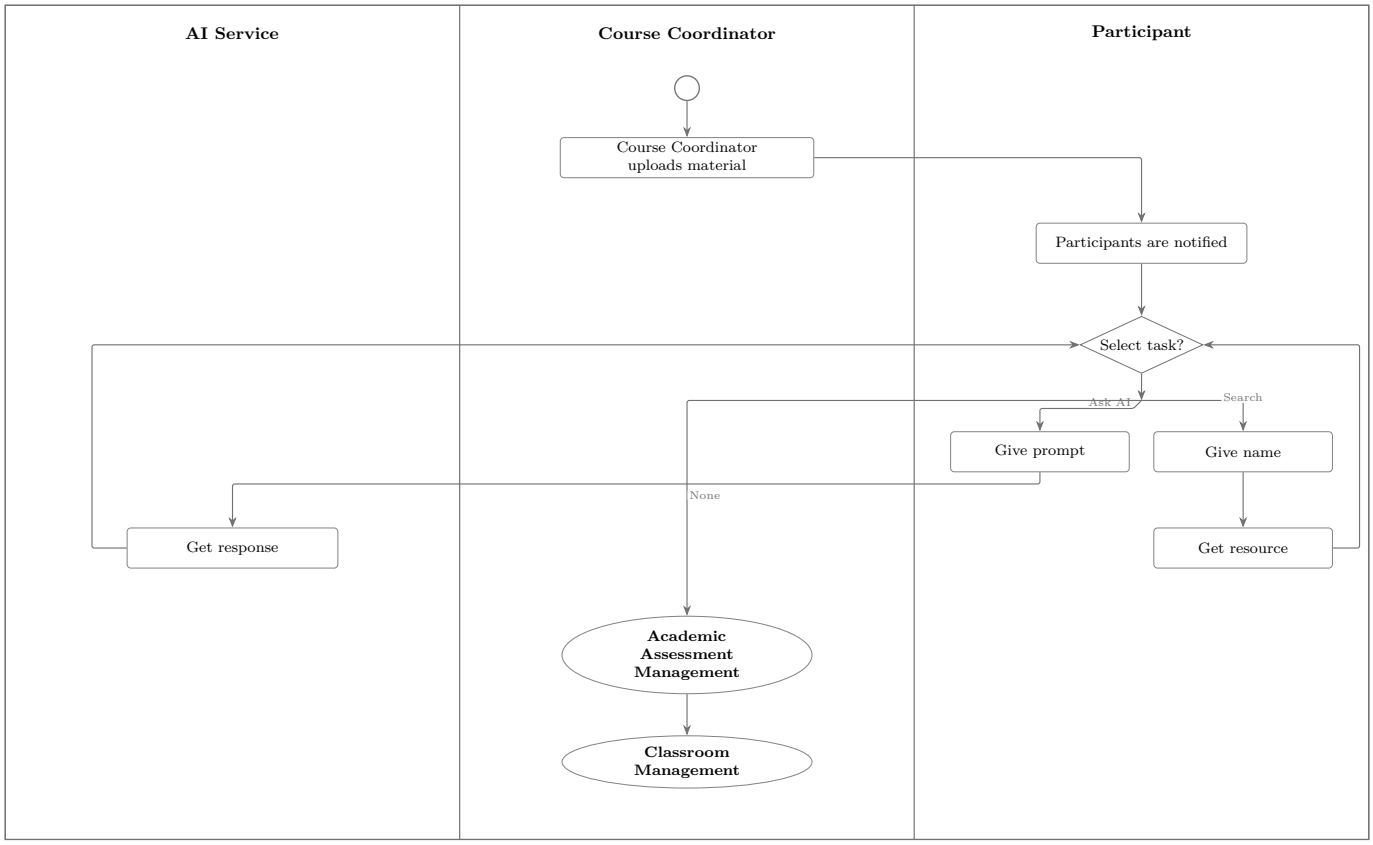


Figure 3: **Level 1.2.3: Course Resource Management**

Level 1.2.4:

Name: Academic Assessment Management
Reference: Activity Diagram Level 1.2.4

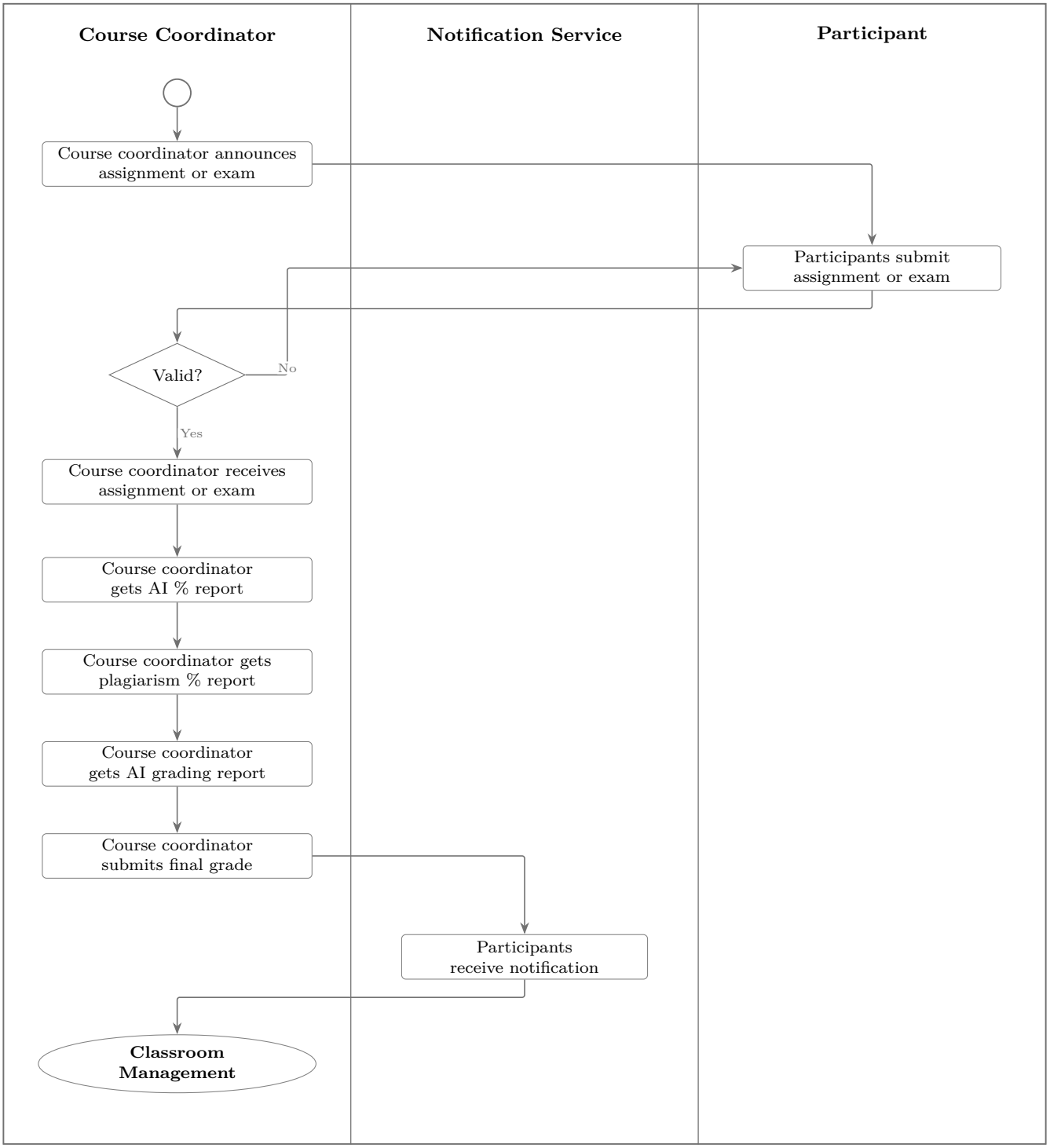


Figure 4: **Level 1.2.4: Academic Assessment Management**

Swimlane Diagram

Level 1.3.1:

Name: Membership and Access
Reference: Activity Diagram Level 1.3.1

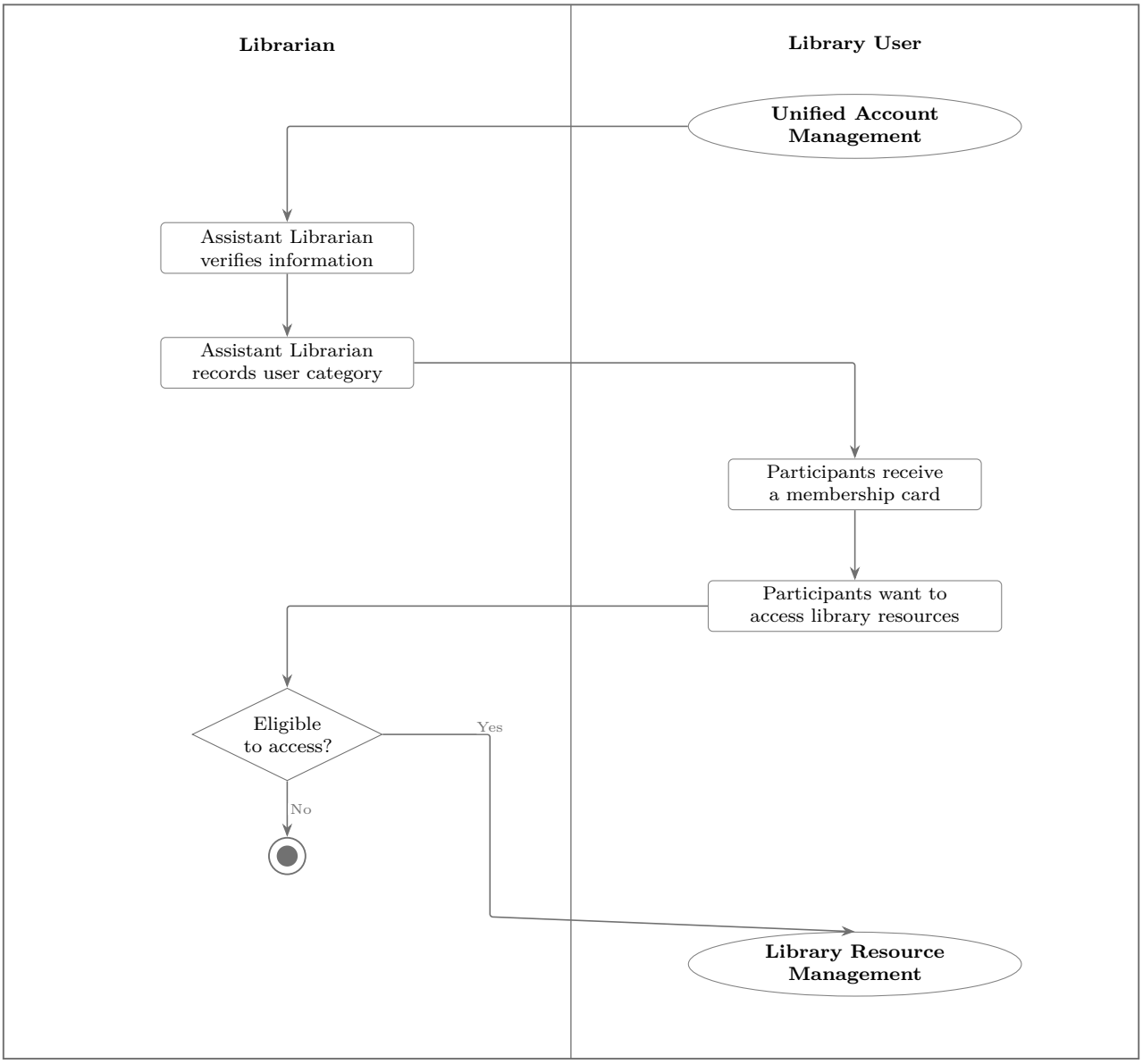


Figure 1: **Level 1.3.1: Membership and Access**

Level 1.3.2:

Name: Library Resource Management
Reference: Activity Diagram Level 1.3.2

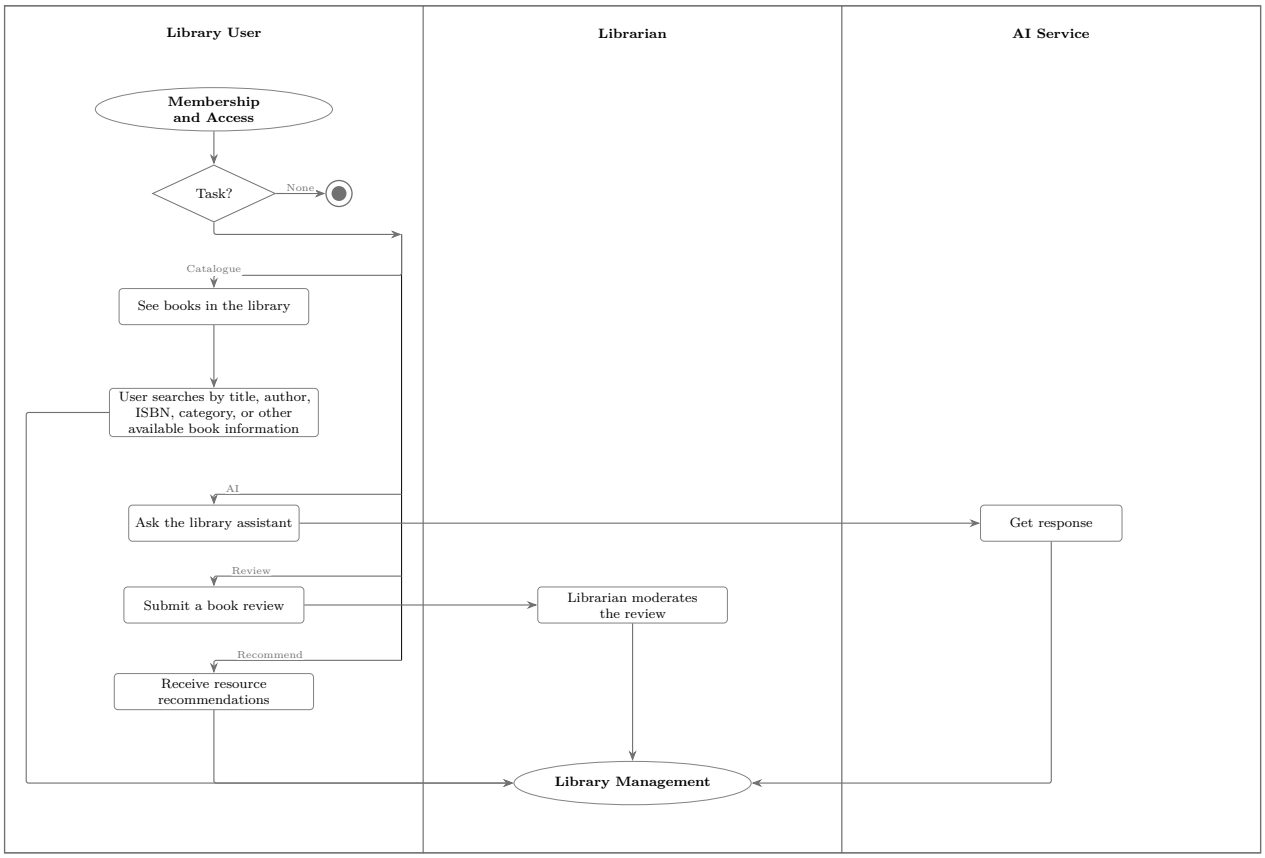


Figure 2: **Level 1.3.2: Library Resource Management**

Level 1.3.3:

Name: Circulation Management
Reference: Activity Diagram Level 1.3.3

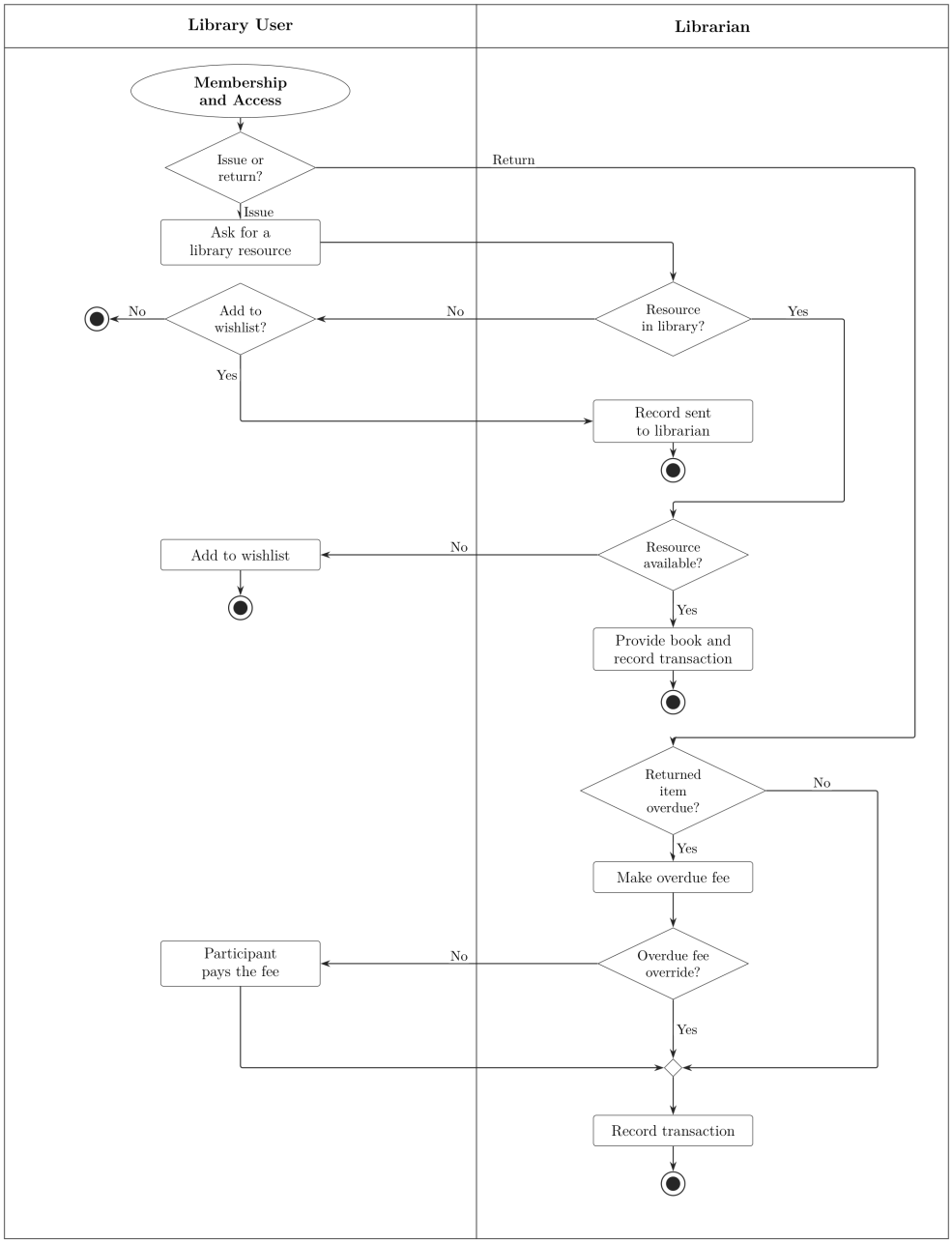


Figure 3: **Level 1.3.3: Circulation Management**

Level 1.3.4:

Name: Digital Library
Reference: Activity Diagram Level 1.3.4

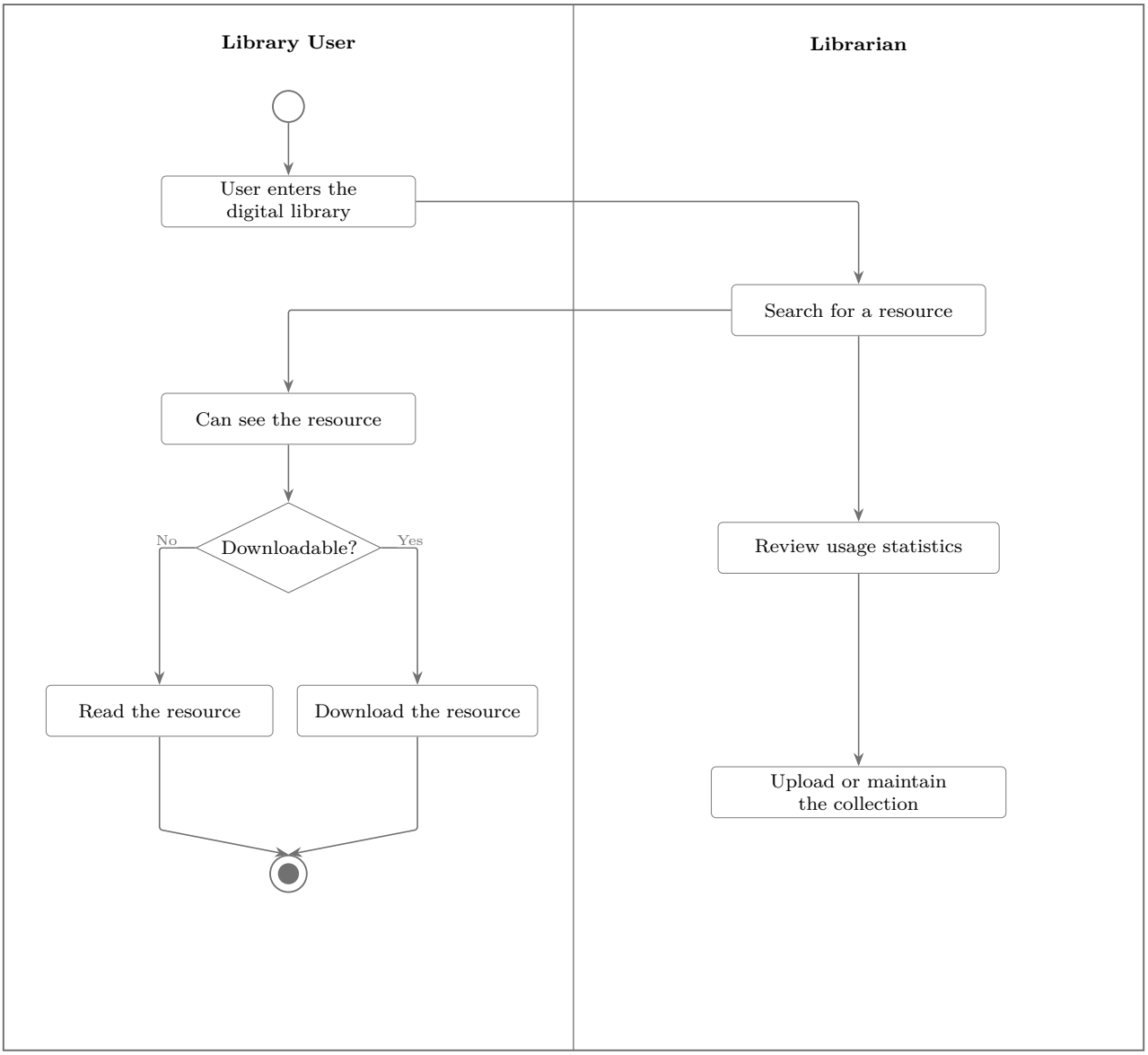


Figure 4: **Level 1.3.4: Digital Library**

Swimlane Diagram

Level 1.4.1:

Name: Dormitory Application
Reference: Activity Diagram Level 1.4.1

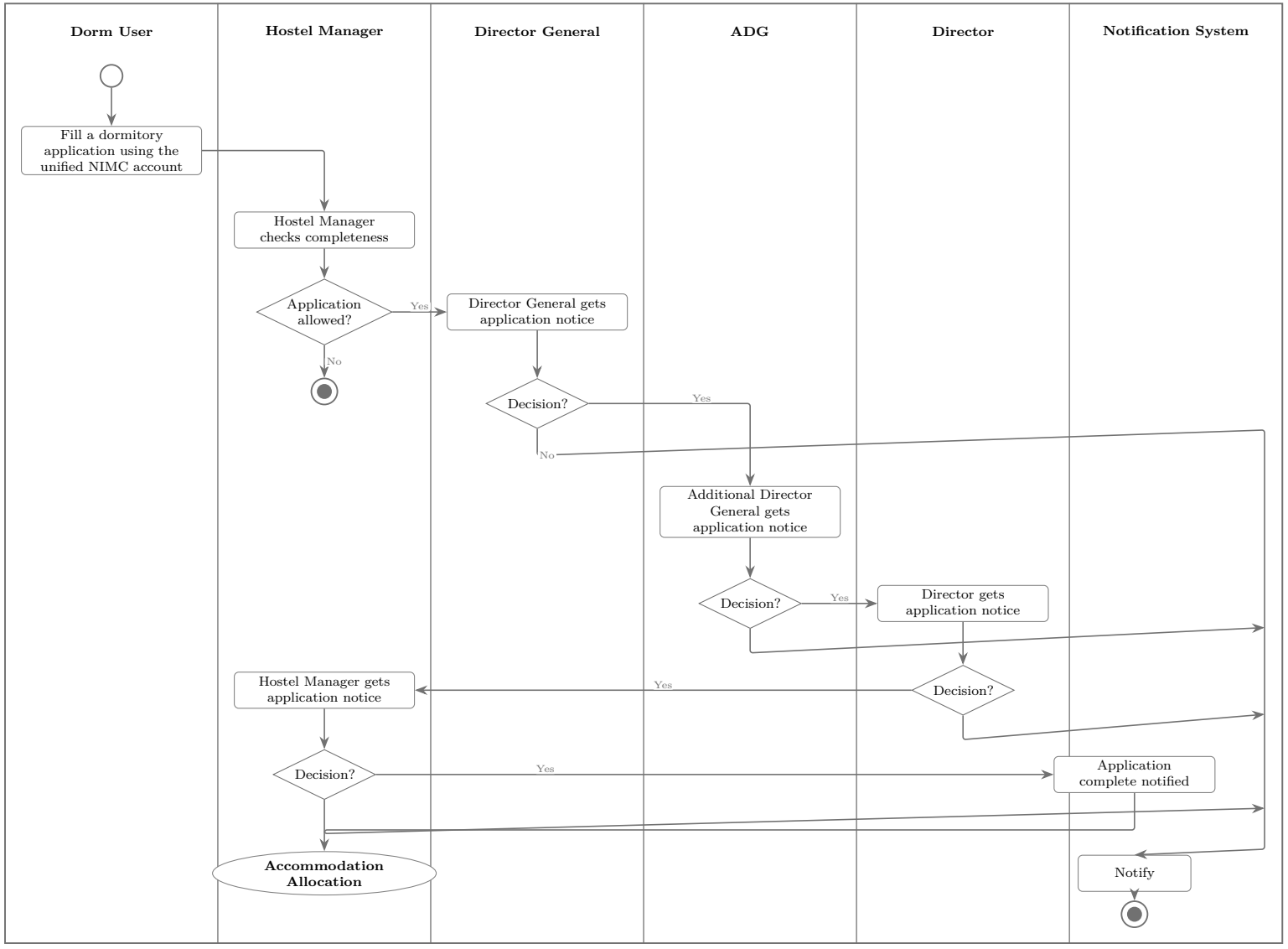


Figure 1: **Level 1.4.1: Dormitory Application**

Level 1.4.2:

Name: Accommodation Allocation
Reference: Activity Diagram Level 1.4.2

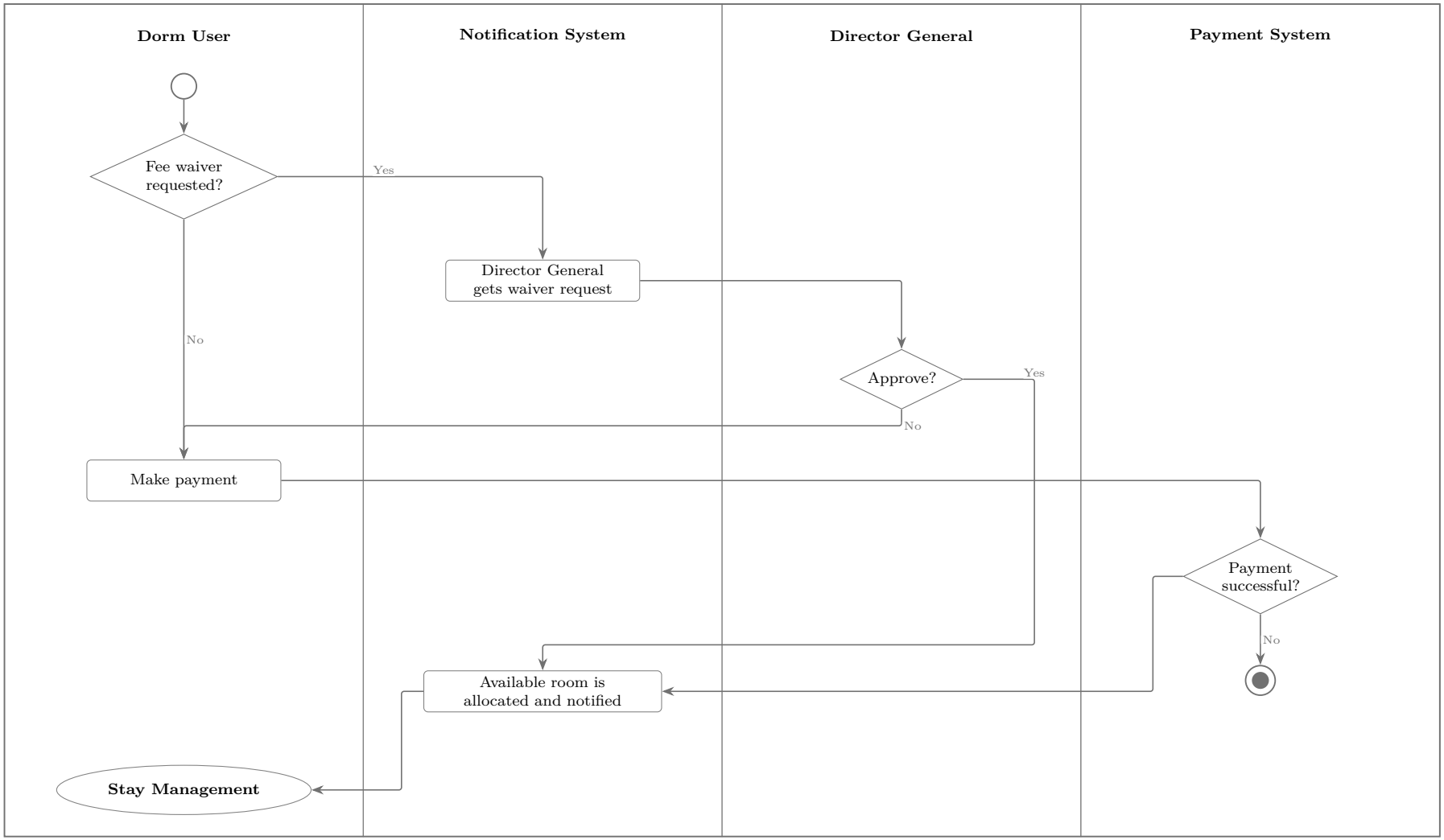


Figure 2: **Level 1.4.2: Accommodation Allocation**

Level 1.4.3:

Name: Stay Management

Reference: Activity Diagram Level 1.4.3

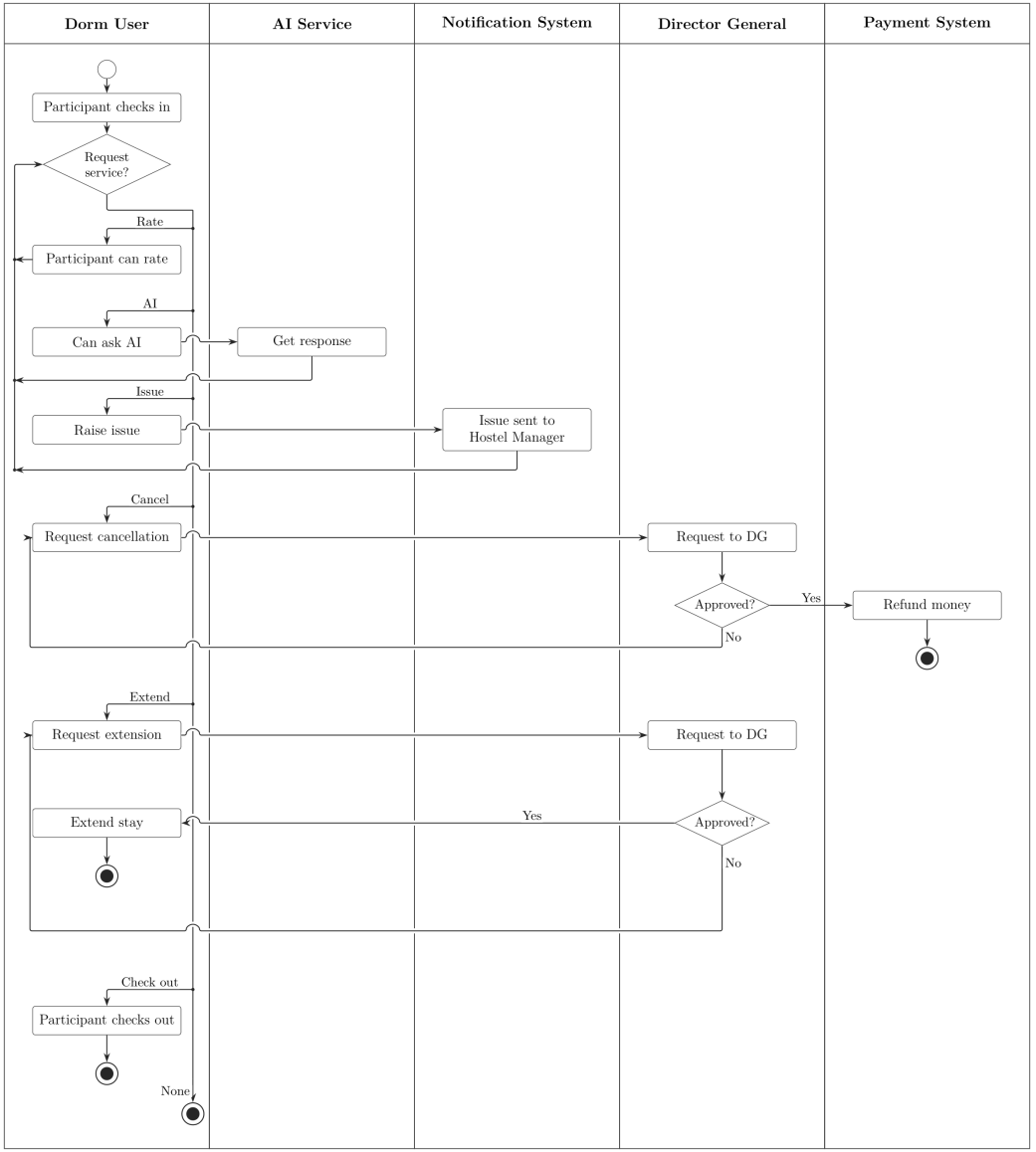


Figure 3: **Level 1.4.3: Stay Management**

Level 1.4.4:

Name: Dormitory Information System
Reference: Activity Diagram Level 1.4.4

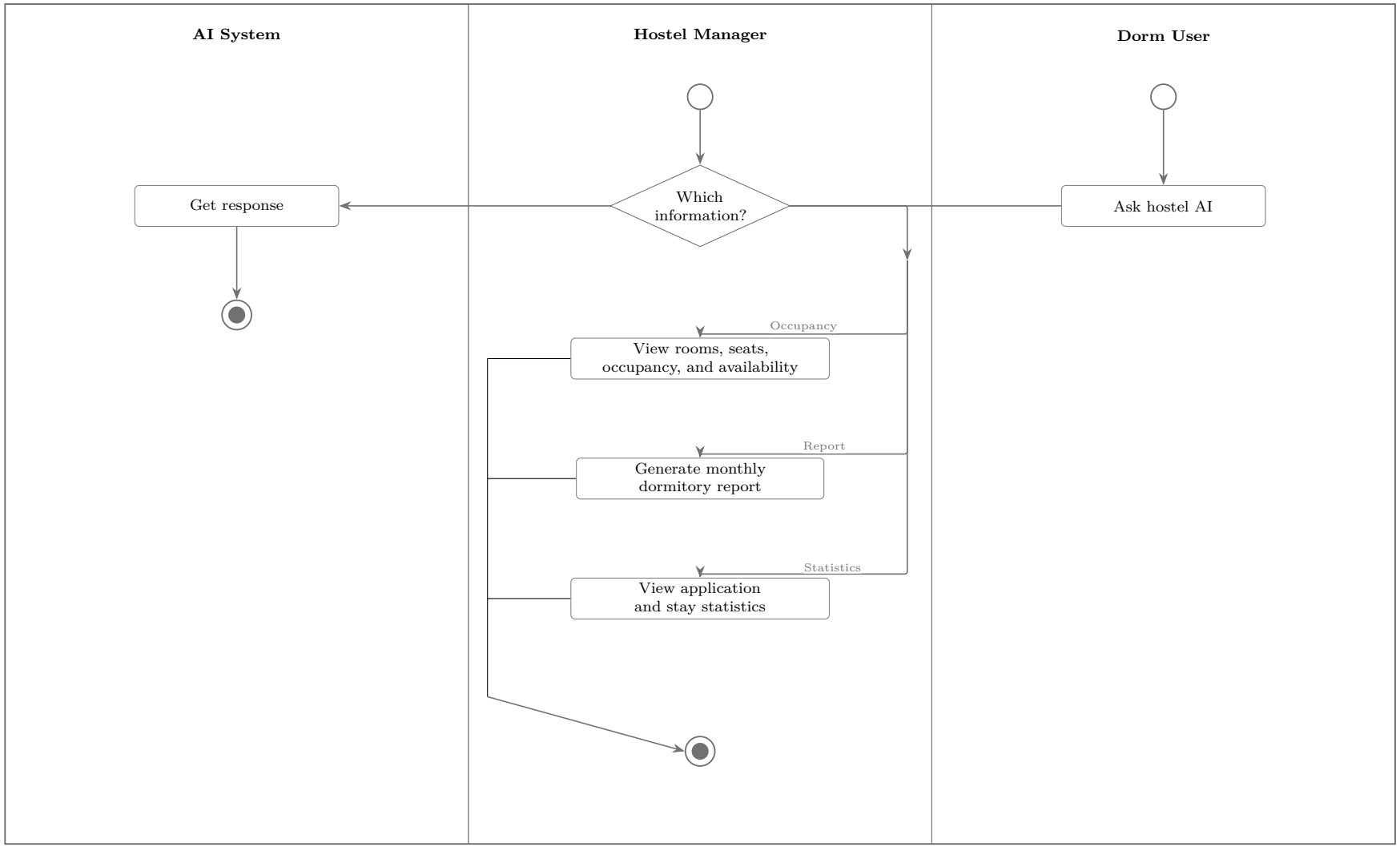


Figure 4: **Level 1.4.4: Dormitory Information System**

List of Nouns

Sl. no.	Noun / noun phrase	Problem Domain / Solution Space	Attributes (sl. no.)
1	NIMC account	Solution Space	8, 10, 11, 12, 13, 120, 121, 122, 154
2	participant	Solution Space	3, 4, 5, 6, 7, 8, 9, 119, 120, 121, 124, 154
3	NIMC employee	Solution Space	
4	name	Solution Space	
5	company	Solution Space	
6	designation	Solution Space	
7	phone number	Solution Space	
8	email address	Solution Space	
9	NID number	Solution Space	
10	email verification	Solution Space	8, 120
11	password	Solution Space	
12	password reset	Solution Space	11, 120
13	existing identity	Solution Space	
14	Classroom service	Solution Space	
15	Library service	Solution Space	
16	Dormitory service	Solution Space	
17	training course	Solution Space	18, 19
18	seminar	Solution Space	
19	workshop	Solution Space	
20	speaker	Solution Space	
21	Course Coordination Team	Solution Space	22
22	Course Coordinator	Solution Space	
23	classroom	Solution Space	24, 126
24	classroom code	Solution Space	
25	session plan	Solution Space	26, 27, 28, 29, 128, 129, 130, 131, 133, 139, 140
26	session topic	Solution Space	
27	date	Solution Space	
28	time	Solution Space	
29	assigned speaker	Solution Space	
30	learning resource	Solution Space	31, 32, 33, 34, 35, 132, 159, 161
31	document	Solution Space	
32	presentation	Solution Space	
33	video	Solution Space	
34	notes	Solution Space	
35	class recording	Solution Space	
36	attendance	Solution Space	37, 135
37	overall attendance	Solution Space	
38	engagement indicator	Solution Space	136, 137
39	notification	Solution Space	141, 142, 143
40	assignment	Solution Space	41, 42, 145
41	examination	Solution Space	
42	assessment result	Solution Space	
43	submission	Solution Space	144
44	AI-based analysis	Solution Space	45, 46, 47, 147
45	similarity indicator	Solution Space	
46	plagiarism indicator	Solution Space	
47	AI-generated-content indicator	Solution Space	
48	speaker evaluation	Solution Space	20, 49, 50, 148, 149
49	rating	Solution Space	
50	comment	Solution Space	
51	course evaluation	Solution Space	50, 52, 151, 152, 153
52	course feedback	Solution Space	
53	certificate	Solution Space	54, 157, 158
54	QR code	Solution Space	
55	LLM-based assistant	Problem Domain	
56	library membership	Solution Space	60, 62, 63, 64, 162, 167, 168, 169
57	visitor	Solution Space	166

Sl. no.	Noun / noun phrase	Problem Domain / Solution Space	Attributes (sl. no.)
58	assistant librarian	Problem Domain	
59	librarian	Problem Domain	
60	membership validity	Solution Space	
61	digital membership card	Solution Space	62, 54
62	membership ID	Solution Space	
63	borrowing eligibility	Solution Space	
64	borrowing limit	Solution Space	
65	physical book	Solution Space	177, 180, 189
66	library catalogue	Solution Space	67, 68, 69, 173, 174, 175, 176, 203, 204, 205
67	ISBN	Solution Space	
68	category	Solution Space	
69	publisher	Solution Space	
70	issue date	Solution Space	
71	due date	Solution Space	
72	return date	Solution Space	
73	overdue status	Solution Space	
74	penalty	Solution Space	
75	circulation information	Solution Space	
76	e-library	Problem Domain	
77	digital resource	Solution Space	78, 79, 80, 81, 178, 190, 191, 192
78	magazine	Solution Space	
79	open-source book	Solution Space	
80	access permission	Solution Space	
81	download permission	Solution Space	
82	usage statistics	Solution Space	
83	LibraryAssistantAI	Problem Domain	
84	book review	Solution Space	49, 197, 199
85	book request	Solution Space	200, 201, 202, 211
86	monthly library report	Problem Domain	
87	dormitory application	Solution Space	88, 89, 90, 91, 213, 214
88	accommodation type	Solution Space	
89	expected stay	Solution Space	
90	single seat	Solution Space	
91	double seat	Solution Space	
92	Hostel Manager	Problem Domain	
93	approval link	Solution Space	
94	Director General	Solution Space	
95	Additional Director General	Solution Space	
96	Director	Solution Space	
97	room	Solution Space	99, 100, 101, 218
98	seat	Solution Space	100, 101, 218
99	room capacity	Solution Space	
100	occupied seats	Solution Space	
101	available seats	Solution Space	
102	dormitory fee	Solution Space	
103	waiver	Solution Space	
104	payment confirmation	Solution Space	
105	room allocation	Solution Space	
106	check-in	Solution Space	107, 114
107	stay status	Solution Space	
108	stay-extension request	Solution Space	109
109	cancellation request	Solution Space	
110	dormitory issue	Solution Space	111, 112, 113
111	Hostel Super	Solution Space	
112	Warden	Solution Space	
113	issue status	Solution Space	
114	check-out	Solution Space	
115	dormitory dashboard	Problem Domain	
116	monthly dormitory report	Problem Domain	
117	Hostel AI	Problem Domain	
118	3D map location	Solution Space	

Sl. no.	Noun / noun phrase	Problem Domain / Solution Space	Attributes (sl. no.)
119	authorized NIMC employee	Solution Space	
120	registered email address	Solution Space	
121	account information	Solution Space	
122	shared NIMC account	Solution Space	
123	module-specific information	Solution Space	14, 15, 16, 154
124	participant information	Solution Space	
125	organization name	Problem Domain	
126	unique classroom code	Solution Space	
127	participant eligibility	Solution Space	
128	initial session plan	Solution Space	
129	schedule change	Solution Space	
130	topic change	Solution Space	
131	speaker change	Solution Space	
132	resource type	Solution Space	
133	online session	Solution Space	
134	authorized participant	Problem Domain	
135	attendance percentage	Solution Space	
136	participation level	Solution Space	
137	participant performance	Solution Space	
138	important classroom activity	Problem Domain	
139	session schedule change	Solution Space	
140	online session information	Solution Space	
141	assignment notice	Solution Space	
142	examination notice	Solution Space	
143	course update	Solution Space	
144	submitted work	Solution Space	
145	assessment information	Solution Space	
146	academic decision	Problem Domain	
147	flagged work	Solution Space	
148	speaker rating	Solution Space	
149	speaker rating history	Solution Space	
150	evaluation process	Problem Domain	
151	overall course evaluation	Solution Space	
152	course effectiveness	Solution Space	
153	area for improvement	Solution Space	
154	authorized user account	Solution Space	
155	participant record	Solution Space	2, 127
156	performance information	Problem Domain	
157	certificate eligibility condition	Solution Space	
158	certificate details	Solution Space	
159	course resource	Solution Space	
160	participant question	Problem Domain	
161	course material	Solution Space	
162	library user category	Solution Space	
163	course participant membership	Problem Domain	
164	office employee	Problem Domain	
165	government employee	Problem Domain	
166	external visitor	Solution Space	
167	membership period	Solution Space	
168	membership category	Solution Space	
169	library user	Solution Space	
170	current borrowed books	Problem Domain	
171	return deadline	Solution Space	
172	borrowing period	Solution Space	
173	book name	Solution Space	
174	author	Solution Space	
175	catalogue information	Solution Space	
176	book information	Solution Space	
177	physical copy	Solution Space	
178	online resource	Solution Space	
179	circulation record	Solution Space	70, 71, 72, 73, 74, 75, 171, 172, 181, 182, 183, 184, 186

Sl. no.	Noun / noun phrase	Problem Domain / Solution Space	Attributes (sl. no.)
180	book status	Solution Space	
181	borrowed-book list	Solution Space	
182	return reminder	Solution Space	
183	overdue book	Solution Space	
184	penalty amount	Solution Space	
185	penalty rule	Solution Space	74, 184, 186
186	penalty adjustment	Solution Space	
187	waiting user	Solution Space	188
188	resource availability	Solution Space	
189	withdrawn book record	Solution Space	
190	online copy	Solution Space	
191	resource access permission	Solution Space	
192	resource download permission	Solution Space	
193	digital resource usage	Solution Space	82
194	book recommendation	Problem Domain	
195	user interest	Problem Domain	
196	learning need	Problem Domain	
197	resource review	Solution Space	
198	related resource	Problem Domain	
199	inappropriate review	Solution Space	
200	book-request list	Solution Space	
201	resource demand	Solution Space	
202	resource acquisition	Solution Space	
203	catalogue details	Solution Space	
204	publication details	Solution Space	
205	physical-copy status	Solution Space	
206	library activity	Problem Domain	
207	newly added book	Problem Domain	
208	active library member	Problem Domain	
209	new membership	Problem Domain	
210	most borrowed book	Problem Domain	
211	popular requested resource	Solution Space	
212	dormitory facility	Problem Domain	
213	accommodation requirement	Solution Space	
214	reason for accommodation	Solution Space	
215	approval process	Solution Space	93, 94, 95, 96, 216, 217
216	responsible person	Solution Space	
217	approval sequence	Solution Space	
218	room and seat chart	Solution Space	

Data Objects and Attributes

Sl.	Data object	Attributes
Unified Account		
1	User	user_id, name, company, designation, phone_number, email_address, NID_number (optional), user_type, password_hash, email_verified, account_status
Classroom		
1	ClassroomUser	user_id (inherited), class_user_status
2	Speaker	speaker_id, user_id, speaker_name, speaker_information
3	Course	course_id, course_name, course_type, start_date, end_date, classroom_code, course_status
4	Session	session_id, course_id, speaker_id, topic, session_date, session_time, online_session_info, session_status
5	LearningResource	resource_id, course_id, session_id, resource_type, resource_title, file_location, is_class_recording, uploaded_at
6	Assignment	assignment_id, course_id, assignment_type, assignment_title, due_at
7	Certificate	certificate_id, user_id, course_id, qr_code, certificate_details, eligibility_status, issue_date
Library		
1	LibraryUser	user_id (inherited), membership_id, user_category, membership_validity, membership_status, borrowing_limit, qr_code
2	LibraryResource	resource_id, book_name, author, ISBN, category, publisher, publication_details, copy_number, resource_status, shelf_location
3	Ebook	ebook_id, resource_id, ebook_title, file_location, access_permission, download_permission
4	BookRequest	request_id, user_id, requested_title, request_status, requested_at
5	Waitlist	waitlist_id, user_id, resource_id, requested_at, waitlist_status
Dormitory		
1	DormUser	user_id (inherited), dormuser_type
2	Application	application_id, user_id, NID_number, application_type, accommodation_type, expected_stay, reason_for_accommodation, application_status, submitted_at
3	Room	room_id, room_number, room_type, room_capacity, wifi_information, map_location, room_status
4	Seat	seat_id, room_id, seat_number, seat_status
5	IoTDevice	device_id, room_id, device_type, device_status, map_location
6	Issue	issue_id, user_id, room_id, issue_description, issue_status, reported_at

Unified Account Management

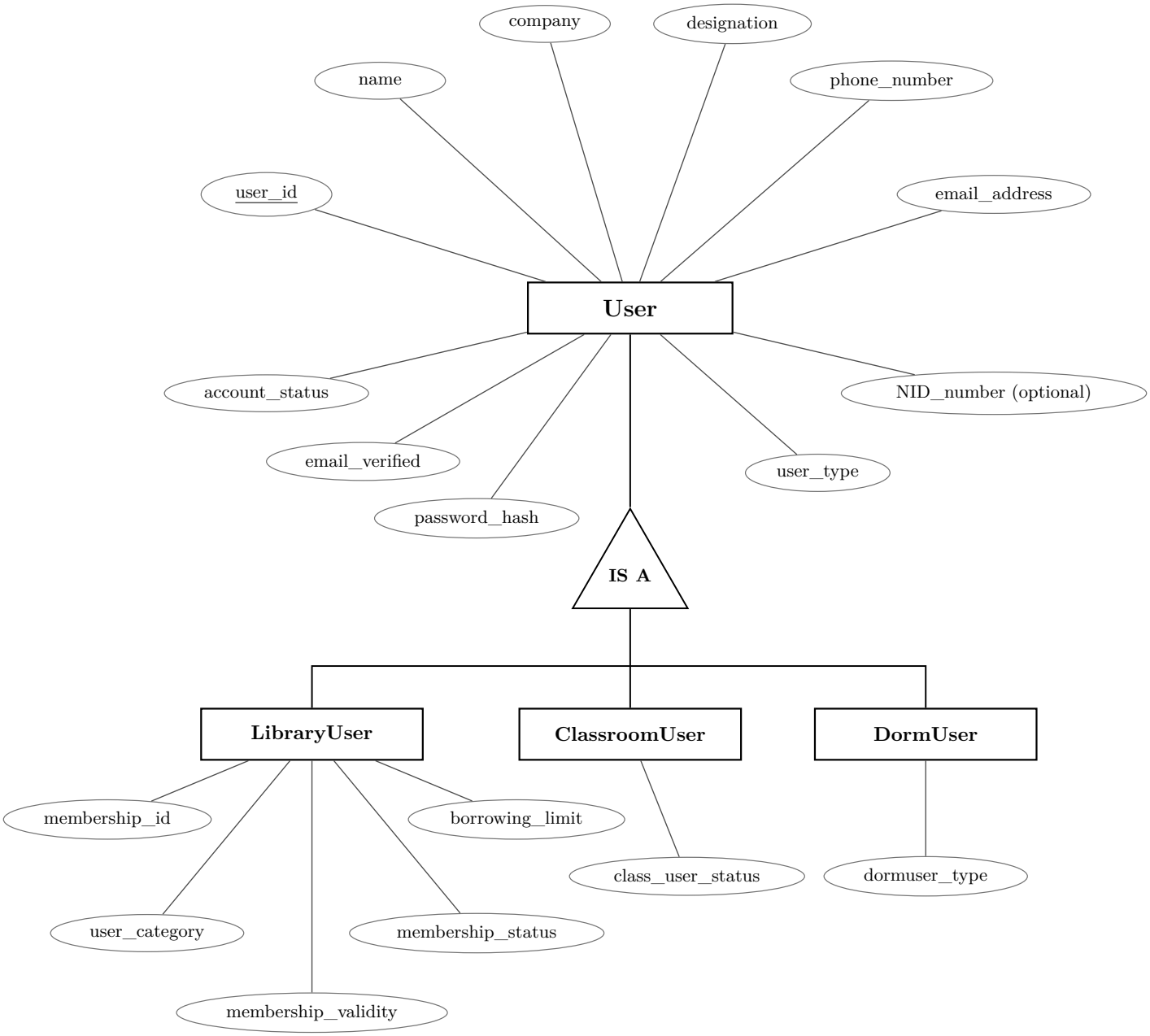


Figure 1: Unified Account Management

User

Attribute	Type	Size
user_id	VARCHAR2	100
name	VARCHAR2	200
company	VARCHAR2	200
designation	VARCHAR2	200
phone_number	VARCHAR2	30
email_address	VARCHAR2	254
NID_number	VARCHAR2	30
user_type	VARCHAR2	100
password_hash	VARCHAR2	255
email_verified	NUMBER	1
account_status	VARCHAR2	100

ClassroomUser

Attribute	Type	Size
user_id (PK, FK)	VARCHAR2	100
class_user_status	VARCHAR2	100

LibraryUser

Attribute	Type	Size
user_id (PK, FK)	VARCHAR2	100
membership_id	VARCHAR2	100
user_category	VARCHAR2	100
membership_validity	DATE	N/A
membership_status	VARCHAR2	100
borrowing_limit	NUMBER	5
qr_code	VARCHAR2	500

DormUser

Attribute	Type	Size
user_id (PK, FK)	VARCHAR2	100
dormuser_type	VARCHAR2	100

Classroom Management

An arrow points to the one side. No arrow is many-to-many. Arrows at both ends is one-to-one.

1. **Entities**

- ClassroomUser
- Speaker
- Course
- Session
- LearningResource
- Assignment
- Certificate

2. **Relations**

- Speaker Rates ClassroomUser (many to many): rating, comment, anonymous flag, session reference, rated at
- Speaker Takes Session (one to many)
- ClassroomUser Participates Course (many to many): joined at, participation status
- ClassroomUser Evaluates Course (many to many): rating, comment, submitted at
- ClassroomUser Submits Assignment (many to many): submitted work, submitted at, assessment result
- ClassroomUser Attends Session (many to many): attendance status, recorded at
- ClassroomUser Receives Certificate (one to many)
- Course Contains Session (one to many)
- Course Has Assignment (one to many)
- Course Has LearningResource (one to many)
- Course Includes Certificate (one to many)
- Session Has LearningResource (one to many)

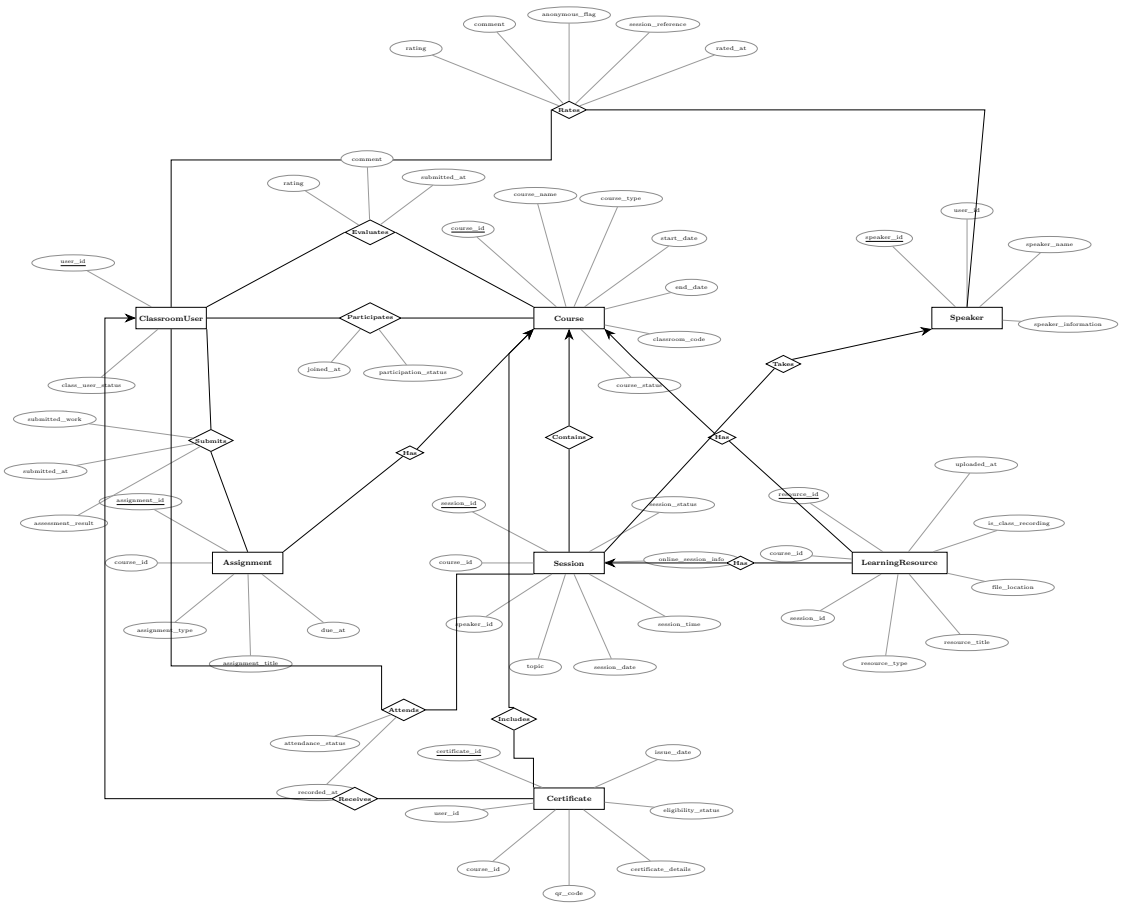


Figure 2: Classroom Management

Speaker

Attribute	Type	Size
speaker_id	VARCHAR2	100
user_id	VARCHAR2	100
speaker_name	VARCHAR2	200
speaker_information	CLOB	N/A

Course

Attribute	Type	Size
course_id	VARCHAR2	100
course_name	VARCHAR2	200
course_type	VARCHAR2	100
start_date	DATE	N/A
end_date	DATE	N/A
classroom_code	VARCHAR2	50
course_status	VARCHAR2	100

Session

Attribute	Type	Size
session_id	VARCHAR2	100
course_id	VARCHAR2	100
speaker_id	VARCHAR2	100
topic	VARCHAR2	300
session_date	DATE	N/A
session_time	VARCHAR2	30
online_session_info	VARCHAR2	500
session_status	VARCHAR2	100

LearningResource

Attribute	Type	Size
resource_id	VARCHAR2	100
course_id	VARCHAR2	100
session_id	VARCHAR2	100
resource_type	VARCHAR2	100
resource_title	VARCHAR2	300
file_location	VARCHAR2	500
is_class_recording	NUMBER	1
uploaded_at	TIMESTAMP	N/A

Assignment

Attribute	Type	Size
assignment_id	VARCHAR2	100
course_id	VARCHAR2	100
assignment_type	VARCHAR2	100
assignment_title	VARCHAR2	300
due_at	TIMESTAMP	N/A

Certificate

Attribute	Type	Size
certificate_id	VARCHAR2	100
user_id	VARCHAR2	100
course_id	VARCHAR2	100
qr_code	VARCHAR2	500
certificate_details	CLOB	N/A
eligibility_status	VARCHAR2	100
issue_date	DATE	N/A

CourseParticipation

Attribute	Type	Size
participation_id	VARCHAR2	100
user_id	VARCHAR2	100
course_id	VARCHAR2	100
joined_at	TIMESTAMP	N/A
participation_status	VARCHAR2	100

SpeakerRating

Attribute	Type	Size
speaker_rating_id	VARCHAR2	100
user_id	VARCHAR2	100
speaker_id	VARCHAR2	100
session_id	VARCHAR2	100
rating	NUMBER	2,1
comment	CLOB	N/A
anonymous_flag	NUMBER	1
rated_at	TIMESTAMP	N/A

AssignmentSubmission

Attribute	Type	Size
submission_id	VARCHAR2	100
user_id	VARCHAR2	100
assignment_id	VARCHAR2	100
submitted_work	VARCHAR2	500
submitted_at	TIMESTAMP	N/A
assessment_result	VARCHAR2	100

SessionAttendance

Attribute	Type	Size
attendance_id	VARCHAR2	100
user_id	VARCHAR2	100
session_id	VARCHAR2	100
attendance_status	VARCHAR2	100
recorded_at	TIMESTAMP	N/A

CourseEvaluation

Attribute	Type	Size
evaluation_id	VARCHAR2	100
user_id	VARCHAR2	100
course_id	VARCHAR2	100
rating	NUMBER	2,1
comment	CLOB	N/A
submitted_at	TIMESTAMP	N/A

Library Management

An arrow points to the one side. No arrow is many-to-many. Arrows at both ends is one-to-one.

1. Entities

- LibraryUser
- LibraryResource
- Ebook
- BookRequest
- Waitlist

2. Relations

- LibraryUser Borrows LibraryResource (many to many): issue date, due date, return date, overdue status, penalty amount
- LibraryUser Reviews LibraryResource (many to many): rating, review text, reviewed at
- LibraryUser Reads Ebook (many to many): accessed at, access action
- LibraryUser Reviews Ebook (many to many): rating, review text, reviewed at
- LibraryUser Submits BookRequest (one to many)
- LibraryUser Has Waitlist (one to many)
- LibraryResource Has Ebook (one to many)
- LibraryResource Has Waitlist (one to many)

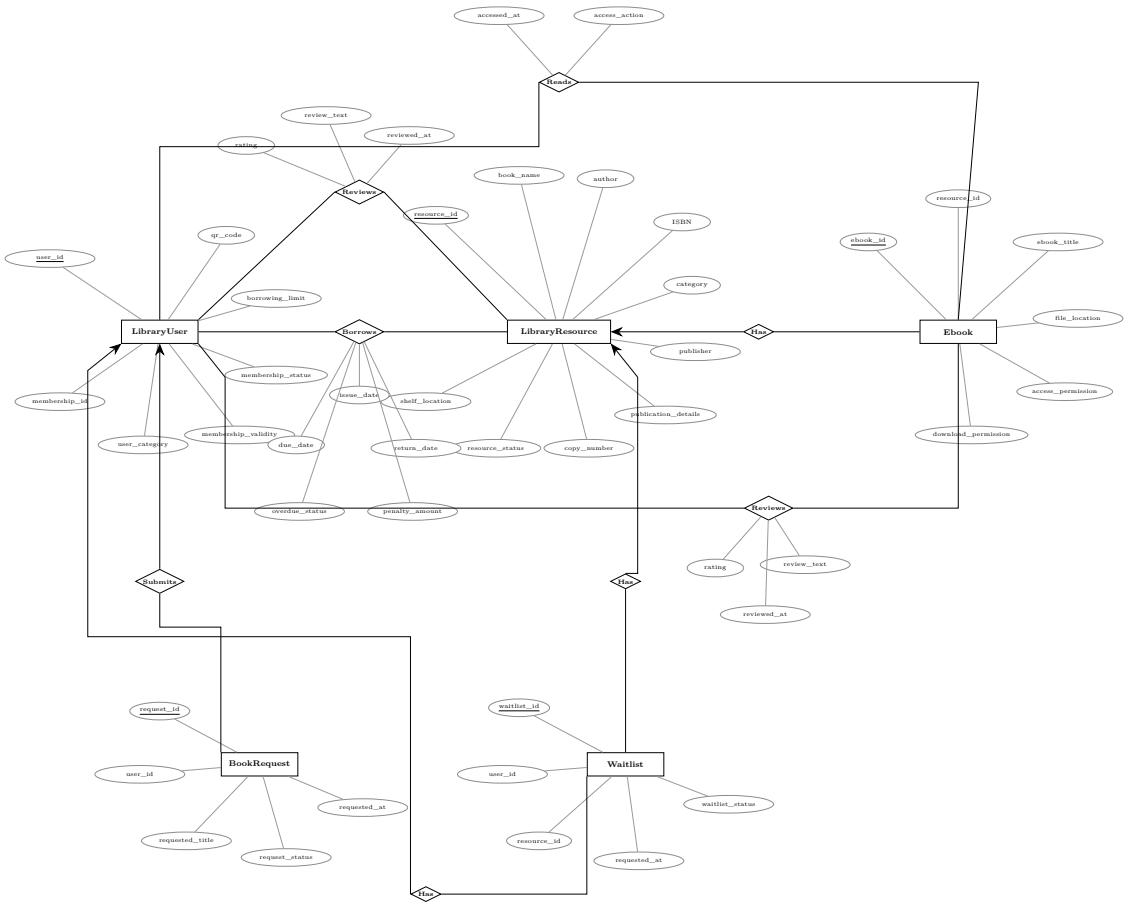


Figure 3: Library Management

LibraryResource

Attribute	Type	Size
resource_id	VARCHAR2	100
book_name	VARCHAR2	300
author	VARCHAR2	200
ISBN	VARCHAR2	20
category	VARCHAR2	100
publisher	VARCHAR2	200
publication_details	CLOB	N/A
copy_number	VARCHAR2	100
resource_status	VARCHAR2	100
shelf_location	VARCHAR2	255

Ebook

Attribute	Type	Size
ebook_id	VARCHAR2	100
resource_id	VARCHAR2	100
ebook_title	VARCHAR2	300
file_location	VARCHAR2	500
access_permission	VARCHAR2	100
download_permission	NUMBER	1

BookRequest

Attribute	Type	Size
request_id	VARCHAR2	100
user_id	VARCHAR2	100
requested_title	VARCHAR2	300
request_status	VARCHAR2	100
requested_at	TIMESTAMP	N/A

Waitlist

Attribute	Type	Size
waitlist_id	VARCHAR2	100
user_id	VARCHAR2	100
resource_id	VARCHAR2	100
requested_at	TIMESTAMP	N/A
waitlist_status	VARCHAR2	100

Borrowing

Attribute	Type	Size
borrowing_id	VARCHAR2	100
user_id	VARCHAR2	100
resource_id	VARCHAR2	100
issue_date	DATE	N/A
due_date	DATE	N/A
return_date	DATE	N/A
overdue_status	NUMBER	1
penalty_amount	NUMBER	12,2

ResourceReview

Attribute	Type	Size
resource_review_id	VARCHAR2	100
user_id	VARCHAR2	100
resource_id	VARCHAR2	100
rating	NUMBER	2,1
review_text	CLOB	N/A
reviewed_at	TIMESTAMP	N/A

EbookReview

Attribute	Type	Size
ebook_review_id	VARCHAR2	100
user_id	VARCHAR2	100
ebook_id	VARCHAR2	100
rating	NUMBER	2,1
review_text	CLOB	N/A
reviewed_at	TIMESTAMP	N/A

EbookRead

Attribute	Type	Size
ebook_read_id	VARCHAR2	100
user_id	VARCHAR2	100
ebook_id	VARCHAR2	100
accessed_at	TIMESTAMP	N/A
access_action	VARCHAR2	100

Dormitory Management

An arrow points to the one side. No arrow is many-to-many. Arrows at both ends is one-to-one.

1. Entities

- DormUser
- Application
- Room
- Seat
- IoTDevice
- Issue

2. Relations

- DormUser Submits Application (one to many)
- DormUser Raises Issue (one to many)
- DormUser Rates Room (many to many): rating, feedback, rated at
- DormUser Inhabits Room (many to many): check in, check out, stay status
- Room Has Seat (one to many)
- Room Has IoTDevice (one to many)
- Room Has Issue (one to many)

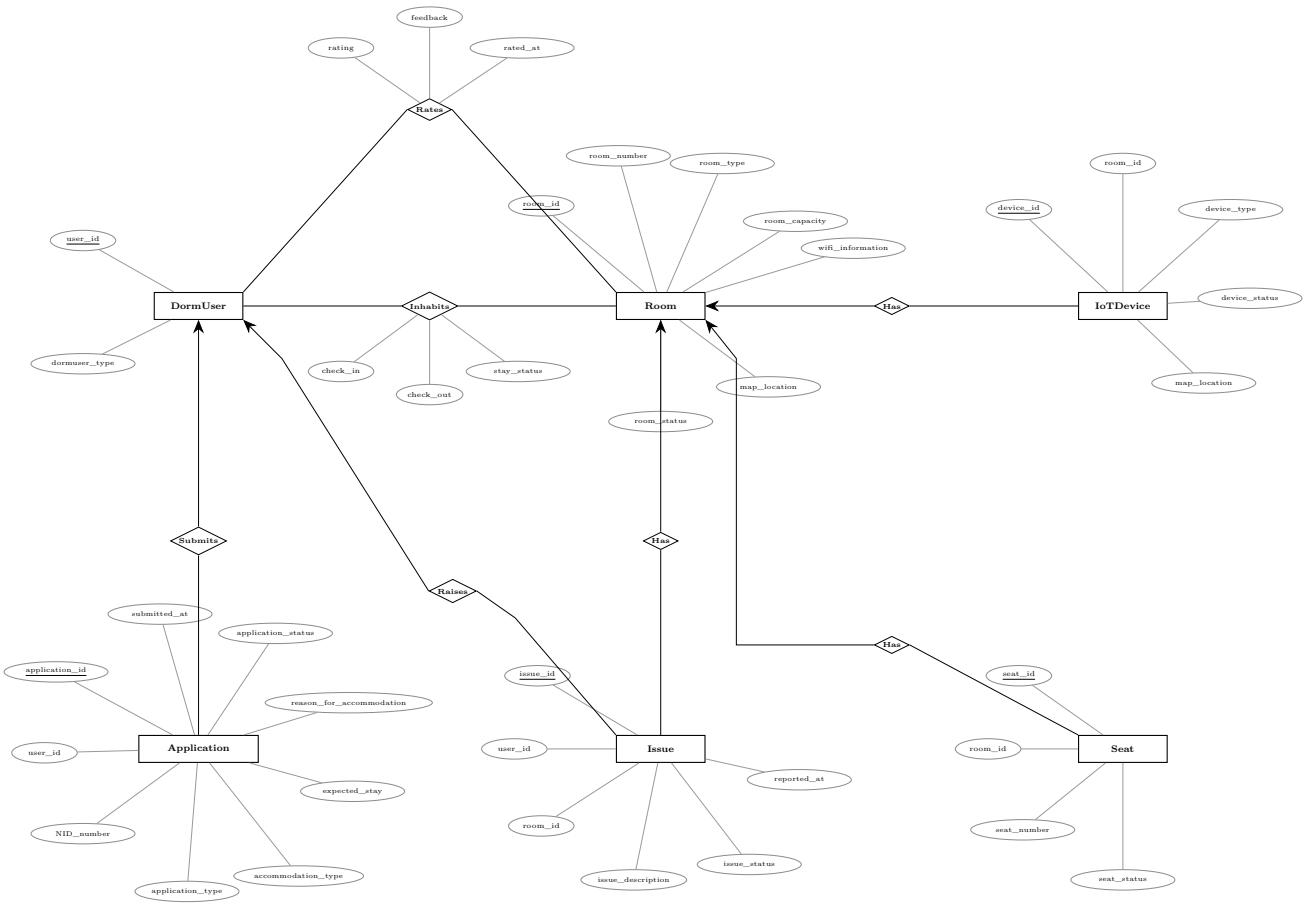


Figure 4: Dormitory Management

Application

Attribute	Type	Size
application_id	VARCHAR2	100
user_id	VARCHAR2	100
NID_number	VARCHAR2	30
application_type	VARCHAR2	100
accommodation_type	VARCHAR2	100
expected_stay	VARCHAR2	200
reason_for_accommodation	CLOB	N/A
application_status	VARCHAR2	100
submitted_at	TIMESTAMP	N/A

Room

Attribute	Type	Size
room_id	VARCHAR2	100
room_number	VARCHAR2	100
room_type	VARCHAR2	100
room_capacity	NUMBER	5
wifi_information	VARCHAR2	255
map_location	VARCHAR2	255
room_status	VARCHAR2	100

Seat

Attribute	Type	Size
seat_id	VARCHAR2	100
room_id	VARCHAR2	100
seat_number	VARCHAR2	100
seat_status	VARCHAR2	100

IoTDevice

Attribute	Type	Size
device_id	VARCHAR2	100
room_id	VARCHAR2	100
device_type	VARCHAR2	100
device_status	VARCHAR2	100
map_location	VARCHAR2	255

Issue

Attribute	Type	Size
issue_id	VARCHAR2	100
user_id	VARCHAR2	100
room_id	VARCHAR2	100
issue_description	CLOB	N/A
issue_status	VARCHAR2	100
reported_at	TIMESTAMP	N/A

RoomRating

Attribute	Type	Size
room_rating_id	VARCHAR2	100
user_id	VARCHAR2	100
room_id	VARCHAR2	100
rating	NUMBER	2,1
feedback	CLOB	N/A
rated_at	TIMESTAMP	N/A

StayRecord

Attribute	Type	Size
stay_id	VARCHAR2	100
user_id	VARCHAR2	100
room_id	VARCHAR2	100
check_in	TIMESTAMP	N/A
check_out	TIMESTAMP	N/A
stay_status	VARCHAR2	100

Class-based modeling

Class-based modeling identifies the static structure of the NIMC Integrated Management System. The steps are noun parsing, general classification, selection criteria, attribute and method selection, analysis, class cards, and CRC diagrams.

A candidate passes general classification when it is an external entity or it receives at least three category numbers. A candidate passes selection when it satisfies at least three criteria, or criterion 6. Passing selection does not prevent a later merge.

Room, Seat, and Room Allocation do not pass selection, so they are not classes. Physical Book and Digital Resource pass selection and are merged into Library Resource.

1. Noun parsing

Nouns and noun phrases are taken from the user story. The problem-domain list is the wording of the story. The solution-space list is the candidate set carried into general classification.

1.1 Problem-domain nouns

1. NIMC account
2. participant
3. NIMC employee
4. name
5. company
6. designation
7. phone number
8. email address
9. NID number
10. email verification
11. password
12. password reset
13. existing identity
14. Classroom service
15. Library service
16. Dormitory service
17. training course
18. seminar
19. workshop
20. speaker
21. Course Coordination Team
22. Course Coordinator
23. classroom
24. classroom code
25. session plan
26. session topic
27. date
28. time
29. assigned speaker
30. learning resource
31. document
32. presentation
33. video
34. notes
35. class recording
36. attendance
37. overall attendance
38. engagement indicator
39. notification
40. assignment
41. examination
42. assessment result
43. submission
44. AI-based analysis
45. similarity indicator
46. plagiarism indicator
47. AI-generated-content indicator
48. speaker evaluation
49. rating
50. comment
51. course evaluation
52. course feedback
53. certificate
54. QR code
55. LLM-based assistant
56. library membership
57. visitor
58. assistant librarian
59. librarian
60. membership validity
61. digital membership card
62. membership ID
63. borrowing eligibility
64. borrowing limit
65. physical book
66. library catalogue
67. ISBN
68. category
69. publisher
70. issue date
71. due date
72. return date
73. overdue status
74. penalty
75. circulation information
76. e-library
77. digital resource
78. magazine
79. open-source book
80. access permission
81. download permission
82. usage statistics
83. LibraryAssistantAI
84. book review
85. book request
86. monthly library report
87. dormitory application
88. accommodation type
89. expected stay
90. single seat
91. double seat
92. Hostel Manager
93. approval link
94. Director General
95. Additional Director General
96. Director
97. room
98. seat
99. room capacity
100. occupied seats
101. available seats
102. dormitory fee
103. waiver
104. payment confirmation
105. room allocation
106. check-in
107. stay status
108. stay-extension request
109. cancellation request
110. dormitory issue
111. Hostel Super
112. Warden
113. issue status
114. check-out
115. dormitory dashboard
116. monthly dormitory report
117. Hostel AI
118. 3D map location
119. authorized NIMC employee
120. registered email address
121. account information
122. shared NIMC account
123. module-specific information
124. participant information
125. organization name
126. unique classroom code
127. participant eligibility
128. initial session plan

129. schedule change
130. topic change
131. speaker change
132. resource type
133. online session
134. authorized participant
135. attendance percentage
136. participation level
137. participant performance
138. important classroom activity
139. session schedule change
140. online session information
141. assignment notice
142. examination notice
143. course update
144. submitted work
145. assessment information
146. academic decision
147. flagged work
148. speaker rating
149. speaker rating history
150. evaluation process
151. overall course evaluation
152. course effectiveness
153. area for improvement
154. authorized user account
155. participant record
156. performance information
157. certificate eligibility condition
158. certificate details
159. course resource
160. participant question
161. course material
162. library user category
163. course participant membership
164. office employee
165. government employee
166. external visitor
167. membership period
168. membership category
169. library user
170. current borrowed books
171. return deadline
172. borrowing period
173. book name
174. author
175. catalogue information
176. book information
177. physical copy
178. online resource
179. circulation record
180. book status
181. borrowed-book list
182. return reminder
183. overdue book
184. penalty amount
185. penalty rule
186. penalty adjustment
187. waiting user
188. resource availability
189. withdrawn book record
190. online copy
191. resource access permission
192. resource download permission
193. digital resource usage
194. book recommendation
195. user interest
196. learning need
197. resource review
198. related resource
199. inappropriate review
200. book-request list
201. resource demand
202. resource acquisition
203. catalogue details
204. publication details
205. physical-copy status
206. library activity
207. newly added book
208. active library member
209. new membership
210. most borrowed book
211. popular requested resource
212. dormitory facility
213. accommodation requirement
214. reason for accommodation
215. approval process
216. responsible person
217. approval sequence
218. room and seat chart

1.2 Solution-space nouns

1. User
2. NIMC Account
3. Email Service
4. SMS Service
5. AI Service
6. Payment System
7. ClassroomParticipant
8. Speaker
9. Course Coordinator
10. Course Coordination Team
11. Course
12. NIMC Classroom
13. Course Enrollment
14. Course Session
15. Learning Resource
16. Attendance Record
17. Assessment
18. Submission
19. Submission Analysis Service
20. Certificate
21. Classroom AI Assistant
22. Library Member
23. Librarian
24. Library Resource
25. Physical Book
26. Digital Resource
27. E-Library
28. Borrowing Record
29. Book Review
30. Book Request
31. Library Assistant AI
32. Dorm User
33. Dormitory Application
34. Dormitory Staff
35. Hostel Manager
36. Approval Authority
37. Approval Record
38. Room
39. Seat
40. Room Allocation
41. Stay Record
42. Stay Change Request
43. Dormitory Issue
44. Hostel AI
45. Classroom Code
46. Session Plan
47. Session Topic
48. Engagement Indicator
49. Notification
50. Rating
51. Course Feedback
52. QR Code
53. Membership Validity
54. Digital Membership Card
55. Borrowing Limit
56. Due Date
57. Penalty
58. Usage Statistics
59. Room Capacity
60. Dormitory Fee
61. Waiver
62. Check-in
63. Check-out
64. Dormitory Dashboard
65. Monthly Library Report
66. Monthly Dormitory Report
67. Name
68. Company
69. Designation
70. Phone Number
71. Email Address
72. NID Number
73. Email Verification
74. Password
75. Password Reset
76. Existing Identity
77. Participant Information
78. Organization Name
79. Participant Eligibility
80. Date
81. Time
82. Assigned Speaker
83. Document
84. Presentation

85. Video
86. Notes
87. Class Recording
88. Attendance
89. Overall Attendance
90. Similarity Indicator
91. Plagiarism Indicator
92. AI-generated-content Indicator
93. Speaker Rating
94. Comment
95. Certificate Eligibility Condition
96. Certificate Details
97. Resource Type
98. Online Session Information
99. Participant Performance
100. Membership ID
101. Borrowing Eligibility
102. Library Catalogue
103. ISBN
104. Category
105. Publisher
106. Issue Date
107. Return Date
108. Overdue Status
109. Circulation Information
110. Access Permission
111. Download Permission
112. Library User Category
113. Membership Period
114. Book Name
115. Author
116. Catalogue Information
117. Book Status
118. Resource Availability
119. Penalty Amount
120. Penalty Rule
121. Penalty Adjustment
122. Publication Details
123. Accommodation Type
124. Expected Stay
125. Single Seat
126. Double Seat
127. Approval Link
128. Occupied Seats
129. Available Seats
130. Payment Confirmation
131. Stay Status
132. Issue Status
133. 3D Map Location
134. Accommodation Requirement
135. Reason for Accommodation
136. Approval Sequence
137. Room and Seat Chart

2. General classification (GC)

General classification uses seven categories: 1. External entity, 2. Things, 3. Occurrence or event, 4. Role, 5. Organizational unit or subsystem, 6. Places, 7. Structures. A noun passes when it has classification 1, or at least three separate classification numbers. Solution-space nouns that fail are kept in this table to show why they are not potential classes.

Sl. no.	Noun / noun phrase	GC passed in	Comment
1	User	4 5 7	Passed
2	NIMC Account	2 5 7	Passed
3	Email Service	1 5 7	Passed
4	SMS Service	1 5 7	Passed
5	AI Service	1 5 7	Passed
6	Payment System	1 5 7	Passed
7	ClassroomParticipant	1 4 7	Passed
8	Speaker	1 4 7	Passed
9	Course Coordinator	1 4 5	Passed
10	Course Coordination Team	1 4 5	Passed
11	Course	2 5 7	Passed
12	NIMC Classroom	2 5 7	Passed
13	Course Enrollment	2 3 7	Passed
14	Course Session	2 3 7	Passed
15	Learning Resource	2 5 7	Passed
16	Attendance Record	2 3 7	Passed
17	Assessment	2 3 7	Passed
18	Submission	2 3 7	Passed
19	Submission Analysis Service	1 5 7	Passed
20	Certificate	2 3 7	Passed
21	Classroom AI Assistant	2 5 7	Passed
22	Library Member	1 4 7	Passed
23	Librarian	1 4 5	Passed
24	Library Resource	2 5 7	Passed
25	Physical Book	1 2 7	Passed
26	Digital Resource	2 5 7	Passed
27	E-Library	2 5 7	Passed
28	Borrowing Record	2 3 7	Passed
29	Book Review	2 3 7	Passed
30	Book Request	2 3 7	Passed
31	Library Assistant AI	2 5 7	Passed
32	Dorm User	1 4 7	Passed
33	Dormitory Application	2 3 7	Passed
34	Dormitory Staff	1 4 5	Passed
35	Hostel Manager	1 4 5	Passed

Sl. no.	Noun / noun phrase	GC passed in	Comment
36	Approval Authority	1 4 5	Passed
37	Approval Record	2 3 7	Passed
38	Room	1 2 6 7	Passed
39	Seat	1 2 6 7	Passed
40	Room Allocation	2 3 7	Passed
41	Stay Record	2 3 7	Passed
42	Stay Change Request	2 3 7	Passed
43	Dormitory Issue	2 3 7	Passed
44	Hostel AI	2 5 7	Passed
45	Classroom Code	2	Failed
46	Session Plan	7	Failed
47	Session Topic	2	Failed
48	Engagement Indicator	2	Failed
49	Notification	3	Failed
50	Rating	2	Failed
51	Course Feedback	3	Failed
52	QR Code	2	Failed
53	Membership Validity	2	Failed
54	Digital Membership Card	2	Failed
55	Borrowing Limit	2	Failed
56	Due Date	2	Failed
57	Penalty	3	Failed
58	Usage Statistics	2	Failed
59	Room Capacity	2	Failed
60	Dormitory Fee	2	Failed
61	Waiver	3	Failed
62	Check-in	3	Failed
63	Check-out	3	Failed
64	Dormitory Dashboard	7	Failed
65	Monthly Library Report	2	Failed
66	Monthly Dormitory Report	2	Failed
67	Name		Failed
68	Company		Failed
69	Designation		Failed
70	Phone Number		Failed
71	Email Address		Failed
72	NID Number		Failed
73	Email Verification	3	Failed
74	Password	2	Failed
75	Password Reset	3	Failed
76	Existing Identity	2	Failed
77	Participant Information	7	Failed
78	Organization Name		Failed
79	Participant Eligibility	2	Failed
80	Date		Failed
81	Time		Failed
82	Assigned Speaker	4	Failed
83	Document	2	Failed
84	Presentation	2	Failed
85	Video	2	Failed
86	Notes	2	Failed
87	Class Recording	2	Failed
88	Attendance	3	Failed
89	Overall Attendance	2	Failed
90	Similarity Indicator	2	Failed
91	Plagiarism Indicator	2	Failed
92	AI-generated-content Indicator	2	Failed
93	Speaker Rating	2	Failed
94	Comment		Failed
95	Certificate Eligibility Condition	7	Failed
96	Certificate Details	7	Failed

Sl. no.	Noun / noun phrase	GC passed in	Comment
97	Resource Type		Failed
98	Online Session Information	2	Failed
99	Participant Performance	2	Failed
100	Membership ID		Failed
101	Borrowing Eligibility	2	Failed
102	Library Catalogue	7	Failed
103	ISBN		Failed
104	Category		Failed
105	Publisher	2	Failed
106	Issue Date		Failed
107	Return Date		Failed
108	Overdue Status	2	Failed
109	Circulation Information	7	Failed
110	Access Permission	2	Failed
111	Download Permission	2	Failed
112	Library User Category		Failed
113	Membership Period	2	Failed
114	Book Name		Failed
115	Author		Failed
116	Catalogue Information	7	Failed
117	Book Status	2	Failed
118	Resource Availability	2	Failed
119	Penalty Amount		Failed
120	Penalty Rule	7	Failed
121	Penalty Adjustment	3	Failed
122	Publication Details		Failed
123	Accommodation Type		Failed
124	Expected Stay	2	Failed
125	Single Seat	2	Failed
126	Double Seat	2	Failed
127	Approval Link	2	Failed
128	Occupied Seats	2	Failed
129	Available Seats	2	Failed
130	Payment Confirmation	3	Failed
131	Stay Status	2	Failed
132	Issue Status	2	Failed
133	3D Map Location	6	Failed
134	Accommodation Requirement	2	Failed
135	Reason for Accommodation		Failed
136	Approval Sequence	7	Failed
137	Room and Seat Chart	7	Failed

2.1 Potential classes

These nouns passed general classification and are the input to selection criteria.

1. User
2. NIMC Account
3. Email Service
4. SMS Service
5. AI Service
6. Payment System
7. ClassroomParticipant
8. Speaker
9. Course Coordinator
10. Course Coordination Team
11. Course
12. NIMC Classroom
13. Course Enrollment
14. Course Session
15. Learning Resource
16. Attendance Record
17. Assessment
18. Submission
19. Submission Analysis Service
20. Certificate
21. Classroom AI Assistant
22. Library Member
23. Librarian
24. Library Resource
25. Physical Book
26. Digital Resource
27. E-Library
28. Borrowing Record
29. Book Review
30. Book Request
31. Library Assistant AI
32. Dorm User
33. Dormitory Application
34. Dormitory Staff
35. Hostel Manager
36. Approval Authority
37. Approval Record
38. Room
39. Seat
40. Room Allocation
41. Stay Record
42. Stay Change Request
43. Dormitory Issue
44. Hostel AI

3. Selection criteria (SC)

Selection criteria are: 1. Retained information, 2. Needed service, 3. Multiple attributes, 4. Common attributes, 5. Common operation, 6. Essential requirements. A candidate passes when it satisfies at least three criteria, or criterion 6.

Noun / noun phrase	Criteria satisfied	Comment
User	1. Retained Information - name,email 2. Needed Service - NIMC Account 3. Multiple Attributes - email„ name	Passed - retained
NIMC Account	1. Retained Information - accountID, passwordHash 2. Needed Service - User 3. Multiple Attributes - accountID, userID	Passed - retained
Email Service	2. Needed Service - NIMC Account 3. Multiple Attributes - messageID, recipientEmail 6. Essential Requirement - Email verification and normal notifications are explicitly	Passed — retained as Notification Service
SMS Service	2. Needed Service - Course Coordinator 3. Multiple Attributes - messageID, recipientPhone 6. Essential Requirement - SMS for special notification cases	Passed — retained as Notification Service
AI Service	2. Needed Service - Course Coordinator 3. Multiple Attributes - serviceID, providerName 6. Essential Requirement - Shared AI capability by analysis and assistant	Passed - retained (external)
Payment System	1. Retained Information - transactionID, userID 2. Needed Service - Dorm User / Hostel Manager 3. Multiple Attributes - transactionID, userID	Passed - retained (external)
ClassroomParticipant	1. Retained Information - participantID, userID 2. Needed Service - User 3. Multiple Attributes - participantID, userID	Passed - retained
Speaker	1. Retained Information - speakerID, name 2. Needed Service - Course Coordinator 3. Multiple Attributes - speakerID, name	Passed - retained
Course Coordinator	1. Retained Information - coordinatorID, userID 2. Needed Service - Course 3. Multiple Attributes - coordinatorID, userID	Passed - retained
Course Coordination Team	2. Needed Service - Course Coordinator	Failed
Course	1. Retained Information - courseID, title 2. Needed Service - Course Coordinator 3. Multiple Attributes - courseID, title	Passed - retained
NIMC Classroom	1. Retained Information - classroomCode 2. Needed Service - Course 3. Multiple Attributes - classroomCode, participant list	Passed - merged into Course during refinement
Course Enrollment	Needed Service - Course	Failed
Course Session	1. Retained Information - sessionID, courseID 2. Needed Service - Course 3. Multiple Attributes - sessionID, courseID	Passed - retained
Learning Resource	1. Retained Information - resourceID, courseID 2. Needed Service - Course Coordinator 3. Multiple Attributes - resourceID, courseID	Passed - retained
Attendance Record	Retained Information - participantID	Failed
Assessment	1. Retained Information - assignment/result information 2. Needed Service - ClassroomParticipant / Course Coordinator 3. Multiple Attributes - submission, result, analysis flags	Passed - merged into ClassroomParticipant and Course Coordinator
Submission		Failed
Submission Analysis Service		Failed
Certificate	1. Retained Information - certificate details 2. Needed Service - ClassroomParticipant / Course 3. Multiple Attributes - issue date, eligibility, QR code	Passed - merged into ClassroomParticipant and Course
Classroom AI Assistant	1. Retained Information - assistantID, courseID 2. Needed Service - ClassroomParticipant 3. Multiple Attributes - assistantID, courseID	Passed - retained
Library Member	1. Retained Information - libraryMemberID, userID 2. Needed Service - User 3. Multiple Attributes - libraryMemberID, userID	Passed - retained
Librarian	1. Retained Information - librarianID, userID 2. Needed Service - Library Member 3. Multiple Attributes - librarianID, userID	Passed - retained; Assistant Librarian merged
Library Resource	1. Retained Information - resourceID, title 2. Needed Service - Library Member 3. Multiple Attributes - resourceID, title	Passed - retained

Noun / noun phrase	Criteria satisfied	Comment
Physical Book	1. Retained Information - resourceID, isbn 2. Needed Service - Library Resource 3. Multiple Attributes - resourceID, isbn	Passed — merged into Library Resource
Digital Resource	1. Retained Information - resourceID, fileLink 2. Needed Service - E-Library 3. Multiple Attributes - resourceID, fileLink	Passed — merged into Library Resource
E-Library	1. Retained Information - eLibraryID, digitalResourceIDs 2. Needed Service - Library Resource 3. Multiple Attributes - eLibraryID, digitalResourceIDs	Passed - retained
Borrowing Record	Needed Service - Resources	Failed
Book Review	1. Retained Information - reviewID, memberID 2. Needed Service - LibraryUser	Failed
Book Request	1. Retained Information - requestID, memberID 2. Needed Service - Librarian	Failed
Library Assistant AI	1. Retained Information - assistantID, userQuery 2. Needed Service - Library Member 3. Multiple Attributes - assistantID, userQuery	Passed - retained
Dorm User	1. Retained Information - dormUserID, userID 2. Needed Service - User 3. Multiple Attributes - dormUserID, userID	Passed - retained
Dormitory Application	1. Retained Information - application and approval/payment information 2. Needed Service - Dorm User / Hostel Manager 3. Multiple Attributes - application status, preference, fee/payment	Passed - merged into Dorm User and Hostel Manager
Dormitory Staff	1. Retained Information - user, name	Failed
Hostel Manager	1. Retained Information - manager role, occupancy/application records 2. Needed Service - Dorm User 3. Multiple Attributes - permissions, occupancySummary	Passed - retained
Approval Authority	1. Retained Information - authority role, decision status 2. Needed Service - Hostel Manager 3. Multiple Attributes - authorityRole, sequenceOrder	Passed - retained
Approval Record	1. Needed Service - Hostel Manager 2. Multiple Attributes - approvalID, applicationID	Failed
Room	1. Retained information — allocatedRoomID on Dorm User 2. Needed service — Hostel Manager	Failed
Seat	1. Retained information — allocatedSeatID on Dorm User 2. Needed service — Hostel Manager	Failed
Room Allocation	1. Retained information — allocatedRoomID, allocatedSeatID on Dorm User 2. Needed service — Hostel Manager / Dorm User	Failed
Stay Record	1. Retained Information - stayID, allocationID 2. Needed service — Dorm User	Failed
Stay Change Request	1. Retained Information - requestID, dormUserID	Failed
Dormitory Issue	1. Retained Information - issueID, dormUser 3. Multiple Attributes - issueID, dormUserID	Failed
Hostel AI	1. Retained Information - assistantID, userQuery 2. Needed Service - Dorm User 3. Multiple Attributes - assistantID, userQuery	Passed - retained

Room satisfies only two criteria: the allocated room identifier is kept on Dorm User, and availability is a service of Hostel Manager. Seat satisfies the same two criteria through `allocatedSeatID`. Room Allocation satisfies two criteria because the allocated room and seat already belong to Dorm User, and Hostel Manager performs the allocation. None of the three reaches three criteria or criterion 6, so none is retained.

Physical Book and Digital Resource each satisfy three criteria. They are not separate classes. Both are merged into Library Resource. `resourceFormat` records whether that resource is physical, digital, or both.

3.1 Selected classes

After assessment and merging, 21 classes are retained.

Shared / common. User; NIMC Account; Notification Service; AI Service; Payment System.

Classroom. ClassroomParticipant; Speaker; Course Coordinator; Course; Course Session; Learning Resource; Classroom AI Assistant.

Library. Library Member; Librarian; Library Resource; E-Library; Library Assistant AI.

Dormitory. Dorm User; Hostel Manager; Approval Authority; Hostel AI.

4. Attribute selection and method selection

4.1 Method selection by verb parsing

Verbs and verb phrases from the user story are evaluated as probable methods. A dash means the verb is a user or interface action and does not create a separate class method.

Verb / verb phrase	Probable method	Class
open	`openResource()`	E-Library
activate	`activateIdentity()`	User
verify	`verifyEmail()`	NIMC Account
sign in	`login()`	NIMC Account
update	`updateProfile()`	User
request	`requestStayExtension()`	Dorm User
reset	`resetPassword()`	NIMC Account
create	`createClassroom()`	Course Coordinator
share	-	-
enter	`joinClassroom(code)`	ClassroomParticipant
remove	`removeParticipant()`	Course Coordinator
contact	-	-
assign	`assignSpeaker()`	Course Session
prepare	`prepareSessionPlan()`	Course Coordinator
change	`updateSession()`	Course Session
view	`viewSessionPlan()`	ClassroomParticipant
upload	`uploadResource()`	Course Coordinator
access	`accessResource()`	ClassroomParticipant
search	`searchInResource()`	Learning Resource
sort	`sortResources()`	Course Session
filter	`filterResources()`	Course Session
store	-	-
record	`recordParticipantAttendance()`	Course Session
calculate	`calculateFee()`	Hostel Manager
generate	`generateCertificate()`	Course
identify	`analyzeSubmission()`	Course Coordinator
notify	`sendNotification()`	Notification Service
submit	`submitAssignment()`	ClassroomParticipant
evaluate	`evaluateSpeaker()`	ClassroomParticipant
analyze	`analyzeSubmission()`	Course Coordinator
flag	-	-
review	`reviewApplication()`	Hostel Manager
determine	`calculateFee()`	Hostel Manager
rate	`evaluateSpeaker()`	ClassroomParticipant
collect	-	-
manage	`manageCatalogue()`	Librarian
issue	`issueResource()`	Librarian
scan	`verifyCertificateQRCode()`	Course
check	`checkCertificateEligibility()`	Course
borrow	`issueResource()`	Librarian
extend	`extendDueDate()`	Librarian
display	`displayResources()`	Course
select	`checkAvailability()`	Hostel Manager
return	`returnResource()`	Librarian
apply	`applyWaiver()`	Payment System
override	`overridePenalty()`	Librarian
read	`openResource()`	E-Library
download	`downloadResource()`	E-Library
recommend	`recommendResources()`	Library Assistant AI
moderate	`moderateReview()`	Librarian
add	`manageCatalogue()`	Librarian

Verb / verb phrase	Probable method	Class
maintain	`updateResource()`	Library Resource
approve	`approve()`	Approval Authority
reject	`reject()`	Approval Authority
forward	`forwardToNextAuthority()`	Approval Authority
allocate	`allocateRoom()`	Hostel Manager
confirm	`confirmPayment()`	Payment System
check in	`recordCheckIn()`	Hostel Manager
check out	`recordCheckOut()`	Hostel Manager
report	`reportIssue()`	Dorm User
resolve	`manageIssue()`	Hostel Manager
release	`manageOccupancy()`	Hostel Manager
track	`viewApplicationStatus()`	Dorm User
configure	`configureCertificateRules()`	Course
retrieve	`retrieveCourseResources()`	Classroom AI Assistant
preserve	-	-

The attributes of each retained class follow. Methods shown with them are the operations kept after verb parsing. Library Resource carries the attributes of both the physical copy and the digital resource.

4.2 Attribute selection

1. **User.** Attributes: userID, Account, name, company, designation, phone, email, nidNumber. Methods: `viewProfile()`, `updateProfile()`, `activateIdentity()`.

2. **NIMC Account.** Attributes: accountID, userID, registeredEmail, passwordHash, emailVerified, accountStatus. Methods: `registerAccount()`, `verifyEmail()`, `login()`, `resetPassword()`, `updateCredentials()`.

3. **Notification Service.** Attributes: messageID, recipientEmail, recipientPhone, subject, messageType, deliveryStatus. Methods: `sendVerificationEmail()`, `sendPasswordReset()`, `sendNotification()`, `sendApprovalLink()`.

4. **AI Service.** Attributes: providerName, modelName, serviceStatus. Methods: `processPrompt()`, `analyzeContent()`.

5. **Payment System.** Attributes: transactionID, userID, amount, paymentType, paymentStatus. Methods: `processPayment()`, `confirmPayment()`, `applyWaiver()`, `getPaymentStatus()`.

6. **ClassroomParticipant.** Attributes: participantID, attendance[], course[], certificateID, certificateIssueDate, certificateEligibilityStatus, certificateQRCode. Methods: `joinClassroom(code)`, `viewSessionPlan()`, `accessResource()`, `viewAttendance()`, `submitAssignment()`, `evaluateSpeaker()`, `evaluateCourse()`, `viewCertificate()`.

7. **Speaker.** Attributes: speakerID, contactInformation, expertise, overallRating, ratingHistory[], coursesTaken[]. Methods: `viewAssignedSessions()`, `provideMaterials()`, `viewOwnSchedule()`, `seeOverallRating()`.

8. **Course Coordinator.** Attributes: name, email. Methods: `createClassroom()`, `addParticipant()`, `removeParticipant()`, `prepareSessionPlan()`, `updateSession()`, `uploadResource()`, `recordAttendance()`, `recordAssessmentResult()`, `issueCertificate()`, `analyzeSubmission()`, `checkPlagerism()`, `checkAI()`, `checkwithAI()`.

9. **Course.** Attributes: title, courseType, startDate, endDate, description, status, feedbackRecord[], session[], classroomCode, participantIDs[], resourceIDs[], certificateRules. Methods: `updateCourse()`, `configureCertificateRules()`, `getSessions()`, `getParticipants()`, `recordCourseFeedback()`, `displayResources()`, `checkCertificateEligibility()`, `generateCertificate()`, `verifyCertificateQRCode()`.

10. **Course Session.** Attributes: topic, sessionDate, sessionTime, speaker, mode, recordingLink, attendanceList, sessionStatus. Methods: `updateSession()`, `assignSpeaker()`, `recordParticipantAttendance()`, `notifyChange()`, `sortResources()`, `filterResources()`.

11. **Learning Resource.** Attributes: title, resourceType, fileLink, uploadDate, accessLevel. Methods: `upload()`, `update()`, `searchInResource()`.

12. **Classroom AI Assistant.** Attributes: assistantID, AIService, course, user. Methods: `askQuestion()`, `retrieveCourseResources()`, `answerFromResources()`.

13. **Library Member.** Attributes: category, membershipID, validFrom, validUntil, borrowingLimit, borrowedResourceIDs, qrCode, membershipStatus. Methods: `activateMembership()`, `viewBorrowedResources()`, `viewDueDates()`, `searchCatalogue()`, `openDigitalResource()`, `requestBook()`, `submitReview()`, `askLibraryAI()`.

14. **Librarian.** Attributes: librarianID, staffRole, permissions. Methods: `verifyMember()`, `assignMembership()`, `manageCatalogue()`, `issueResource()`, `returnResource()`, `extendDueDate()`, `applyPenalty()`, `overridePenalty()`, `moderateReview()`, `reviewRequests()`, `generateMonthlyReport()`.

15. **Library Resource.** Attributes: title, author, isbn, category, publisher, publicationInfo, resourceFormat, availabilityStatus, accessPermission, downloadAllowed, currentBorrowerID, issueDate, dueDate, returnDate, circulationHistory, reviews, wailtList[]. Methods: `updateResource()`, `search()`, `markBorrowed()`, `markReturned()`, `setAccessPermission()`, `recordUsage()`, `addReview()`, `registerRequest()`. This class is the merged physical copy and digital resource.

16. **E-Library.** Attributes: eResources[], usageStatistics. Methods: `browse()`, `searchDigitalResources()`, `openResource()`, `downloadResource()`, `recordUsage()`.

17. **Library Assistant AI.** Attributes: assistantID, user, AIService. Methods: `recommendResources()`, `answerPrompt()`, `summarizeReviews()`, `suggestAlternative()`.

18. **Dorm User.** Attributes: participantCategory, applicationID, nidNumber, preferredAccommodation, expectedStart, expectedEnd, seatType, reason, applicationStatus, currentApprovalStage, feeAmount, waiverStatus, paymentStatus, allocatedRoomID, allocatedSeatID, stayStatus, reportedIssues. Methods: `submitDormitoryApplication()`, `viewApplicationStatus()`, `viewAllocation()`, `requestStayExtension()`, `requestCancellation()`, `requestWaiver()`, `reportIssue()`, `viewIssueStatus()`, `viewStayDetails()`, `askHostelAI()`.

19. **Hostel Manager.** Attributes: name, email, permissions, occupancySummary, applicationRecords[], approvalHistory[]. Methods: `reviewApplication()`, `updateApplicationStatus()`, `recordApproval()`, `calculateFee()`, `markPendingForAvailability()`, `confirmPaymentStatus()`, `checkAvailability()`, `allocateRoom()`, `recordCheckIn()`, `recordCheckOut()`, `manageOccupancy()`, `approveStayChange()`, `manageIssue()`, `generateMonthlyReport()`.

20. **Approval Authority.** Attributes: name, email, authorityRole, sequenceOrder, decisionStatus. Methods: `reviewApplication()`, `approve()`, `reject()`, `addRemark()`, `forwardToNextAuthority()`.

21. **Hostel AI.** Attributes: assistantID, userQuery, roomContext, generatedResponse, actionStatus, mapReference. Methods: `answerDormitoryQuestion()`, `retrieveRoomInfo()`, `provideNavigation()`, `controlSupportedDevice()`, `escalateToManager()`.

5. Analysis

User has subclasses ClassroomParticipant, Dorm User, and Library Member. Speaker, Course Coordinator, Librarian, Hostel Manager, and Approval Authority are role-specific users.

NIMC Classroom is merged into Course. Assessment is merged into ClassroomParticipant and Course Coordinator. Certificate is merged into ClassroomParticipant and Course. Dormitory Application is merged into Dorm User and Hostel Manager. Email Service and SMS Service are retained together as Notification Service.

Physical Book and Digital Resource are merged into Library Resource. Room, Seat, and Room Allocation fail selection and are not classes. A dormitory stay keeps `allocatedRoomID` and `allocatedSeatID` on Dorm User. Hostel Manager keeps `occupancySummary` and allocates that room and seat.

6. Class cards

CRC cards use a monochrome format. The Collaborator column contains only class names.

Class Name: User	
Attributes	**Methods**
-userID -Account -name -company -designation -phone -email -nidNumber	+viewProfile() +updateProfile() +activateIdentity()
Responsibility	**Collaboration**
Provide the shared identity and profile used to authenticate the person.	NIMCAccount
Provide the common user information required for the classroom role.	ClassroomParticipant
Provide the common user information required for library membership.	LibraryMember
Provide the common profile and required identity details for dormitory use.	DormUser

Class Name: NIMCAccount	
Attributes	**Methods**
-accountID -userID -registeredEmail -passwordHash -emailVerified -accountStatus	+registerAccount() +verifyEmail() +login() +resetPassword() +updateCredentials()
Responsibility	**Collaboration**
Link credentials to the person identity.	User
Use the registered email for verification and password-reset communication.	NotificationService
Provide the authenticated account used for authorized Classroom access.	ClassroomParticipant
Provide the authenticated account used for authorized Library access.	LibraryMember
Provide the authenticated account used for authorized Dormitory access.	DormUser

Class Name: NotificationService	
Attributes	**Methods**
-messageID -recipientEmail -recipientPhone -subject -messageType -deliveryStatus	+sendVerificationEmail() +sendPasswordReset() +sendNotification() +sendApprovalLink()
Responsibility	**Collaboration**
Send email-verification and password-reset messages for account access.	NIMCAccount
Send normal classroom notices such as schedule, speaker, assessment, and course updates.	CourseCoordinator
Send special or urgent classroom updates when an additional notification method is required.	CourseCoordinator
Send library reminders and alerts about due dates, requests, and resource availability.	Librarian
Send dormitory approval links, application and status notices, and urgent dormitory updates.	DormUser, HostelManager, ApprovalAuthority

Class Name: AIService	
Attributes	**Methods**
-providerName -modelName -serviceStatus	+processPrompt() +analyzeContent()
Responsibility	**Collaboration**
Analyze participant assignment and examination submissions and return similarity, plagiarism, or AI-content indicators.	CourseCoordinator
Generate classroom answers from retrieved course material.	ClassroomAI
Generate library recommendations and responses from catalogue data.	LibraryAssistantAI
Generate dormitory assistance from authorized dormitory information.	HostelAI

Class Name: PaymentSystem	
Attributes	**Methods**
-transactionID -userID -amount -paymentType -paymentStatus	+processPayment() +confirmPayment() +applyWaiver() +getPaymentStatus()
Responsibility	**Collaboration**
Process the applicable dormitory fee or approved waiver for the user and return payment status.	DormUser
Provide payment confirmation required before final room and seat allocation is completed.	HostelManager
Provide the participant with payment and waiver information for the accommodation request.	DormUser

Class Name: ClassroomParticipant	
Attributes	**Methods**
-participantID -attendance[] -course[] -certificateID -certificateIssueDate -certificateEligibilityStatus -certificateQRCode	+joinClassroom(code) +viewSessionPlan() +accessResource() +viewAttendance() +submitAssignment() +evaluateSpeaker() +evaluateCourse() +viewCertificate()
Responsibility	**Collaboration**
Use the shared NIMC identity as the participant profile.	User
Join the course space using its classroom code and access the proper course.	Course
View session plans and personal session-attendance information.	CourseSession
Search, filter, open, and use authorized course materials and recordings.	LearningResource
Submit assignment or examination work and view recorded results.	CourseCoordinator
Submit a speaker rating and comment after the session.	Speaker
Receive and view certificate details when the course eligibility requirements are met.	Course
Submit overall course feedback after course completion.	Course

Class Name: Speaker	
Attributes	**Methods**
-speakerID -contactInformation -expertise -overallRating -ratingHistory[] -coursesTaken[]	+viewAssignedSessions() +provideMaterials() +viewOwnSchedule() +seeOverallRating()
Responsibility	**Collaboration**
Receive course-session assignments and coordination information.	CourseCoordinator
Conduct the assigned session and remain linked to its topic, date, and time.	CourseSession
Provide documents, presentations, videos, notes, or other session materials.	LearningResource
Retain participant ratings and calculate the speaker overall rating.	ClassroomParticipant

Class Name: CourseCoordinator	
Attributes	**Methods**
-name -email	+createClassroom() +addParticipant() +removeParticipant() +prepareSessionPlan() +updateSession() +uploadResource() +recordAttendance() +recordAssessmentResult() +issueCertificate() +analyzeSubmission() +checkPlagerism() +checkAI() +checkwithAI()
Responsibility	**Collaboration**
Create and manage the course and its configurable rules.	Course
Create and control the course classroom space, manage participant access, and maintain the classroom code.	Course
Contact and assign speakers to appropriate sessions.	Speaker
Prepare and update the session plan and record participant attendance.	CourseSession
Upload and organize learning materials and class recordings.	LearningResource
Receive participant assignment and examination submissions and record the resulting assessment information.	ClassroomParticipant
Analyze submissions and check plagiarism and AI indicators using the shared AI capability.	AIService
Review course feedback and coordinate certificate eligibility and issuing through the course rules.	Course
View restricted speaker rating history.	Speaker

Class Name: Course	
Attributes	**Methods**
-title -courseType -startDate -endDate -description -status -feedbackRecord[] -session[] -classroomCode -participantIDs[] -resourceIDs[] -certificateRules	+updateCourse() +configureCertificateRules() +getSessions() +getParticipants() +recordCourseFeedback() +displayResources() +checkCertificateEligibility() +generateCertificate() +verifyCertificateQRCode()
Responsibility	**Collaboration**
Provide the main course information, classroom code, access context, and configurable rules managed by the coordinator.	CourseCoordinator
Maintain the participant list for the course space and let eligible participants join using the classroom code.	ClassroomParticipant
Group and organize all planned sessions under the same course.	CourseSession
Expose the course learning resources and recordings through the course space.	LearningResource
Apply the configured certificate rules to check eligibility and generate or verify participant certificate information.	ClassroomParticipant
Retain overall course feedback submitted by participants.	ClassroomParticipant

Class Name: CourseSession	
Attributes	**Methods**
-topic -sessionDate -sessionTime -speaker -mode -recordingLink -attendanceList -sessionStatus	+updateSession() +assignSpeaker() +recordParticipantAttendance() +notifyChange() +sortResources() +filterResources()
Responsibility	**Collaboration**
Belong to the relevant course and contribute to its overall session plan.	Course
Link the assigned speaker to the session topic, date, and time.	Speaker
Provide participants with schedule information and store their session attendance.	ClassroomParticipant
Allow the coordinator to schedule, update, and record attendance for the session.	CourseCoordinator
Trigger normal notification when the session schedule or speaker changes.	NotificationService
Trigger special or urgent notification for important session changes.	NotificationService

Class Name: LearningResource	
Attributes	**Methods**
-title -resourceType -fileLink -uploadDate -accessLevel	+upload() +update() +searchInResource()
Responsibility	**Collaboration**
Allow the coordinator to upload and maintain course material.	CourseCoordinator
Appear in the appropriate course space for centralized resource access.	Course
Link materials or recordings to the session in which they were used or delivered.	CourseSession
Allow authorized participants to search, sort, filter, and access resources.	ClassroomParticipant
Provide the approved knowledge source used to answer classroom questions.	ClassroomAI

Class Name: ClassroomAI	
Attributes	**Methods**
-assistantID -AIService -course -user	+askQuestion() +retrieveCourseResources() +answerFromResources()
Responsibility	**Collaboration**
Receive participant questions and return course-related answers.	ClassroomParticipant
Retrieve relevant documents, notes, presentations, videos, and recordings as evidence.	LearningResource
Use the common AI capability to generate a response grounded in retrieved course content.	AIService

Class Name: LibraryMember	
Attributes	**Methods**
-category -membershipID -validFrom -validUntil -borrowingLimit -borrowedResourceIDs -qrCode -membershipStatus	+activateMembership() +viewBorrowedResources() +viewDueDates() +searchCatalogue() +openDigitalResource() +requestBook() +submitReview() +askLibraryAI()
Responsibility	**Collaboration**
Use the shared NIMC identity as the basis for library membership.	User
Receive membership assignment, verification, and borrowing service from library staff.	Librarian
Search, borrow, open, request, and review library resources as permitted.	LibraryResource
Read or download digital resources according to their access permissions.	E-Library
Ask for recommendations, availability, related resources, and review summaries.	LibraryAssistantAI

Class Name: Librarian	
Attributes	**Methods**
-librarianID -staffRole -permissions	+verifyMember() +assignMembership() +manageCatalogue() +issueResource() +returnResource() +extendDueDate() +applyPenalty() +overridePenalty() +moderateReview() +reviewRequests() +generateMonthlyReport()
Responsibility	**Collaboration**
Verify users, assign membership, and manage issuing and returning resources for members.	LibraryMember
Maintain catalogue details, circulation status, penalties, reviews, and request demand for both physical and digital resources.	LibraryResource
Upload and manage digital resources, access permissions, and e-library usage information.	E-Library
Send due-date reminders and other important library notifications.	NotificationService

Class Name: LibraryResource	
Attributes	**Methods**
-title -author -isbn -category -publisher -publicationInfo -resourceFormat -availabilityStatus -accessPermission -downloadAllowed -currentBorrowerID -issueDate -dueDate -returnDate -circulationHistory -reviews -wailtList[]	+updateResource() +search() +markBorrowed() +markReturned() +setAccessPermission() +recordUsage() +addReview() +registerRequest()
Responsibility	**Collaboration**
Represent the merged physical copy and digital resource. resourceFormat records physical, digital, or both.	LibraryMember
Provide searchable resource information and permitted borrowing or access to members.	LibraryMember
Allow library staff to maintain metadata, availability, circulation, reviews, and request demand.	Librarian
Provide the digital form of a resource to the e-library when resourceFormat is digital.	E-Library
Provide catalogue, availability, and review data for AI recommendations.	LibraryAssistantAI

Class Name: E-Library	
Attributes	**Methods**
-eResources[] -usageStatistics	+browse() +searchDigitalResources() +openResource() +downloadResource() +recordUsage()
Responsibility	**Collaboration**
Host and expose the digital resources held by Library Resource. E-Library is the collection, not a separate ebook class.	LibraryResource
Allow members and public users to read or download resources according to access permission.	LibraryMember
Allow the librarian to manage digital access and review usage statistics.	Librarian

Class Name: LibraryAssistantAI	
Attributes	**Methods**
-assistantID -user -AIService	+recommendResources() +answerPrompt() +summarizeReviews() +suggestAlternative()
Responsibility	**Collaboration**
Receive the member's interests, course context, and recommendation questions.	LibraryMember
Search catalogue data, availability, reviews, and related resources when producing suggestions.	LibraryResource
Refer the user to library staff when a question requires human judgment or unavailable information.	Librarian
Use the common AI capability to generate recommendation and follow-up responses.	AIService

Class Name: DormUser	
Attributes	**Methods**
-participantCategory -applicationID -nidNumber -preferredAccommodation -expectedStart -expectedEnd -seatType -reason -applicationStatus -currentApprovalStage -feeAmount -waiverStatus -paymentStatus -allocatedRoomID -allocatedSeatID -stayStatus -reportedIssues	+submitDormitoryApplication() +viewApplicationStatus() +viewAllocation() +requestStayExtension() +requestCancellation() +requestWaiver() +reportIssue() +viewIssueStatus() +viewStayDetails() +askHostelAI()
Responsibility	**Collaboration**
Use the shared NIMC profile and provide required identity information such as NID when needed.	User
Store and submit the accommodation request and track its approval, fee, waiver, and payment status.	HostelManager
View the allocated room identifier, seat identifier, approved stay period, check-in, and check-out information.	HostelManager
Request a stay extension or cancellation and report dormitory issues for staff action.	HostelManager
Ask dormitory questions and request supported navigation or device assistance.	HostelAI

Class Name: HostelManager	
Attributes	**Methods**
-name -email -permissions -occupancySummary -applicationRecords[] -approvalHistory[]	+reviewApplication() +updateApplicationStatus() +recordApproval() +calculateFee() +markPendingForAvailability() +confirmPaymentStatus() +checkAvailability() +allocateRoom() +recordCheckIn() +recordCheckOut() +manageOccupancy() +approveStayChange() +manageIssue() +generateMonthlyReport()
Responsibility	**Collaboration**
Receive and review dormitory requests, maintain application status, and manage pending availability.	DormUser
Record check-in and check-out, process stay changes, and manage reported dormitory issues.	DormUser
Check capacity from the occupancy summary while processing the accommodation request.	DormUser
Select an available room and seat by identifier and release them when the stay ends. Room, Seat, and Room Allocation are not classes.	DormUser
Calculate the applicable fee, retain approval history, and confirm payment or waiver status before allocation.	PaymentSystem
Receive the formal approval outcome before completing final allocation.	ApprovalAuthority

Class Name: ApprovalAuthority	
Attributes	**Methods**
-name -email -authorityRole -sequenceOrder -decisionStatus	+reviewApplication() +approve() +reject() +addRemark() +forwardToNextAuthority()
Responsibility	**Collaboration**
Review the dormitory request managed by the hostel office, add remarks, and record approve or reject decisions.	HostelManager
Forward the completed approval flow to the manager for availability and allocation.	HostelManager
Use emailed direct approval links and approval notifications when required.	NotificationService

Class Name: HostelAI	
Attributes	**Methods**
-assistantID -userQuery -roomContext -generatedResponse -actionStatus -mapReference	+answerDormitoryQuestion() +retrieveRoomInfo() +provideNavigation() +controlSupportedDevice() +escalateToManager()
Responsibility	**Collaboration**
Receive dormitory questions and supported navigation or device requests from the participant.	DormUser
Use authorized room-specific information, such as Wi-Fi details or map location guidance, from the allocated stay.	HostelManager
Escalate unsupported questions, judgments, or service problems to dormitory management.	HostelManager
Use the common AI capability to interpret requests and generate dormitory responses.	AIService

7. CRC diagrams

Four CRC collaboration diagrams are used: Authentication, Classroom, Library, and Dormitory. Each arrow states the collaboration between two retained classes. These are the four figures at the end of the class-based modeling document. They do not show Room, Seat, Room Allocation, Physical Book, or Digital Resource as classes.

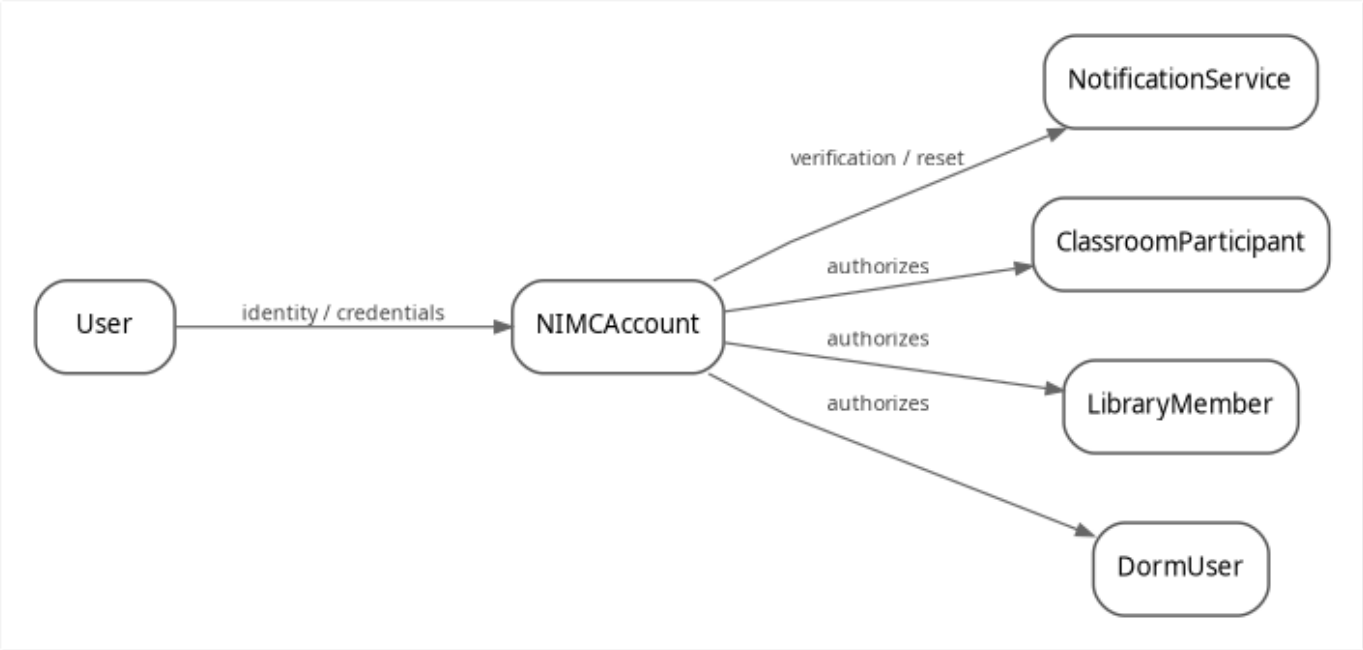


Figure 1: Authentication system collaboration diagram.

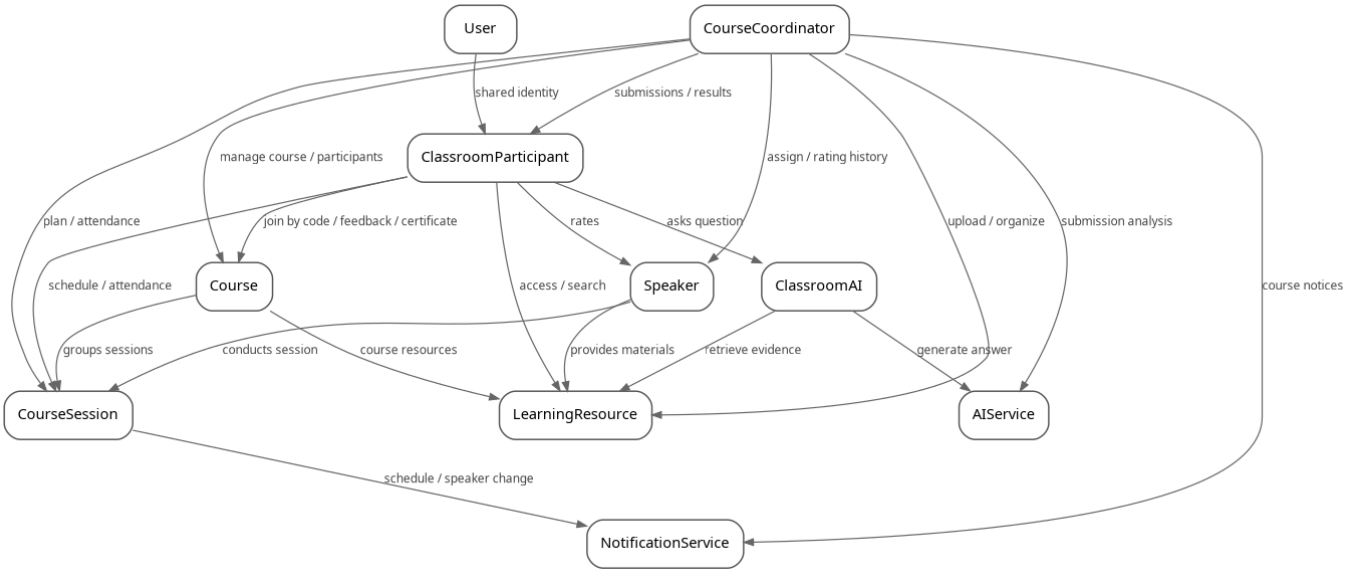


Figure 2: Classroom module collaboration diagram.

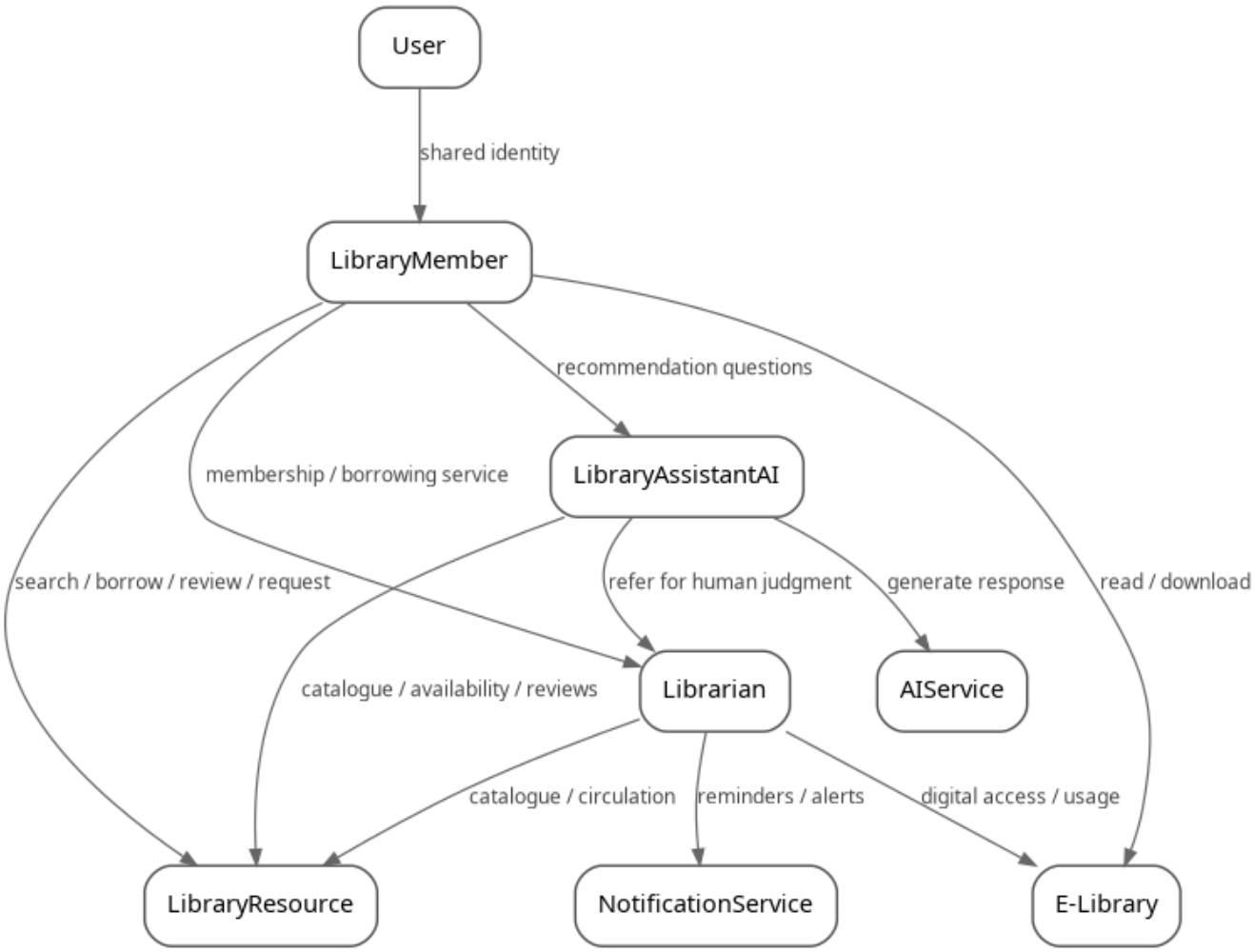


Figure 3: Library module collaboration diagram.

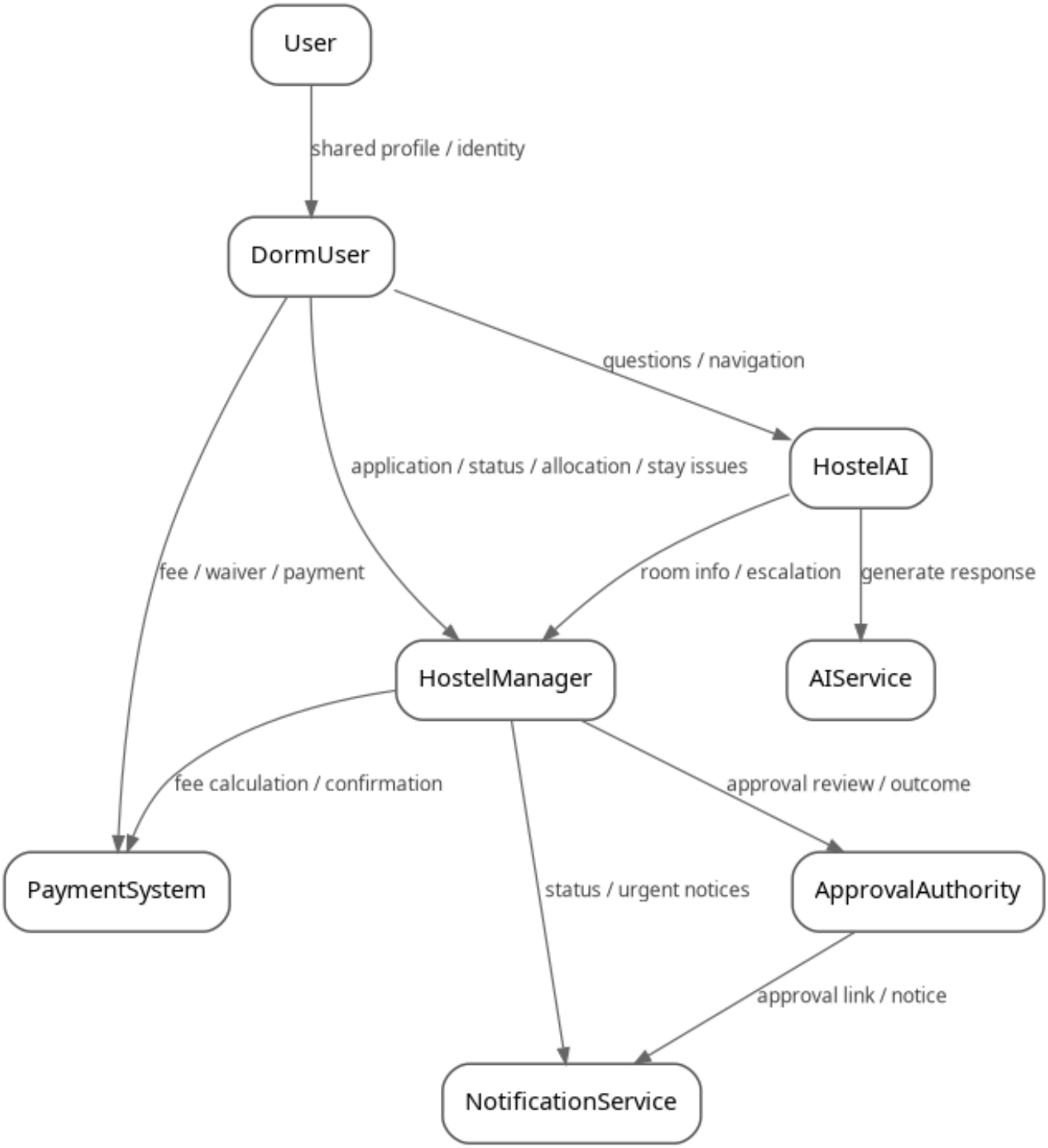


Figure 4: Dormitory module collaboration diagram.

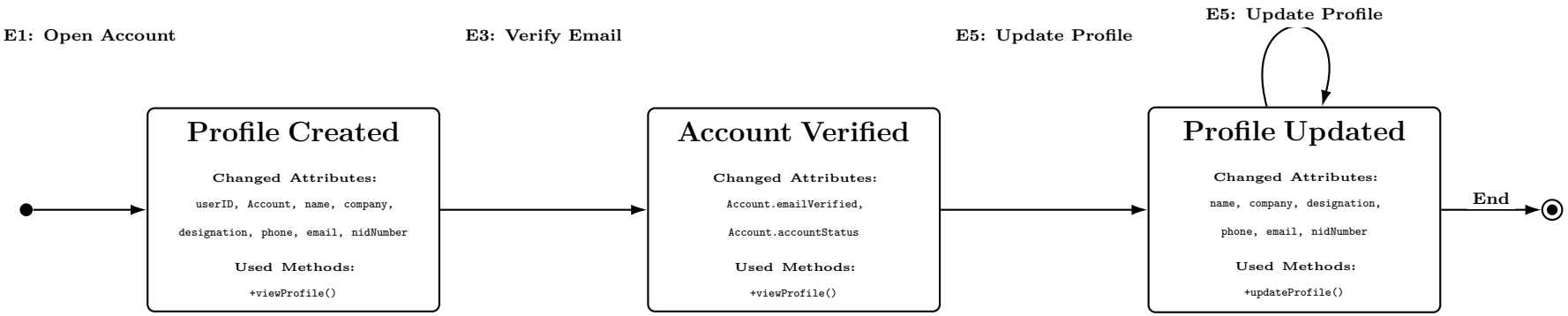
Event List

ID	Event (Story Event)	Event Alias	Initiator Class	Collaborator Class
E1	Participants and NIMC employees open a NIMC account using their identity information and credentials.	Open Account	User	NIMCAccount
E2	NIMC sends a verification email to the registered email address.	Send Notification	NIMCAccount	NotificationService
E3	The account holder responds to the verification email and verifies the email address.	Verify Email	User	NIMCAccount
E4	The account holder signs in using the registered email address and password.	Sign In	User	NIMCAccount
E5	The account holder updates account/profile information when details change.	Update Profile	User	
E6	The account holder requests a password reset through the registered email address.	Request Password Reset	User	NIMCAccount
E7	NIMC sends the password-reset email to the registered email address.	Send Reset Email	NIMCAccount	NotificationService
E8	The account holder completes the password reset.	Reset Password	User	NIMCAccount
E9	The Course Coordinator enters a course name and creates its classroom.	Create Classroom	CourseCoordinator	Course
E10	The Course Coordinator shares the unique classroom code.	Share Classroom Code	CourseCoordinator	ClassroomParticipant
E11	A participant enters the classroom using the classroom code.	Join Classroom	ClassroomParticipant	Course
E12	The Course Coordinator removes an ineligible participant.	Remove Participant	CourseCoordinator	ClassroomParticipant
E13	The Course Coordinator contacts and assigns a speaker.	Assign Speaker	CourseCoordinator	Speaker
E14	The Course Coordinator prepares the initial session plan.	Prepare Session Plan	CourseCoordinator	CourseSession
E15	The session plan is changed for a schedule, topic, or speaker change.	Update Session	CourseCoordinator	CourseSession
E16	The coordination team uploads course learning resources.	Upload Learning Resource	CourseCoordinator	LearningResource
E17	The system stores an online class recording.	Store Session Recording	CourseCoordinator	LearningResource
E18	A participant asks the classroom assistant a question.	Ask Classroom AI	ClassroomParticipant	ClassroomAI
E19	The classroom assistant retrieves relevant course resources.	Retrieve Course Resources	ClassroomAI	LearningResource
E20	The classroom assistant sends the retrieved request to the shared AI service.	Get AI Request	ClassroomAI	AIService
E21	The coordinator or authorized team member records session attendance.	Record Attendance	CourseCoordinator	CourseSession
E22	The system sends an important classroom notification by email or special-case SMS.	Send Classroom Notice	CourseCoordinator	NotificationService
E23	A scheduled course session is completed.	Complete Session	CourseCoordinator	CourseSession
E24	A participant submits a rating and optional comment for a speaker.	Submit Speaker Evaluation	ClassroomParticipant	Speaker
E25	The system updates the speaker's overall rating and rating history.	Calculate Speaker Rating	ClassroomParticipant	Speaker
E26	A participant submits an assignment or examination.	Submit Assessment	ClassroomParticipant	CourseCoordinator
E27	The coordination team evaluates submitted work and records the result.	Record Assessment Result	CourseCoordinator	ClassroomParticipant
E28	The coordination team requests AI analysis of a submission.	Analyze Submission	CourseCoordinator	AIService
E29	The AI service returns similarity, plagiarism, or AI-content indicators.	Return AI Flags	AIService	CourseCoordinator
E30	A participant submits overall course feedback.	Submit Course Feedback	ClassroomParticipant	Course
E31	The course and its required assessments are completed.	Complete Course	CourseCoordinator	Course
E32	The Course Coordinator configures certificate eligibility rules.	Configure Certificate Rules	CourseCoordinator	Course
E33	An eligible participant receives a certificate with a QR code.	Issue Certificate	CourseCoordinator	ClassroomParticipant
E34	A certificate QR code is scanned to view certificate details.	Verify Certificate QR	User	Course
E35	Library staff verifies the account and assigns library membership.	Assign Membership	Librarian	LibraryMember
E36	Library staff identifies a member by account, membership ID, or QR code.	Identify Member	Librarian	LibraryMember

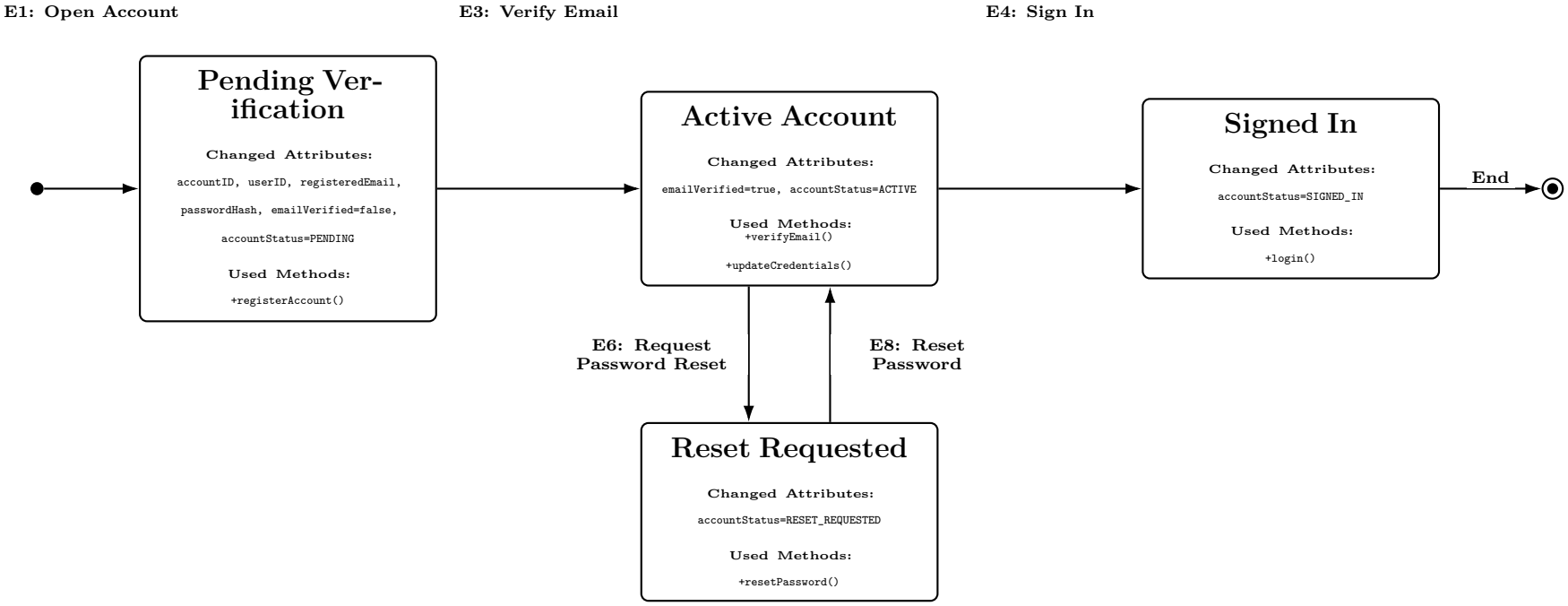
ID	Event (Story Event)	Event Alias	Initiator Class	Collaborator Class
E37	Library staff verifies active membership and borrowing-limit eligibility.	Check Borrowing Eligibility	Librarian	LibraryMember
E38	The assistant librarian extends a borrowing period and due date.	Extend Due Date	Librarian	LibraryResource
E39	A library user searches the catalogue.	Search Catalogue	LibraryMember	LibraryResource
E40	Library staff issues a selected physical resource to an eligible member.	Issue Resource	Librarian	LibraryResource
E41	The system sends a reminder before a borrowed resource's due date.	Send Due Reminder	Librarian	NotificationService
E42	Library staff records the return of a resource.	Return Resource	Librarian	LibraryResource
E43	A resource kept beyond its due date is marked overdue.	Mark Overdue		LibraryResource
E44	Library staff applies the configured overdue penalty.	Apply Penalty	Librarian	LibraryResource
E45	The librarian overrides a penalty in a special case.	Override Penalty	Librarian	LibraryResource
E46	The system notifies a waiting user that a resource is available.	Notify Resource Available	Librarian	NotificationService
E47	A user opens a permitted digital resource.	Open Digital Resource	LibraryMember	E-Library
E48	A user downloads a digital resource when downloading is permitted.	Download Digital Resource	LibraryMember	E-Library
E49	The e-library records digital-resource usage statistics.	Record Digital Usage	E-Library	
E50	A user asks LibraryAssistantAI for recommendations or follow-up information.	Ask Library AI	LibraryMember	LibraryAssistantAI
E51	LibraryAssistantAI generates recommendations from catalogue and review data.	Generate Recommendation	LibraryAssistantAI	AIService
E52	A library member submits a written review and rating.	Submit Book Review	LibraryMember	LibraryResource
E53	The librarian moderates or removes an inappropriate review.	Moderate Review	Librarian	LibraryResource
E54	A registered user requests a book not currently available.	Request New Book	LibraryMember	LibraryResource
E55	The librarian approves a requested book for addition.	Approve Book Request	Librarian	LibraryResource
E56	The system notifies the user that a requested book was approved.	Notify Request Approval	Librarian	NotificationService
E57	Library staff adds or updates a catalogue resource and its metadata.	Maintain Catalogue Resource	Librarian	LibraryResource
E58	Library staff marks a physical copy as withdrawn.	Withdraw Resource	Librarian	LibraryResource
E59	Library staff uploads a permitted digital resource to the e-library.	Upload Digital Resource	Librarian	E-Library
E60	The librarian sets whether a digital resource can be downloaded.	Set Download Permission	Librarian	LibraryResource
E61	The librarian generates the monthly library report.	Generate Library Report	Librarian	
E62	A participant submits a dormitory application.	Submit Dorm Application	DormUser	HostelManager
E63	The Hostel Manager starts the formal approval process.	Start Approval	HostelManager	ApprovalAuthority
E64	The system sends a direct approval link to a responsible authority.	Send Approval Link	HostelManager	NotificationService
E65	An approval authority approves the application.	Approve Application	ApprovalAuthority	HostelManager
E66	An approval authority rejects the application.	Reject Application	ApprovalAuthority	HostelManager
E67	A completed review is forwarded to the next approval authority.	Forward Approval	ApprovalAuthority	ApprovalAuthority
E68	The Hostel Manager checks current room and seat availability.	Check Room Availability	HostelManager	DormUser
E69	An approved application is kept pending because no suitable seat is available.	Mark Pending Availability	HostelManager	DormUser
E70	The applicable dormitory fee is calculated from participant and seat type.	Calculate Dorm Fee	HostelManager	PaymentSystem
E71	An eligible government employee requests a fee waiver.	Request Waiver	DormUser	PaymentSystem
E72	The responsible authority approves the fee waiver.	Approve Waiver	ApprovalAuthority	PaymentSystem
E73	NIMC Accounts processes the applicable dormitory fee.	Process Payment	DormUser	PaymentSystem
E74	Payment or an approved waiver is confirmed.	Confirm Payment	PaymentSystem	HostelManager
E75	The Hostel Manager allocates an available room and seat.	Allocate Room	HostelManager	DormUser
E76	Dormitory staff records the participant's check-in.	Record Check-In	HostelManager	DormUser
E77	A participant submits a stay-extension request.	Request Stay Extension	DormUser	HostelManager
E78	A participant submits a cancellation request.	Request Cancellation	DormUser	HostelManager
E79	The responsible dormitory authority approves a stay change.	Approve Stay Change	HostelManager	DormUser

ID	Event (Story Event)	Event Alias	Initiator Class	Collaborator Class
E80	A participant reports a dormitory issue.	Report Dorm Issue	DormUser	HostelManager
E81	Responsible staff updates the reported issue until it is resolved.	Update Issue Status	HostelManager	DormUser
E82	Dormitory staff records check-out and releases the occupied seat.	Record Check-Out	HostelManager	DormUser
E83	The Hostel Manager generates the monthly dormitory report.	Generate Dorm Report	HostelManager	
E84	The system sends an important dormitory status notification.	Send Dormitory Notice	HostelManager	NotificationService
E85	A participant submits dormitory service or room feedback.	Submit Dorm Feedback	DormUser	HostelManager
E86	A participant asks HostelAI a dormitory or navigation question.	Ask Hostel AI	DormUser	HostelAI
E87	HostelAI generates a response using authorized dormitory information.	Generate Dormitory Response	HostelAI	AIService
E88	HostelAI sends a supported command to an integrated IoT device.	Control Supported Device	HostelAI	
E89	HostelAI escalates an unsupported request to dormitory management.	Escalate to Manager	HostelAI	HostelManager
E90	The shared AI service completes processing and produces a result.	Produce AI Result	AIService	
E91	NotificationService dispatches a prepared notification by email or SMS.	Dispatch Notification	NotificationService	
E92	Successful delivery is recorded in the notification delivery status.	Confirm Delivery		NotificationService
E93	Unsuccessful delivery is recorded in the notification delivery status.	Record Delivery Failure		NotificationService
E94	A failed notification is submitted for another delivery attempt.	Retry Delivery	NotificationService	
E95	An uploaded classroom learning resource is updated.	Update Learning Resource	CourseCoordinator	LearningResource
E96	The assistant librarian adjusts a member's membership validity.	Extend Membership Validity	Librarian	LibraryMember

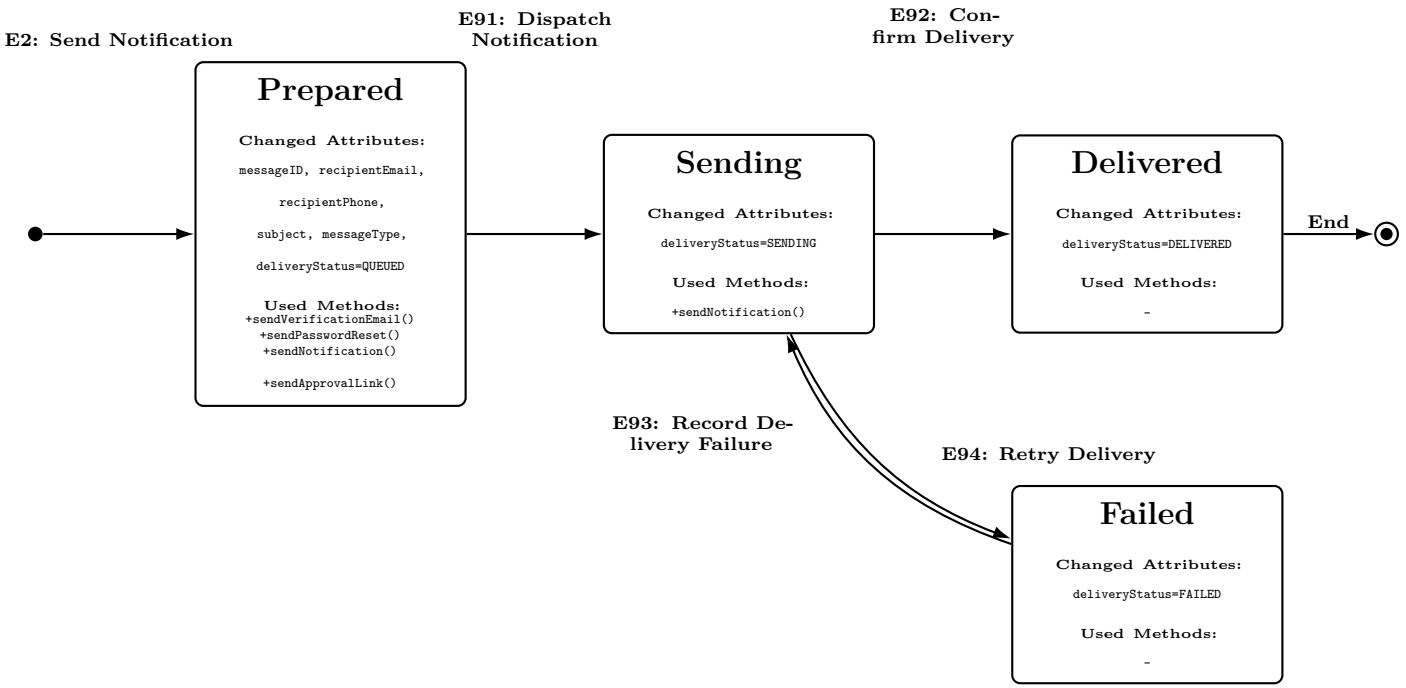
State-transition diagram: User



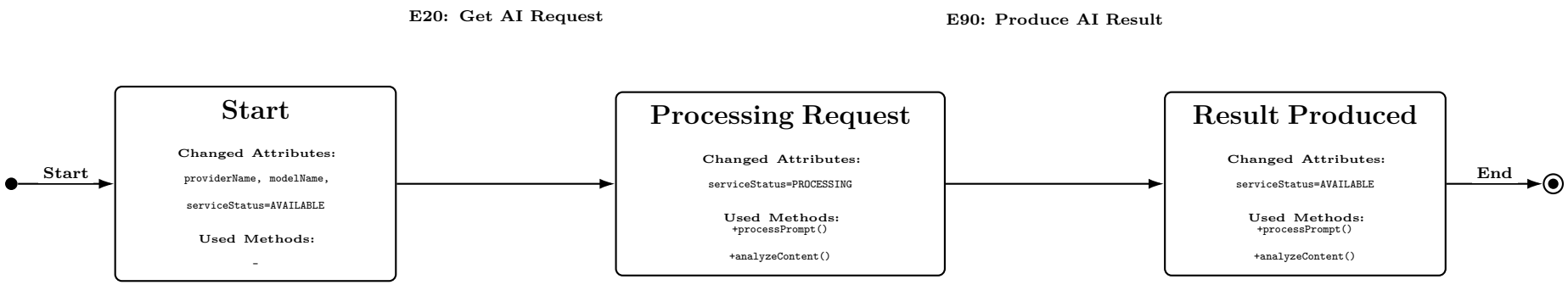
State-transition diagram: NIMCAccount



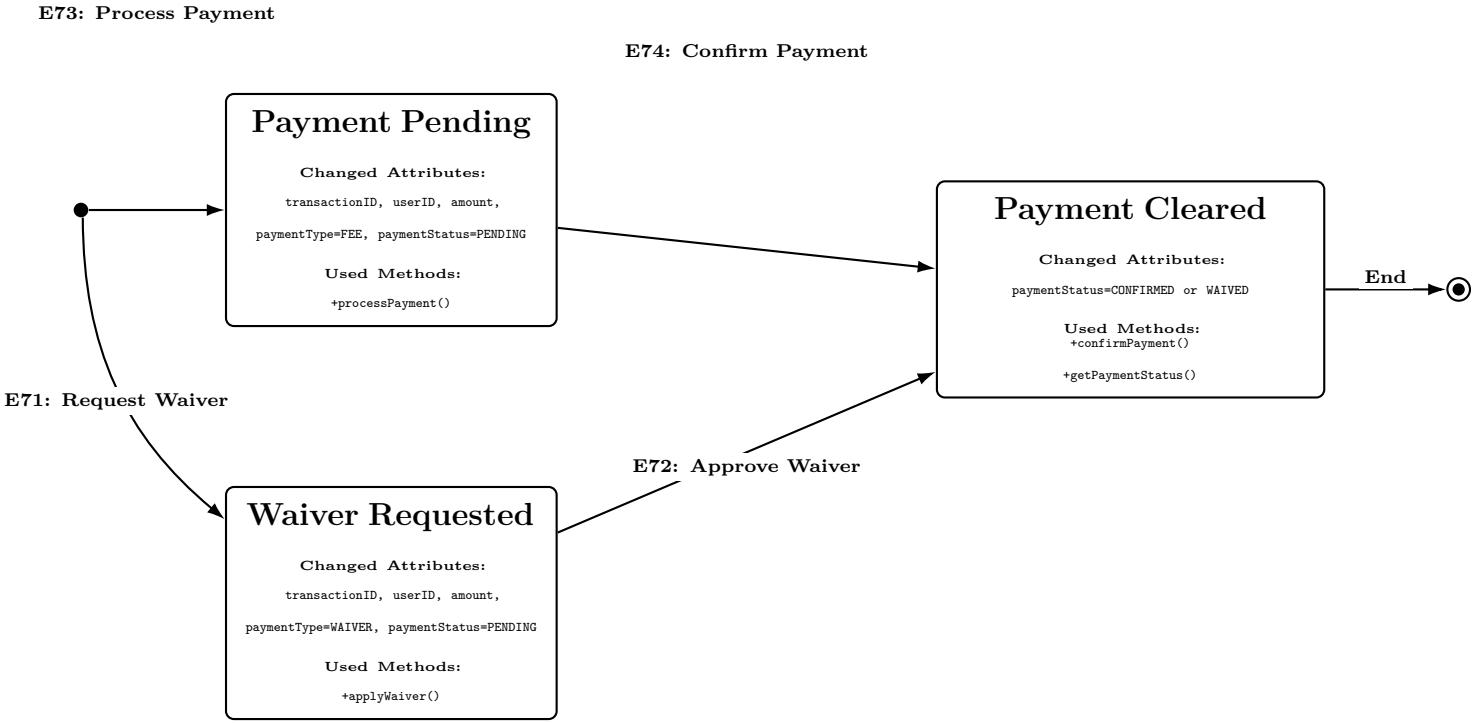
State-transition diagram: NotificationService



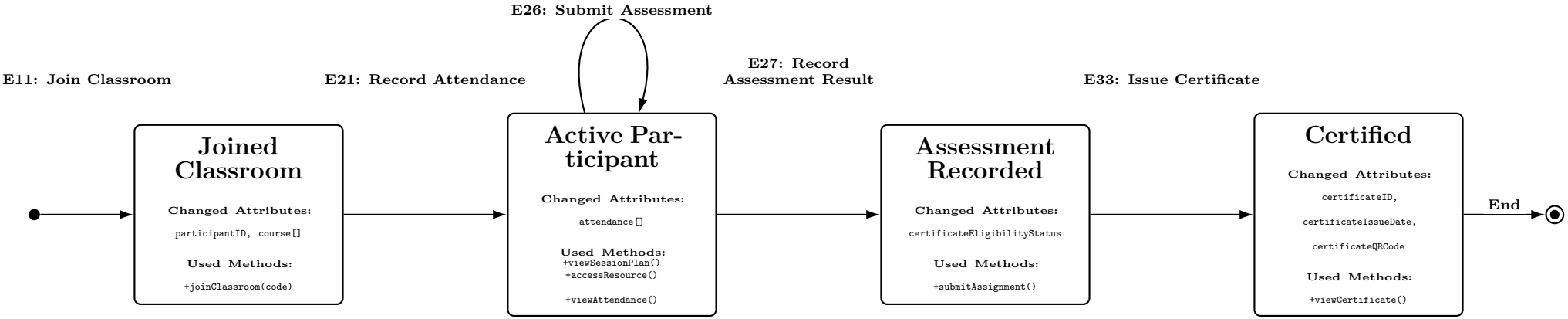
State-transition diagram: AIService



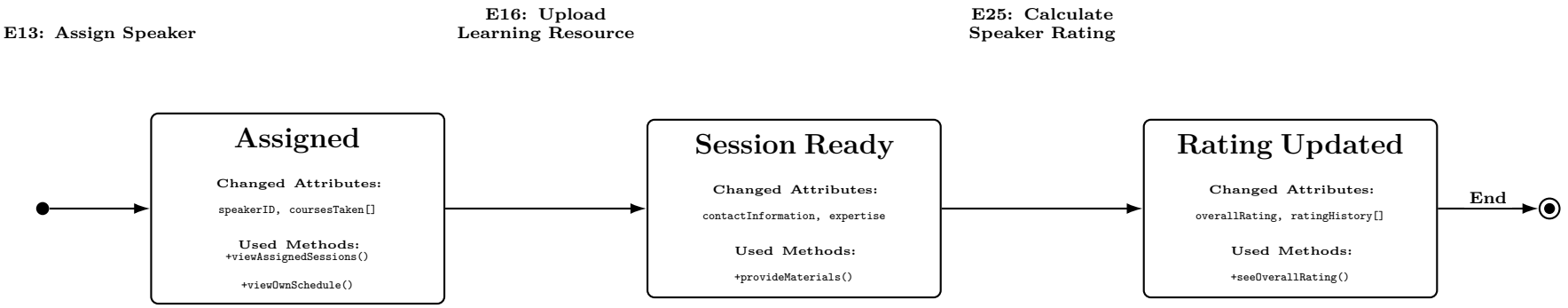
State-transition diagram: PaymentSystem



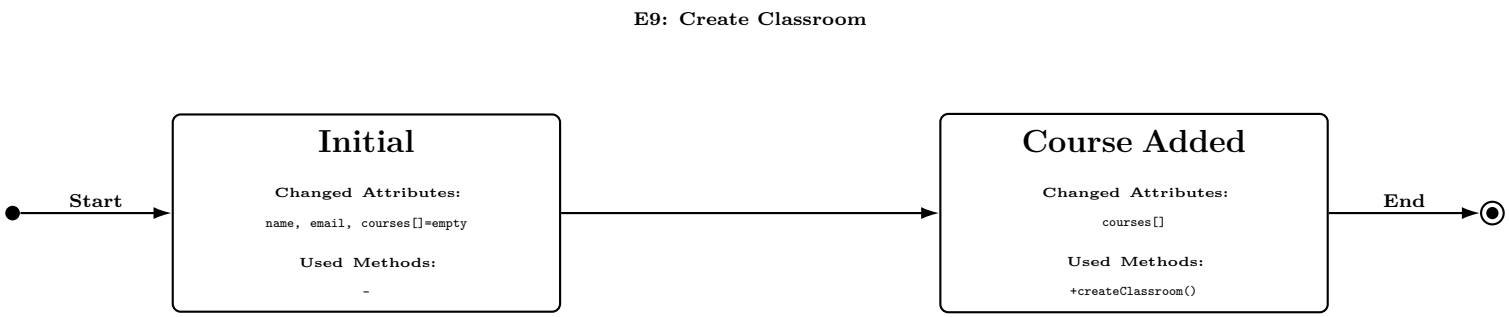
State-transition diagram: ClassroomParticipant



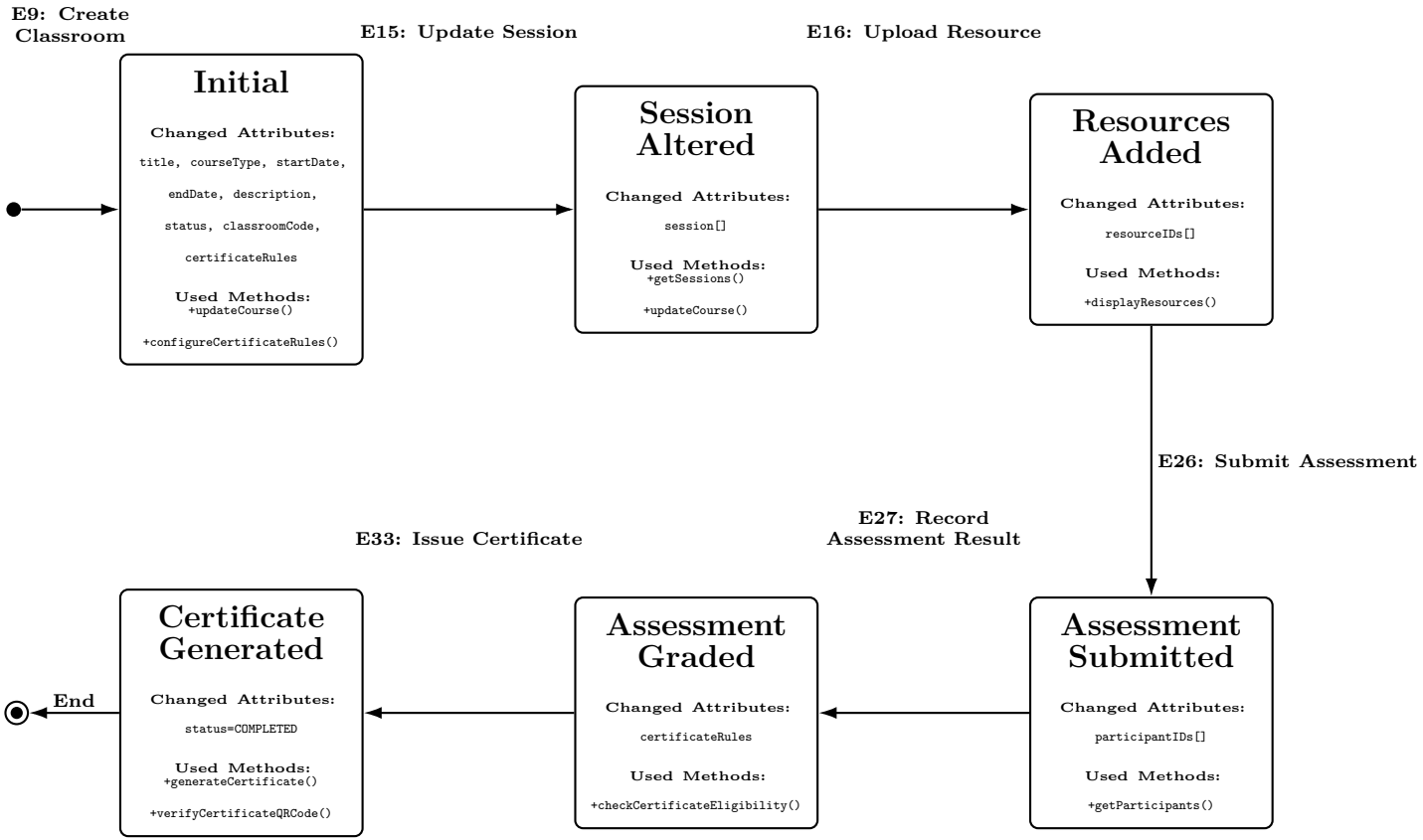
State-transition diagram: Speaker



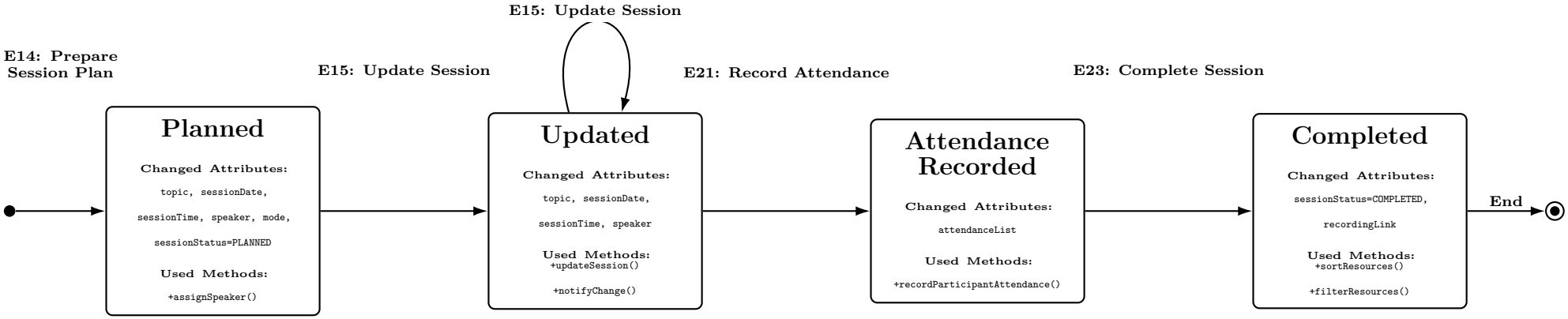
State-transition diagram: CourseCoordinator



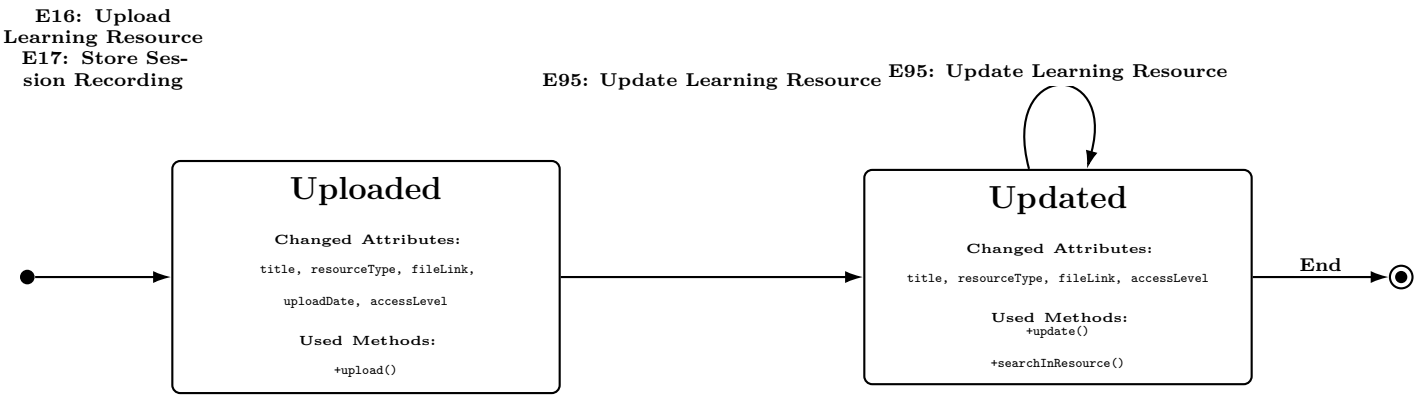
State-transition diagram: Course



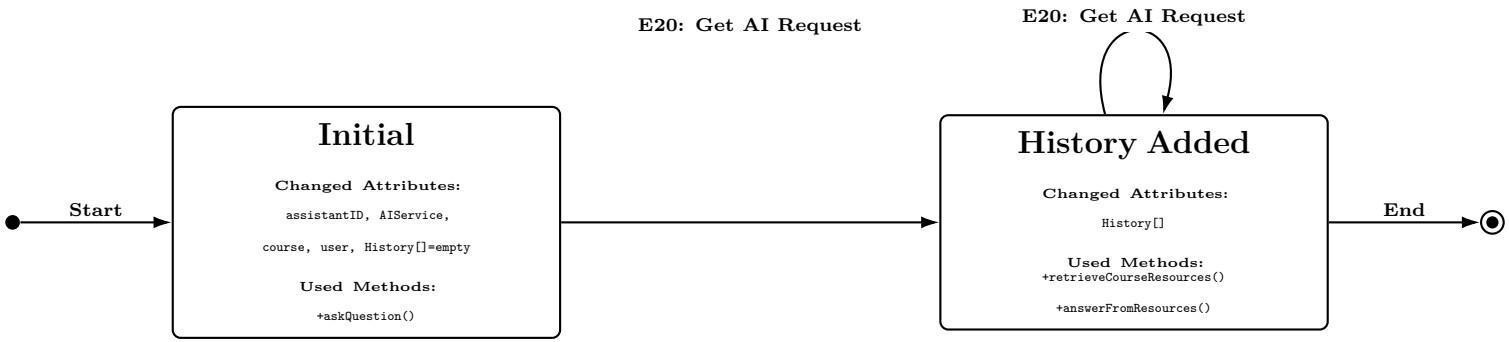
State-transition diagram: CourseSession



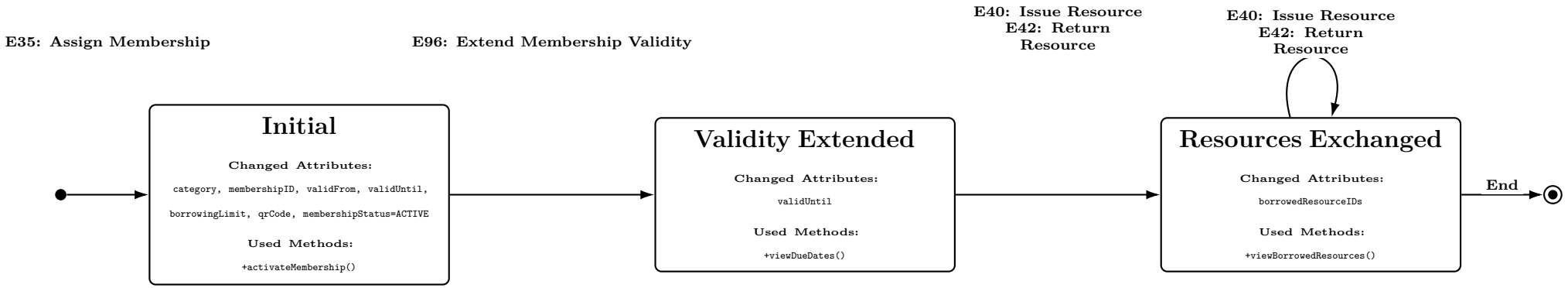
State-transition diagram: LearningResource



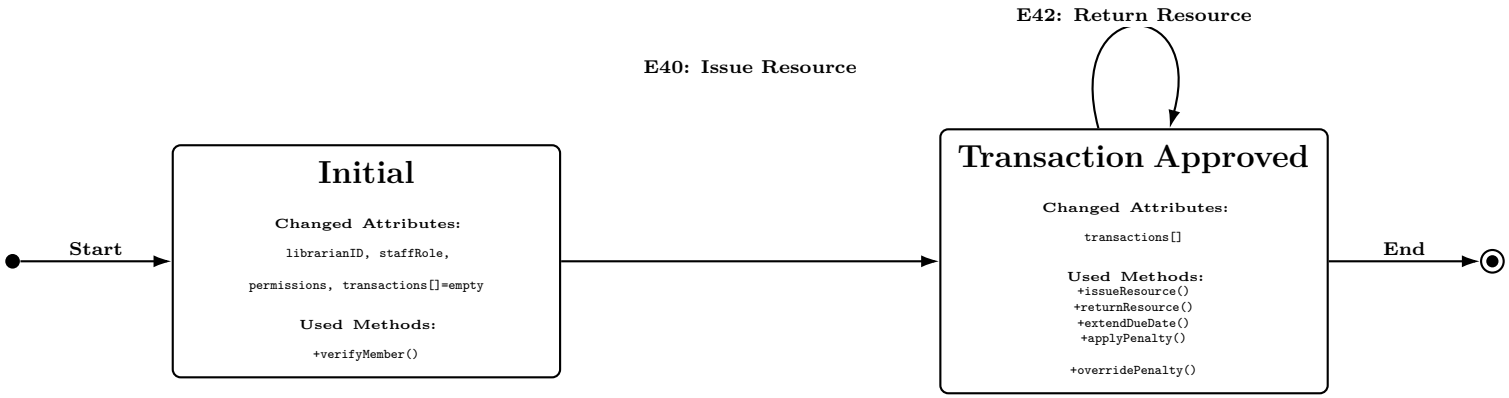
State-transition diagram: ClassroomAI



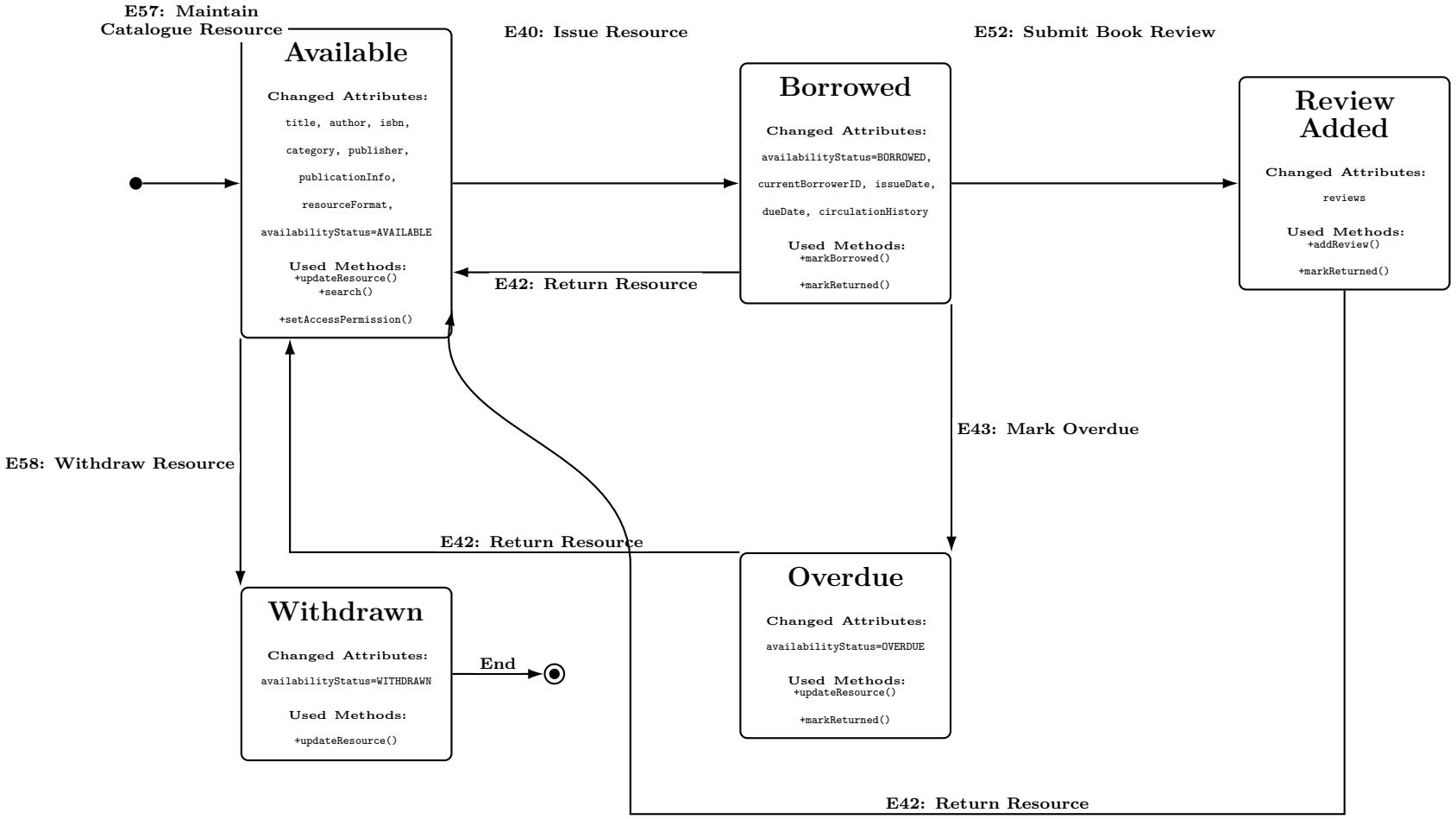
State-transition diagram: LibraryMember



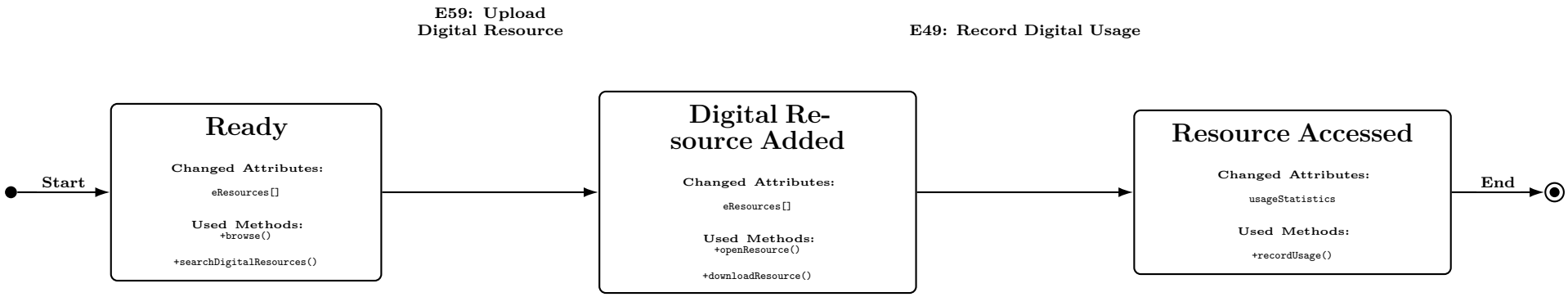
State-transition diagram: Librarian



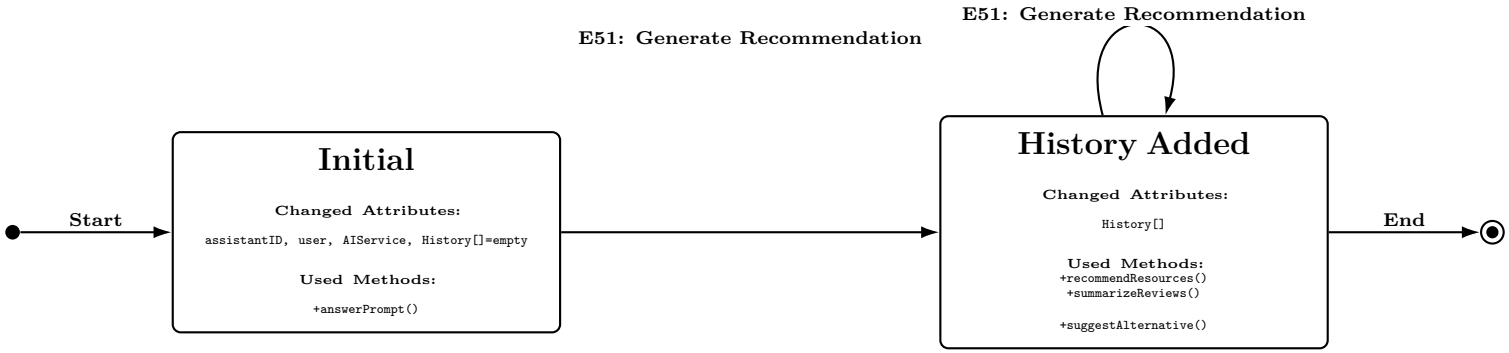
State-transition diagram: LibraryResource



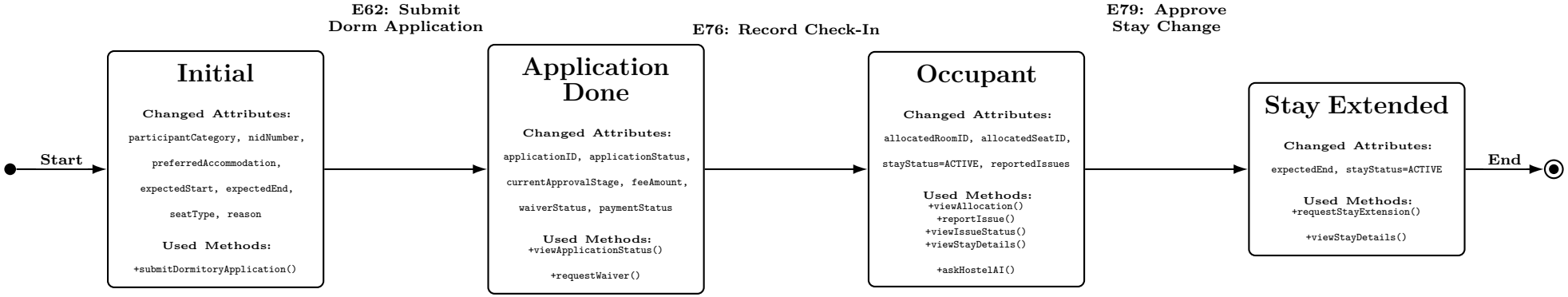
State-transition diagram: E-Library



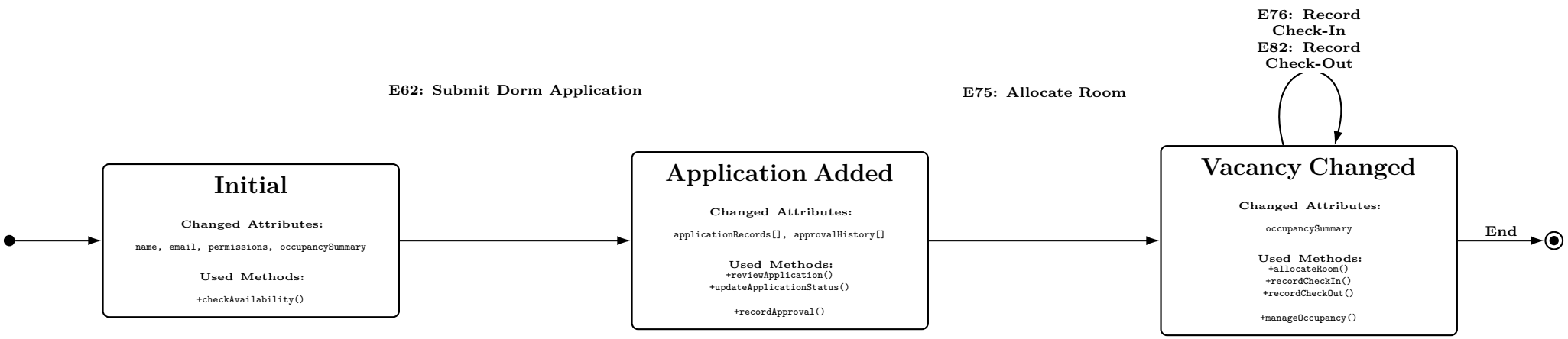
State-transition diagram: LibraryAssistantAI



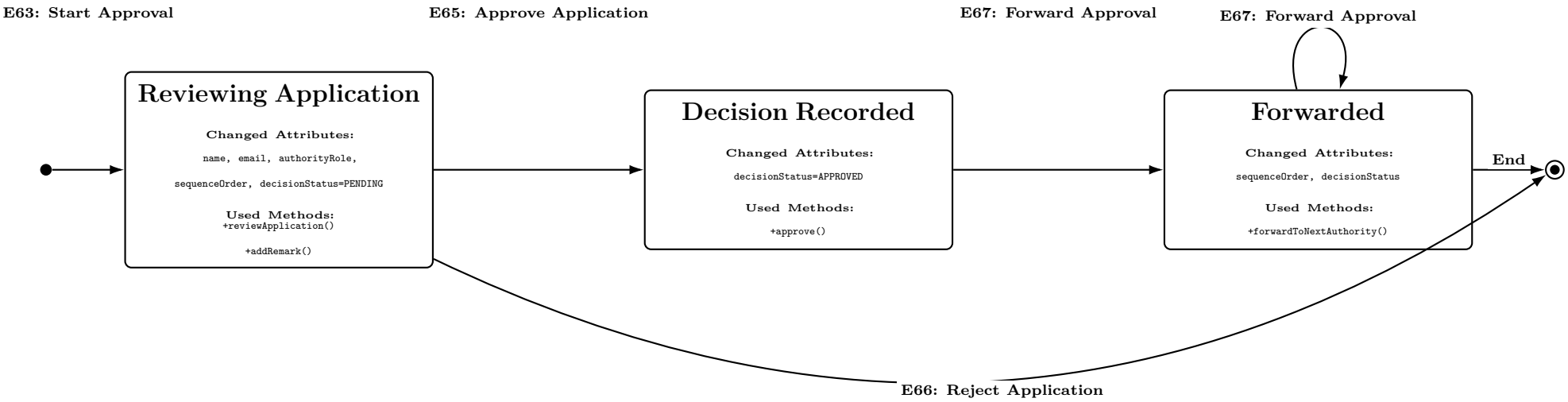
State-transition diagram: DormUser



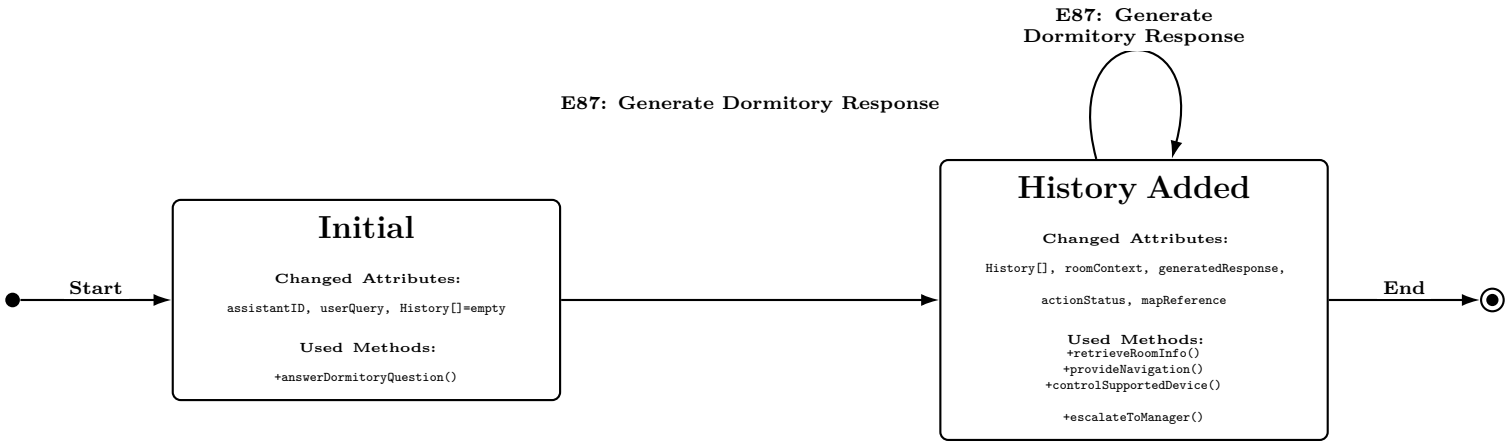
State-transition diagram: HostelManager



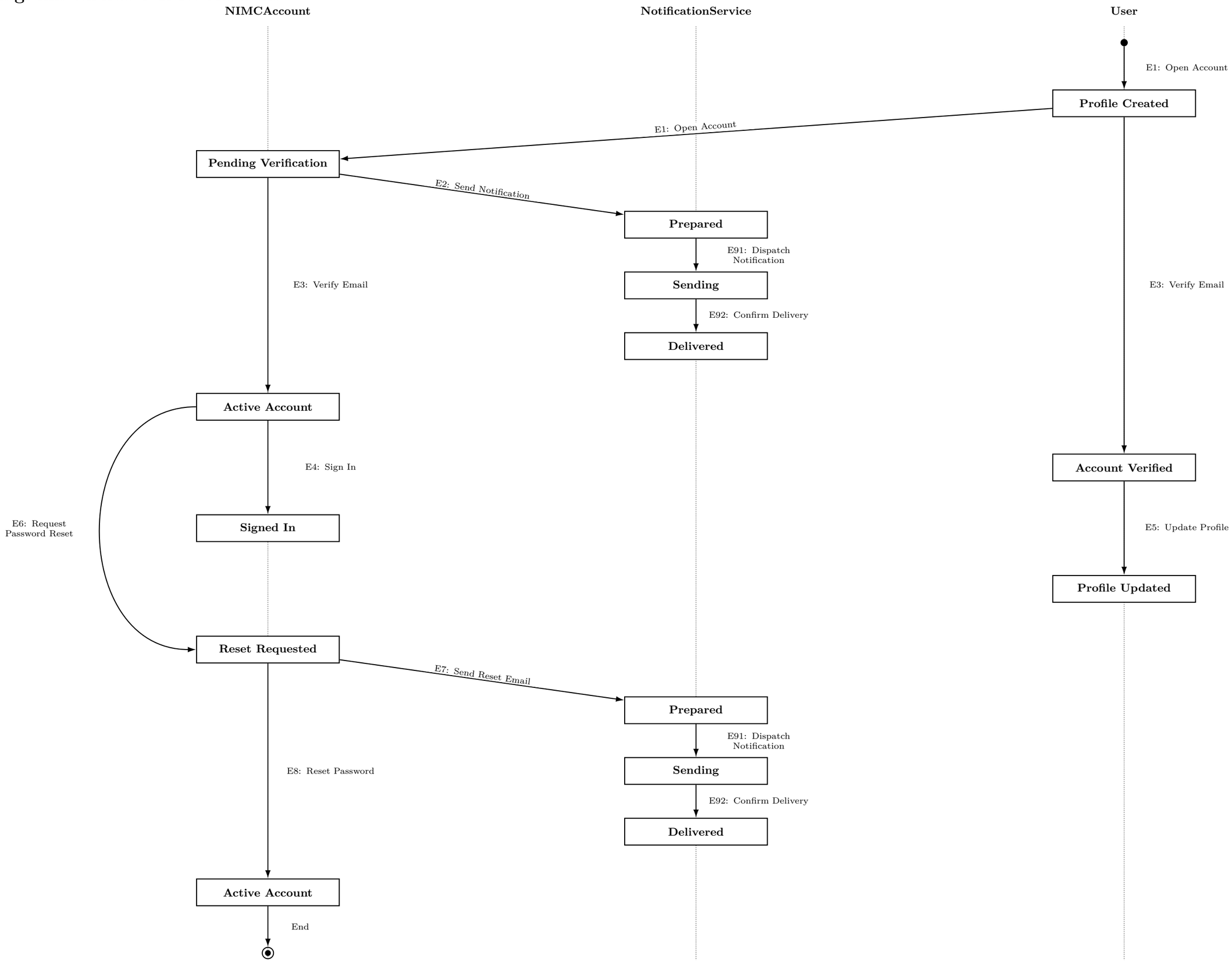
State-transition diagram: ApprovalAuthority



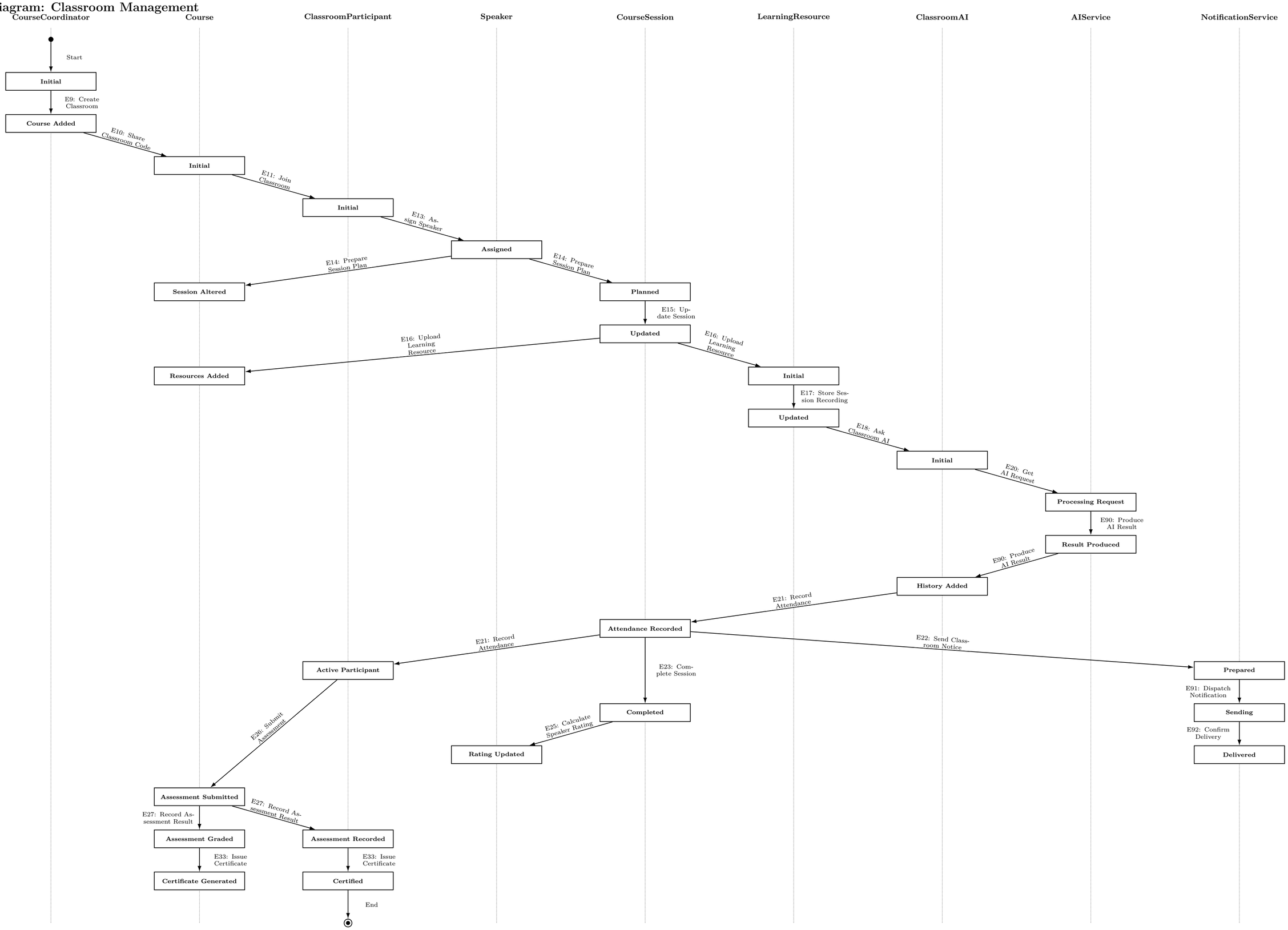
State-transition diagram: HostelAI



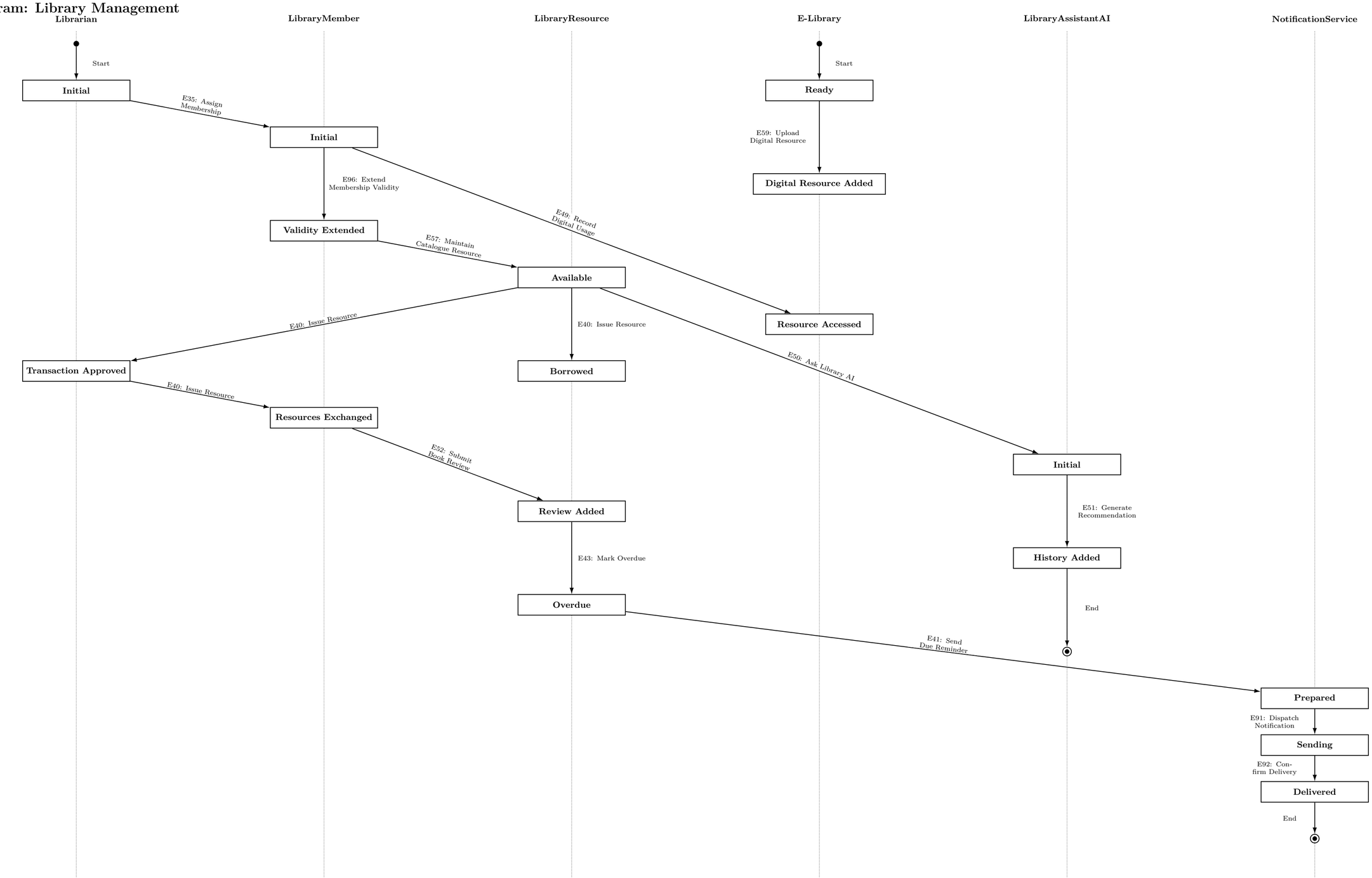
Sequence diagram: Shared NIMC Account



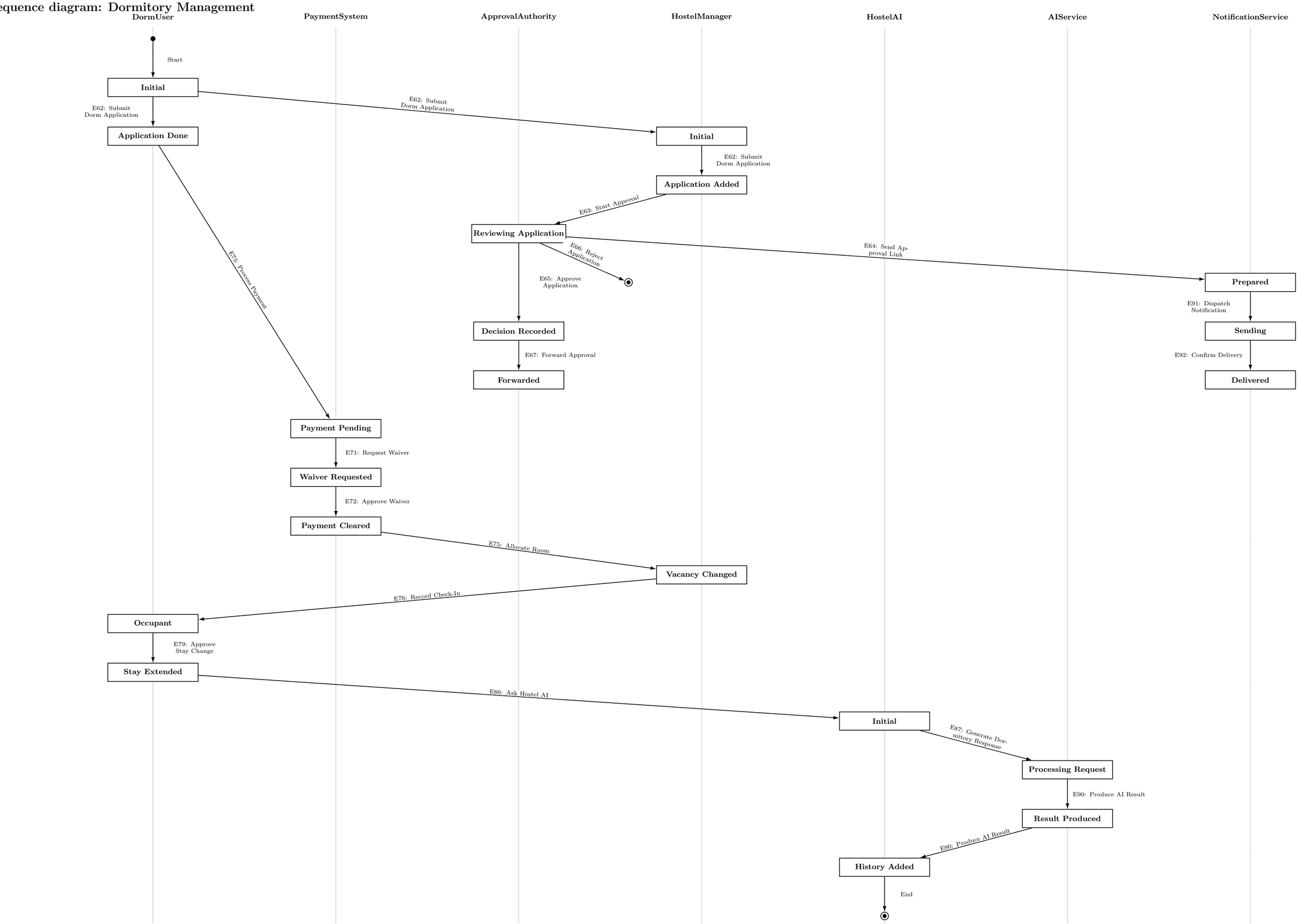
Sequence diagram: Classroom Management



Sequence diagram: Library Management

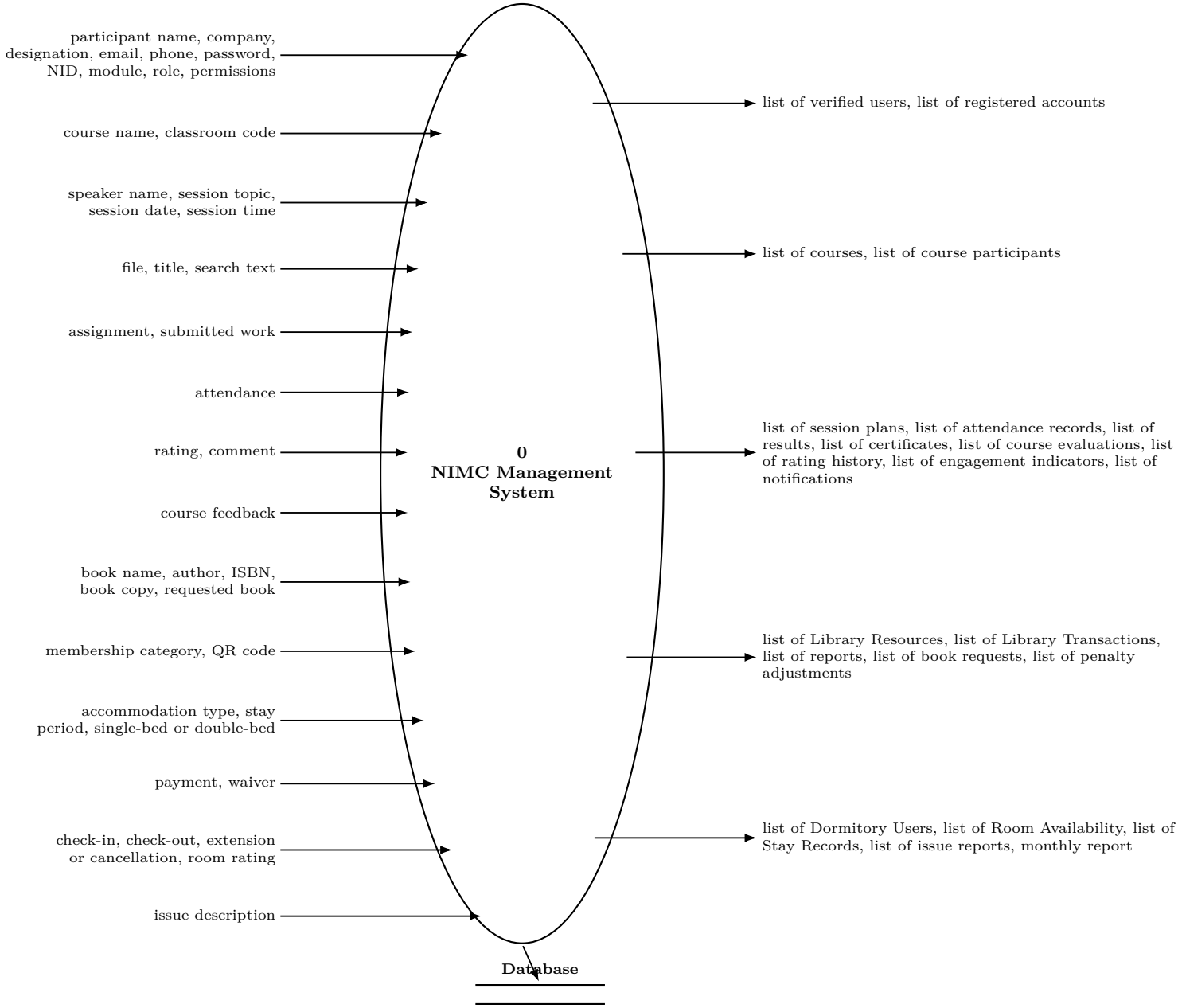


Sequence diagram: Dormitory Management



Level 0:

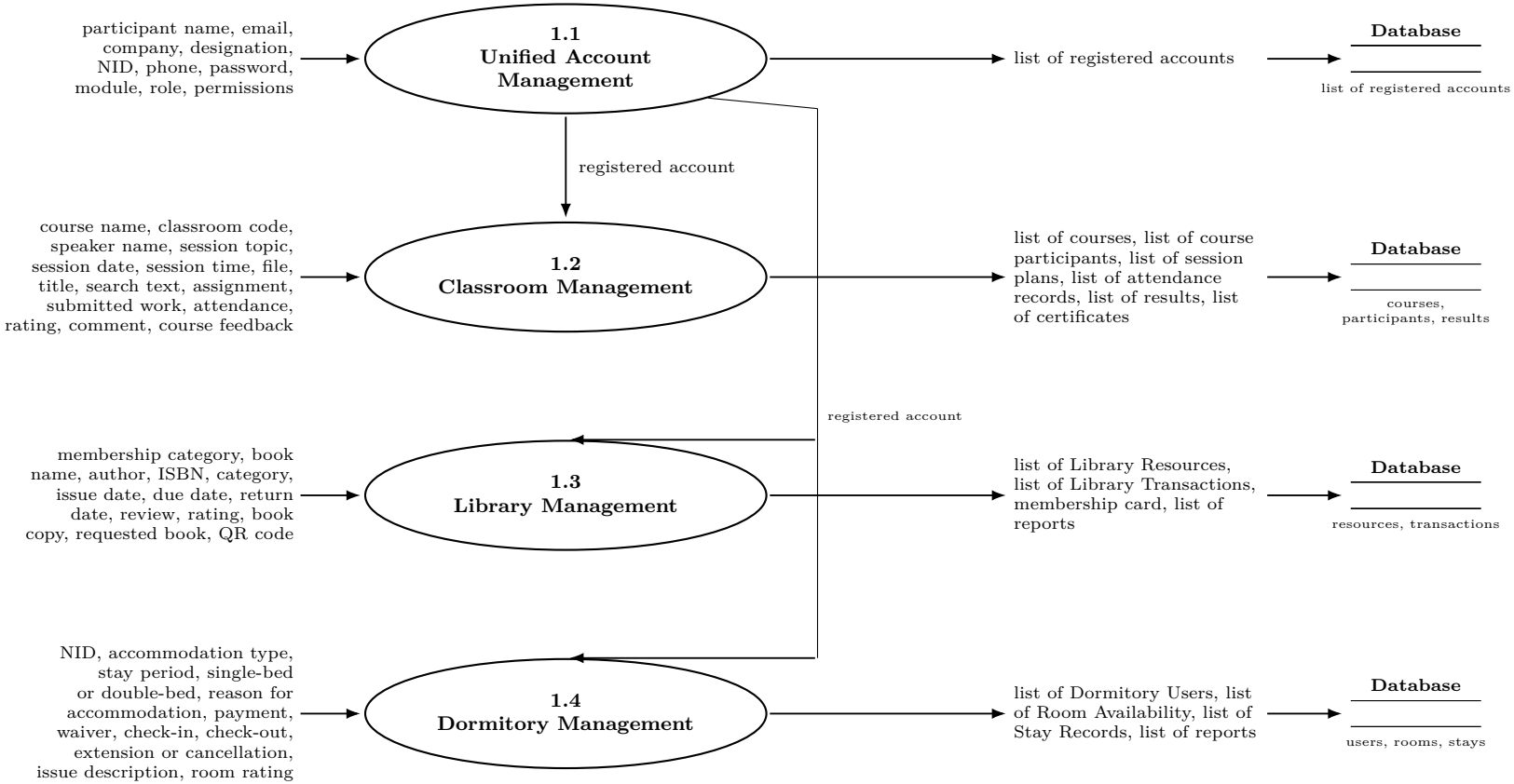
Name: NIMC Management System



Level 0: NIMC Management System

Level 1:

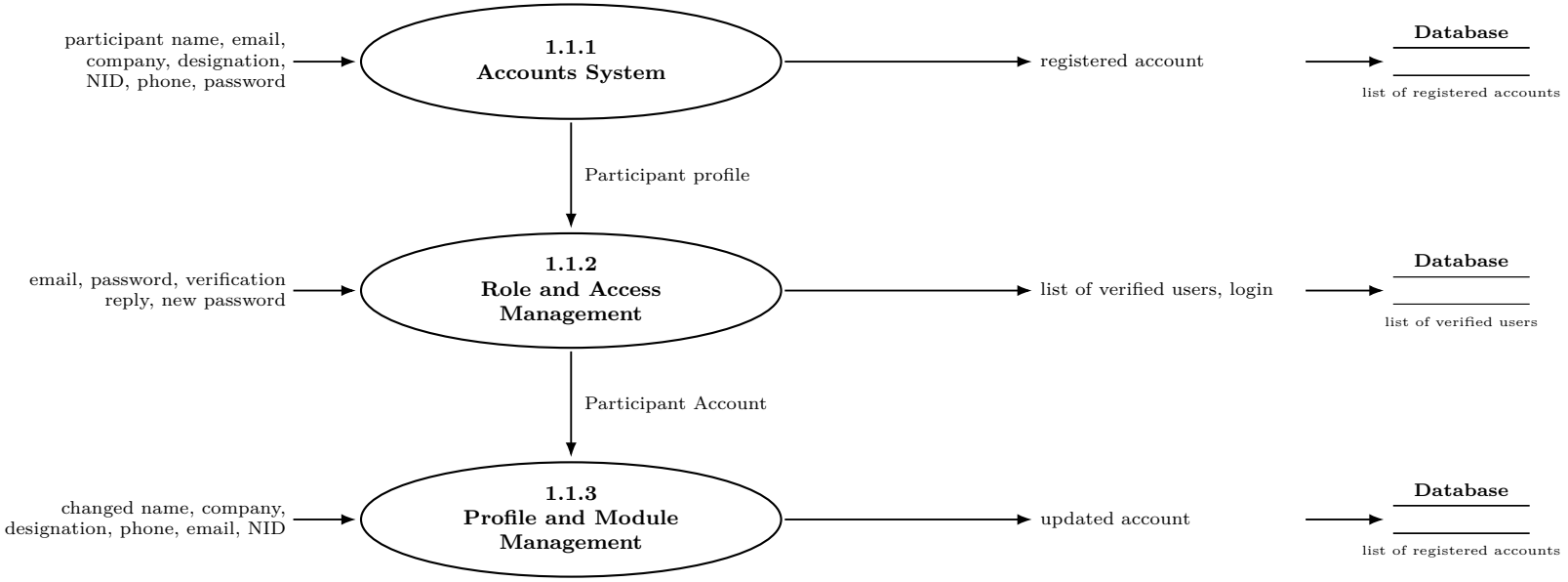
Name: NIMC Management System — Major Modules



Level 1: NIMC Management System — Major Modules

Level 1.1:

Name: Unified Account Management



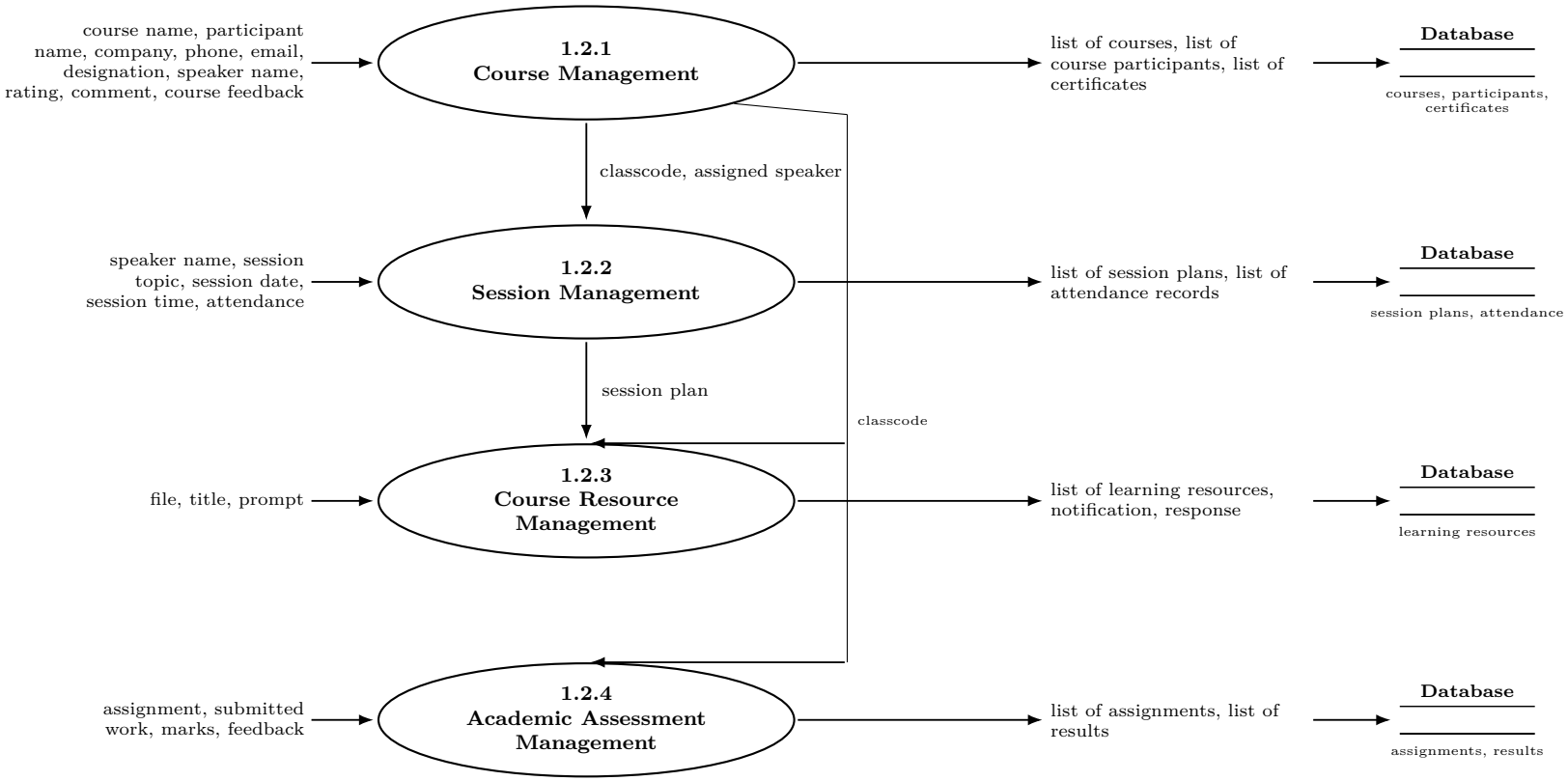
Level 1.1 — Function List

Sl.	Function	Parameters	Output
1	`registerAccount()`	`name, company, designation, phone, email, NID, password`	`accountNo, accountStatus`
2	`activateExistingIdentity()`	`email, password, existingIdentityFlag`	`accountNo, activationStatus`
3	`sendEmailVerification()`	`accountNo, email`	`boolean`
4	`verifyEmail()`	`accountNo, verificationCode`	`verificationStatus`
5	`authenticateUser()`	`email, password`	`loginStatus`
6	`signOut()`	`accountNo`	`boolean`
7	`requestPasswordReset()`	`email`	`boolean`
8	`resetPassword()`	`email, verificationCode, newPassword`	`passwordResetStatus`
9	`assignOrUpdateRole()`	`accountNo, moduleName, role, permissions`	`rolePermissionStatus`
10	`getProfile()`	`accountNo`	`profileInfo`
11	`updateInformation()`	`accountNo, changeField, newValue`	`updateStatus`
12	`requestModuleAccess()`	`accountNo, moduleName, requiredModuleData`	`moduleAccessStatus, moduleRoutingInfo`

Level 1.1: Unified Account Management

Level 1.2:

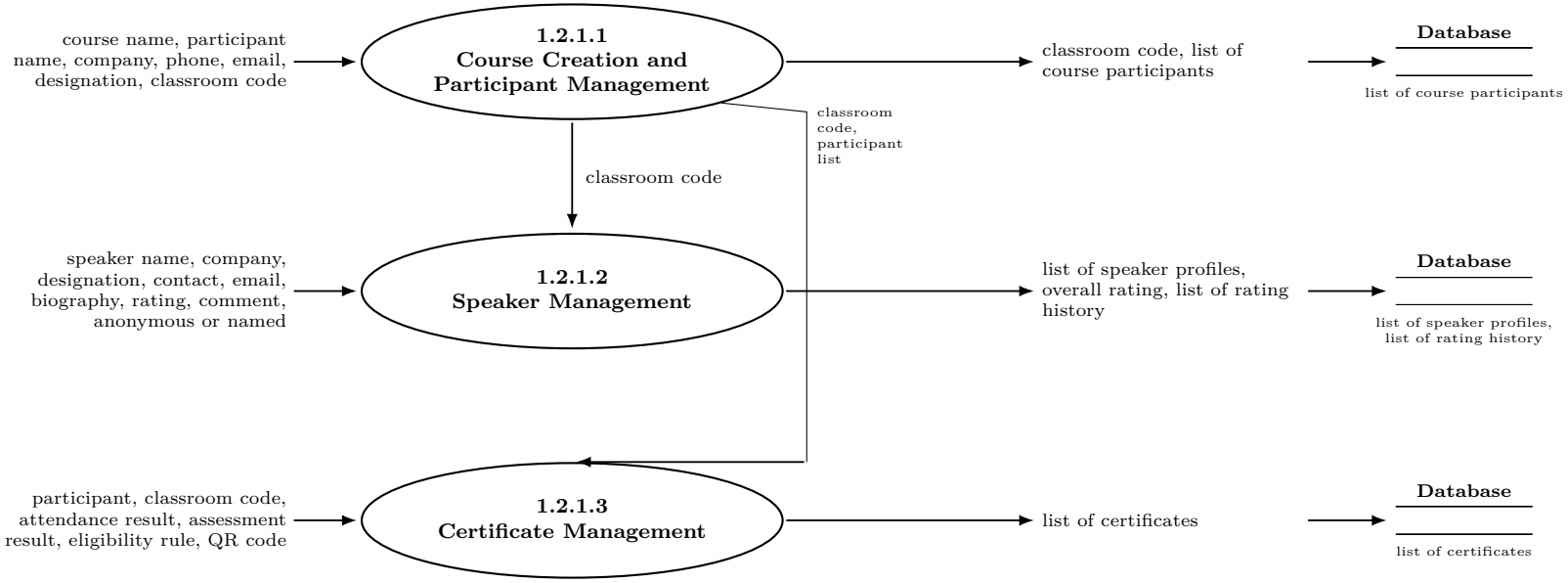
Name: Classroom Management



Level 1.2: Classroom Management

Level 1.2.1:

Name: Course Management



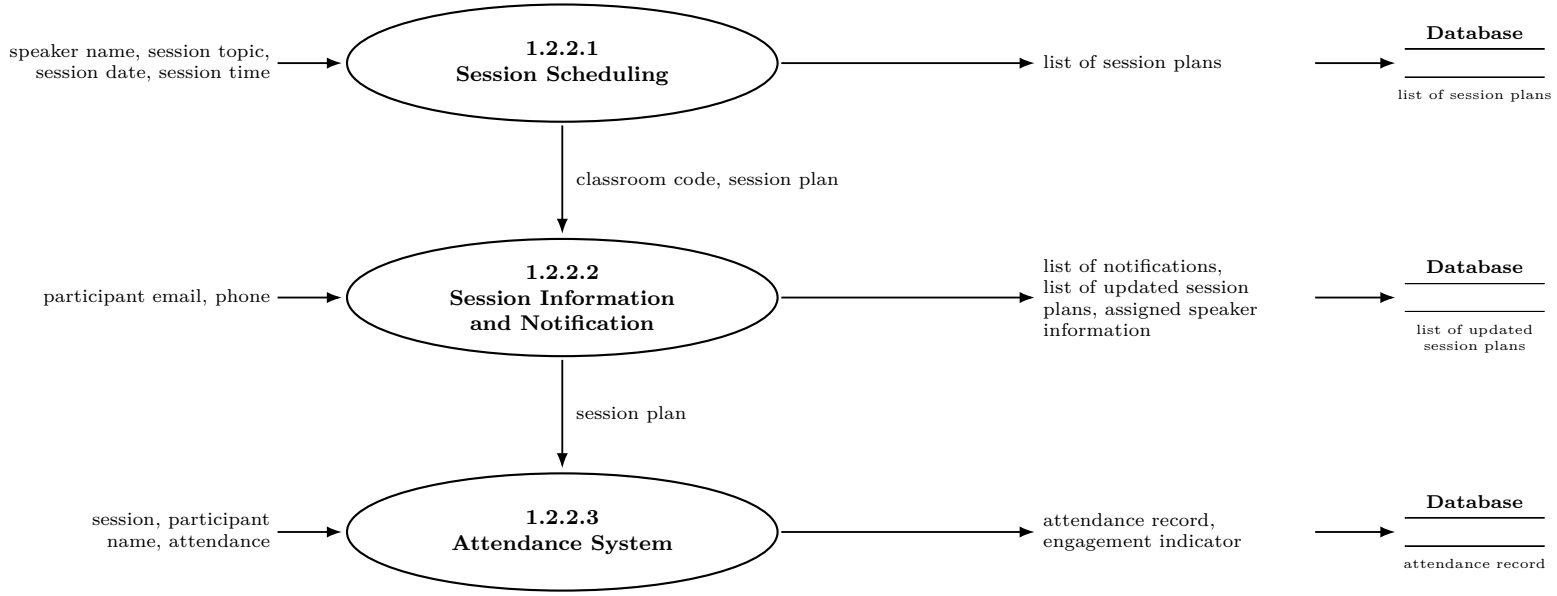
Level 1.2.1 — Function List

Sl.	Function	Parameters	Output
1	`createClass()`	`courseName`	`classroomCode`
2	`requestEnrollment()`	`participantID, classroomCode, participantInfo`	`enrollmentStatus`
3	`reviewParticipant()`	`participantID, classroomCode, eligibilityDecision`	`enrollmentStatus`
4	`removeParticipant()`	`participantID, classroomCode`	`boolean`
5	`getParticipantList()`	`classroomCode`	`participantList`
6	`submitCourseEvaluation()`	`participantID, classroomCode, courseRating, courseFeedback`	`boolean`
7	`getCourseEvaluation()`	`classroomCode`	`courseEvaluationSummary`
8	`saveSpeakerProfile()`	`speakerID, name, company, designation, contact, email, biography`	`boolean`
9	`getSpeakerProfile()`	`speakerID`	`speakerProfile`
10	`submitSpeakerRating()`	`sessionNo, rating, comment, anonymityChoice`	`boolean`
11	`getSpeakerPerformanceHistory()`	`speakerID`	`overallRating, ratingHistory`
12	`issueCertificate()`	`participantID, classroomCode, attendanceResult, assessmentResult, eligibilityRule`	`certificate, QRCode`
13	`getCertificate()`	`qrCode`	`certificate`

Level 1.2.1: Course Management

Level 1.2.2:

Name: Session Management



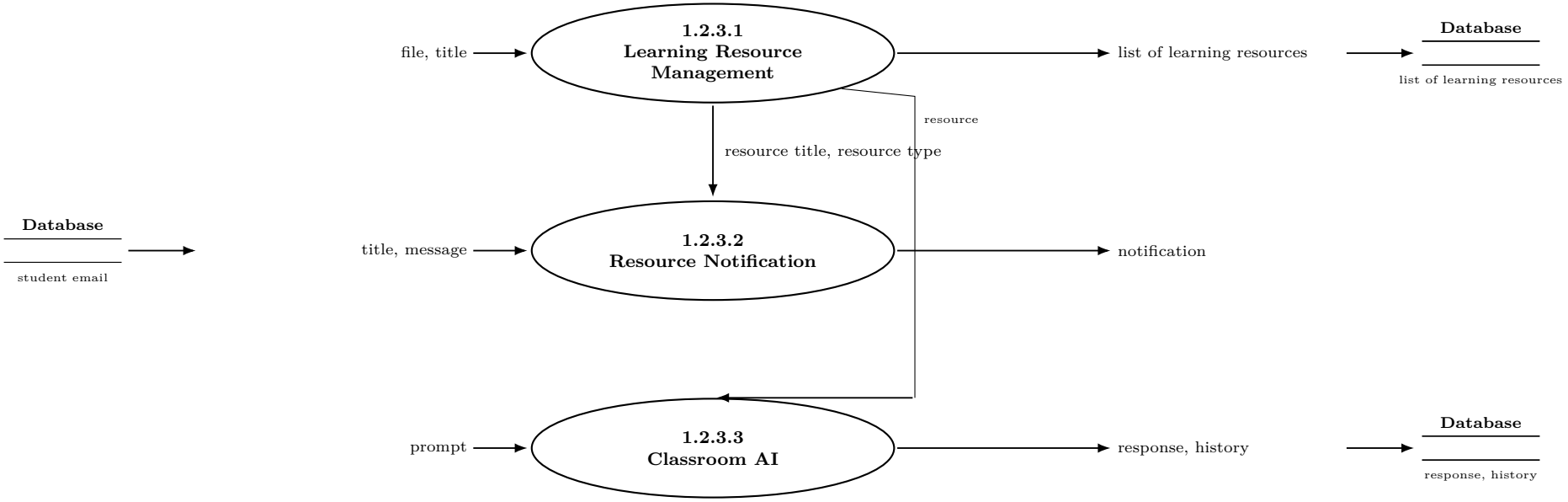
Level 1.2.2 — Function List

Sl.	Function	Parameters	Output
1	`scheduleSession()`	`classroomCode, topic, date, time, speakerID`	`sessionPlan`
2	`rescheduleSession()`	`sessionNo, changedTopic, changedDate, changedTime, changedSpeakerID`	`boolean`
3	`getSessionInformation()`	`classroomCode, sessionNo`	`sessionPlan, assignedSpeakerInfo`
4	`sendSessionNotification()`	`sessionNo, recipientEmail, recipientPhone, notificationType, message, channel`	`notificationDeliveryStatus`
5	`recordAttendance()`	`classroomCode, sessionNo, participantID, attendanceStatus`	`boolean`
6	`getAttendance()`	`participantID, sessionNo`	`attendanceRecord`
7	`calculateOverallAttendance()`	`participantID, classroomCode`	`overallAttendance`
8	`reviewEngagement()`	`classroomCode, attendanceRecord, assessmentSummary, participationData`	`engagementIndicator`

Level 1.2.2: Session Management

Level 1.2.3:

Name: Course Resource Management



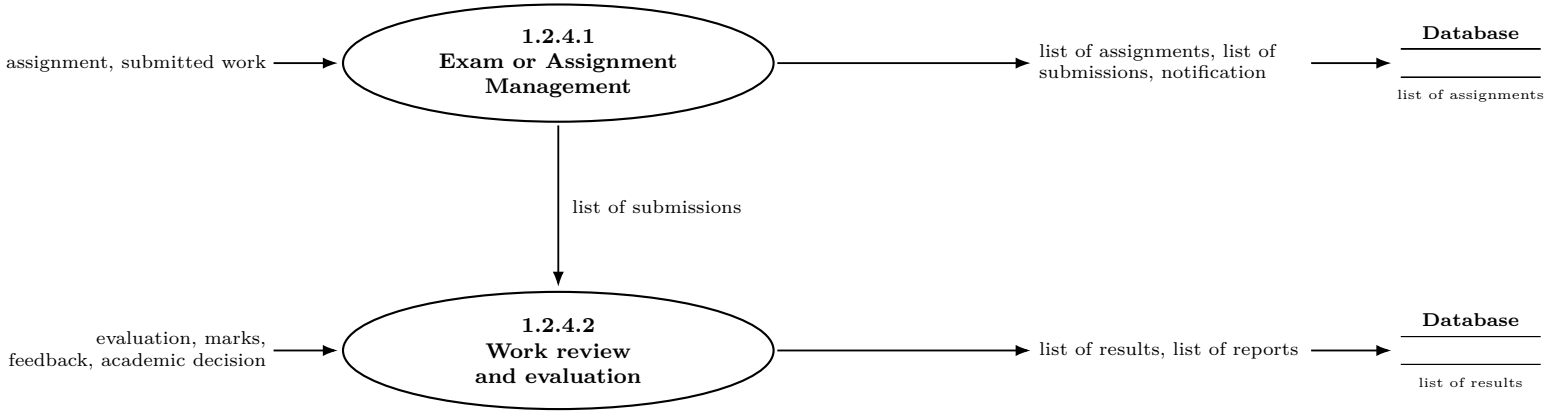
Level 1.2.3 — Function List

Sl.	Function	Parameters	Output
1	`uploadLearningResource()`	`classroomCode, sessionNo, file, title, resourceType, description`	`resourceID`
2	`findCourseResources()`	`classroomCode, searchText, sortBy, filters`	`resourceList`
3	`accessCourseResource()`	`participantID, resourceID`	`resourceContent`
4	`notifyResourceAvailable()`	`resourceID, recipientList, message`	`notificationDeliveryStatus`
5	`askClassroomAI()`	`classroomCode, participantID, question, explanationRequest`	`answer, history`

Level 1.2.3: Course Resource Management

Level 1.2.4:

Name: Academic Assessment Management



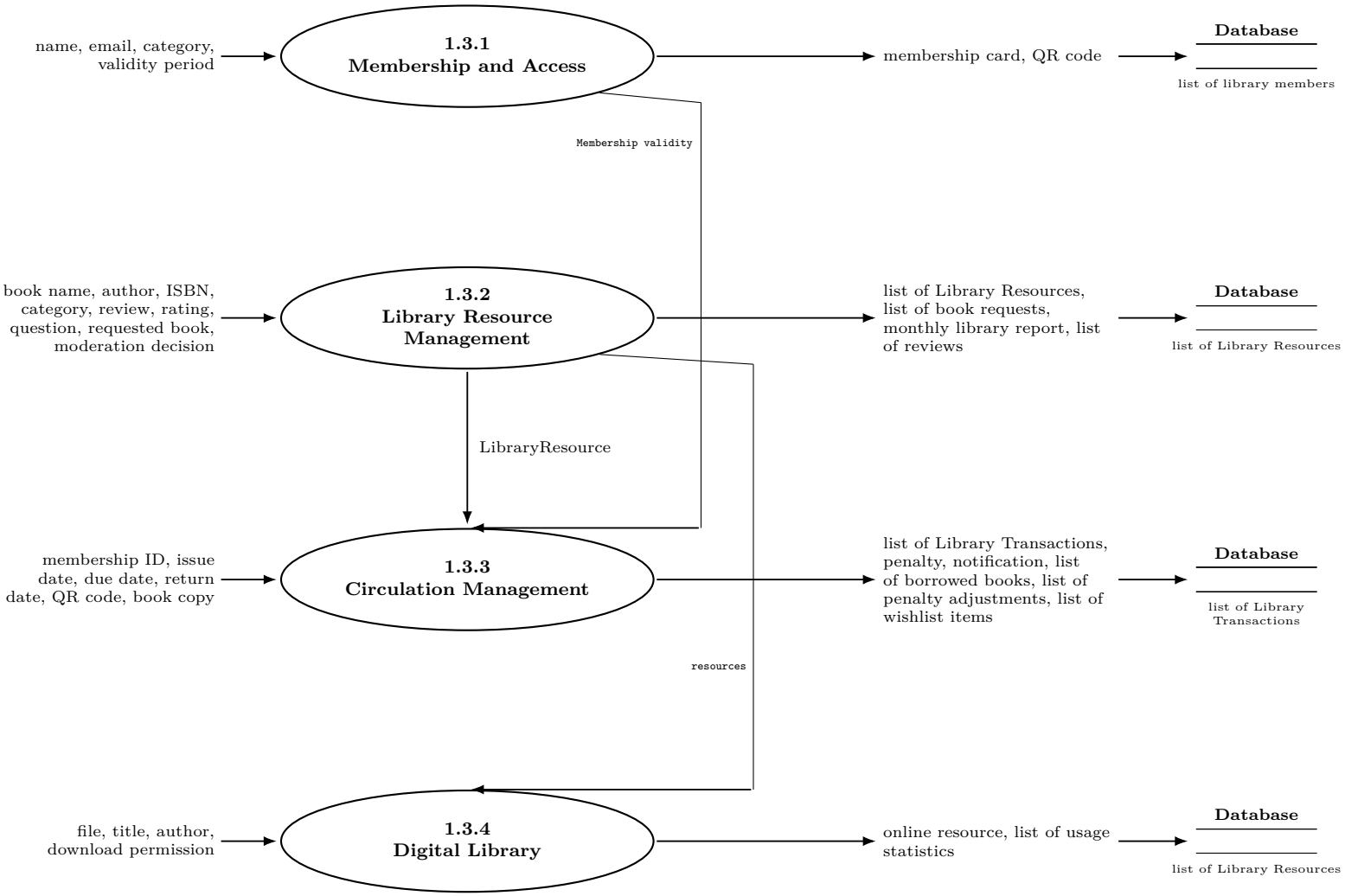
Level 1.2.4 — Function List

Sl.	Function	Parameters	Output
1	`publishAssessment()`	`classroomCode, assessmentType, requirements, dueDate`	`assessmentID, notification`
2	`getAssessmentRequirements()`	`participantID, assessmentID`	`assignmentList`
3	`submitWork()`	`participantID, assessmentID, submittedWork, submissionType`	`submissionStatus`
4	`analyzeAIContent()`	`submissionID, submittedWork`	`AIContentIndicator`
5	`analyzePlagiarism()`	`submissionID, submittedWork`	`similarityIndicator, plagiarismIndicator`
6	`evaluateWork()`	`submissionID, marks, feedback, academicDecision`	`result`
7	`getAssessmentResult()`	`assessmentID, participantID`	`marks, feedback, resultStatus`

Level 1.2.4: Academic Assessment Management

Level 1.3:

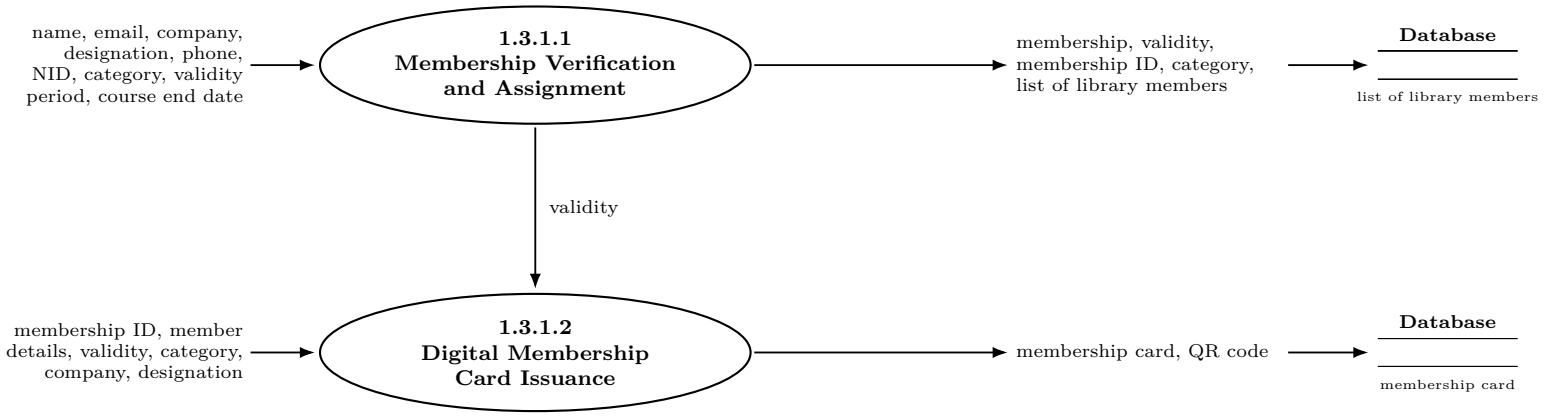
Name: NIMCO Library Management System



Level 1.3: NIMCO Library Management System

Level 1.3.1:

Name: Membership and Access



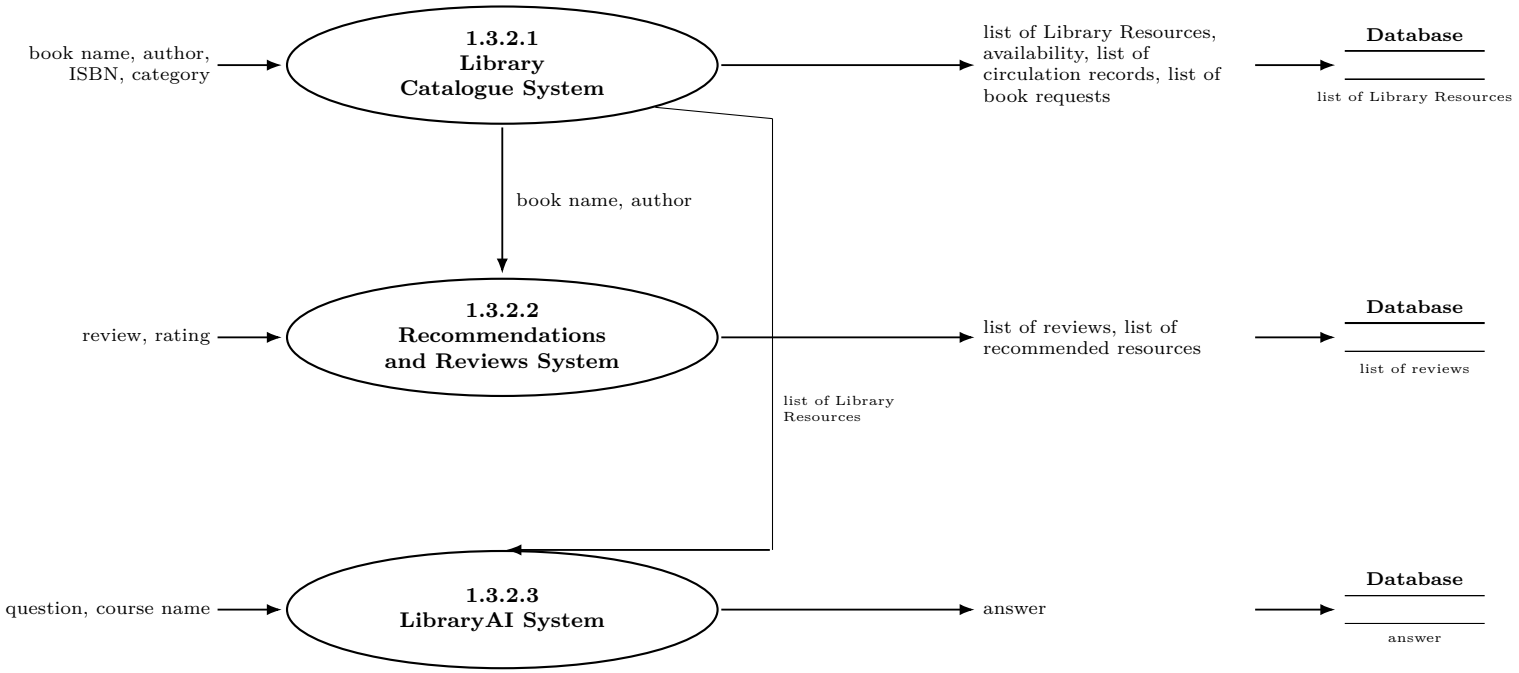
Level 1.3.1 — Function List

Sl.	Function	Parameters	Output
1	`verifyAndAssignMembership()`	`accountNo, name, company, designation, contact, email, NID, category, validityPeriod`	`membershipID, membershipStatus`
2	`adjustMembershipValidity()`	`membershipID, validFrom, validUntil, courseEndDate, validityAdjustment`	`validityPeriod`
3	`issueDigitalMembershipCard()`	`membershipID`	`digitalCard, QRCode, cardStatus`

Level 1.3.1: Membership and Access

Level 1.3.2:

Name: Library Resource Management



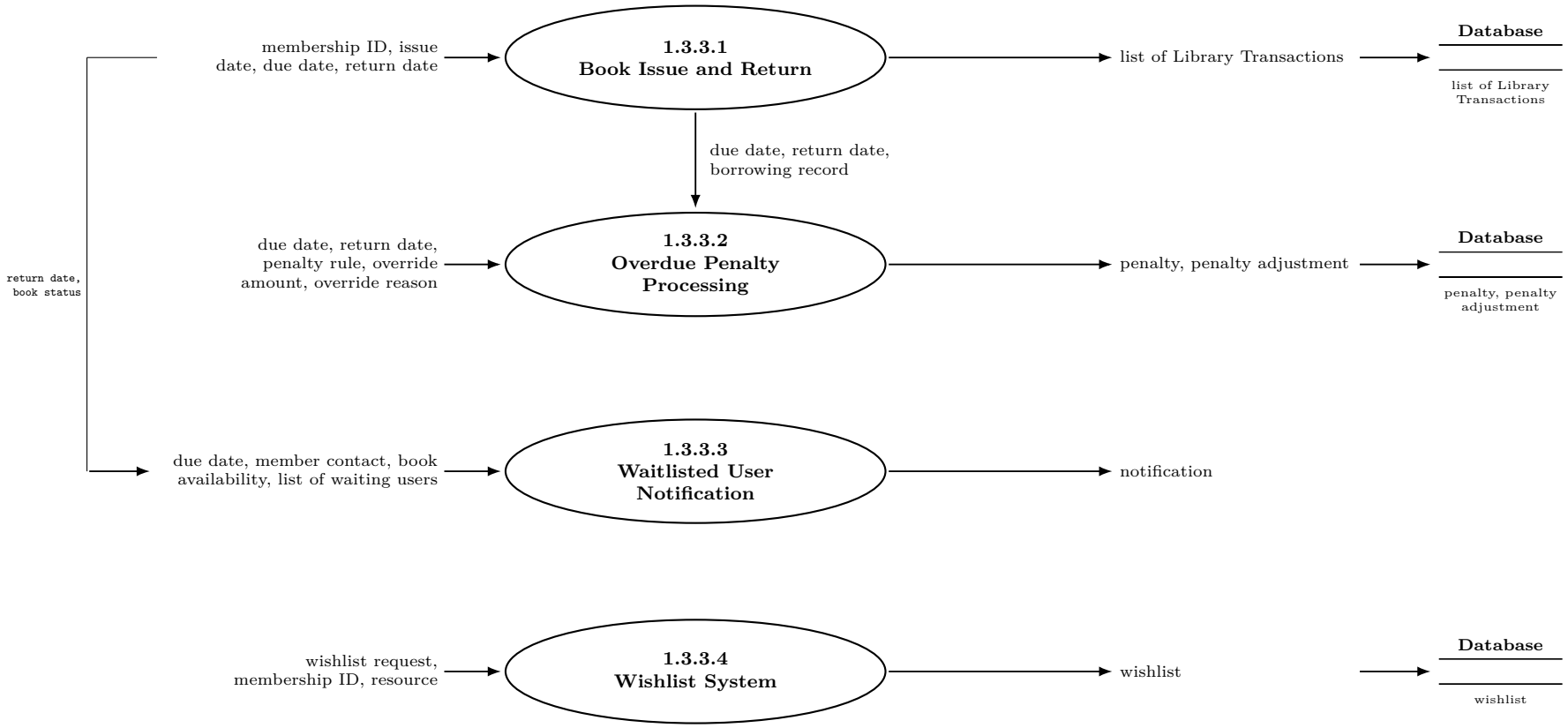
Level 1.3.2 — Function List

Sl.	Function	Parameters	Output
1	`searchCatalogue()`	`title, author, ISBN, category, catalogueInfo, searchText`	`resourceList, availability`
2	`getCirculationRecord()`	`resourceID`	`borrower, issueDate, dueDate, circulationStatus`
3	`manageCatalogueResource()`	`catalogueAction, resourceMetadata, copyStatus`	`catalogueUpdateStatus`
4	`requestNewBook()`	`userID, bookRequestData`	`bookRequestID, bookRequestStatus`
5	`reviewBookRequests()`	`requestFilter`	`bookRequestList, demandSummary`
6	`submitBookReview()`	`userID, resourceID, rating, reviewText`	`reviewID, reviewStatus`
7	`moderateReview()`	`reviewID, moderationDecision`	`reviewStatus`
8	`getRecommendations()`	`userID, preferences, borrowingHistory`	`recommendedResources`
9	`askLibraryAI()`	`userID, question, interest, courseName, learningNeed, followUpType`	`answer, relatedResources, referralFlag`
10	`generateMonthlyLibraryReport()`	`reportMonth, reportYear`	`monthlyLibraryReport`

Level 1.3.2: Library Resource Management

Level 1.3.3:

Name: Circulation Management



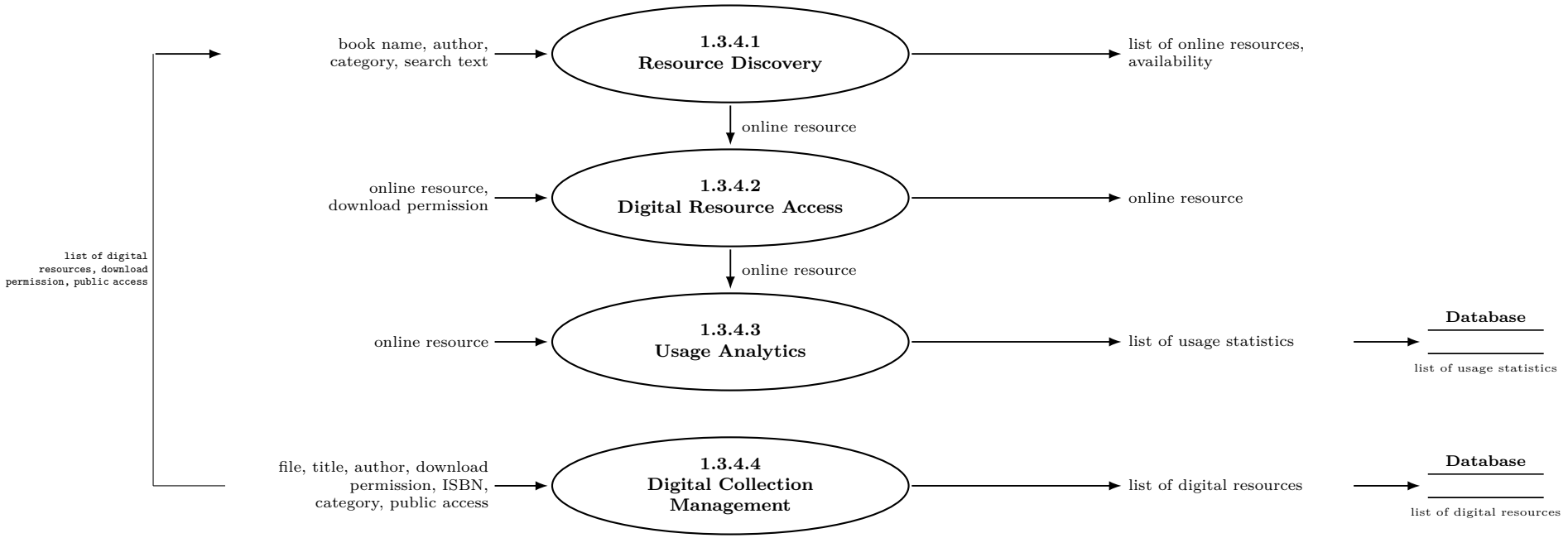
Level 1.3.3 — Function List

Sl.	Function	Parameters	Output
1	`checkBorrowingEligibility()`	`memberID, membershipStatus, currentBorrowCount, borrowingLimit`	`eligibilityStatus`
2	`issueBook()`	`memberID, copyID, issueDate, selectedDueDate`	`borrowingRecordID, issueDate, dueDate, bookStatus`
3	`extendBorrowingPeriod()`	`borrowingRecordID, extensionRequest, newDueDate`	`dueDate`
4	`returnBook()`	`borrowingRecordID, returnDate`	`bookStatus, overdueDays`
5	`processOverduePenalty()`	`borrowingRecordID, overdueDays, penaltyRule`	`penaltyAmount, penaltyStatus`
6	`overridePenalty()`	`penaltyID, adjustedAmount, overrideReason`	`penaltyStatus, adjustmentRecord`
7	`notifyLibraryUser()`	`resourceID, memberID, notificationType, dueDate, availabilityStatus`	`notificationDeliveryStatus`
8	`addToWishlist()`	`memberID, resourceID`	`wishlistStatus`
9	`removeFromWishlist()`	`memberID, resourceID`	`wishlistStatus`

Level 1.3.3: Circulation Management

Level 1.3.4:

Name: Digital Library



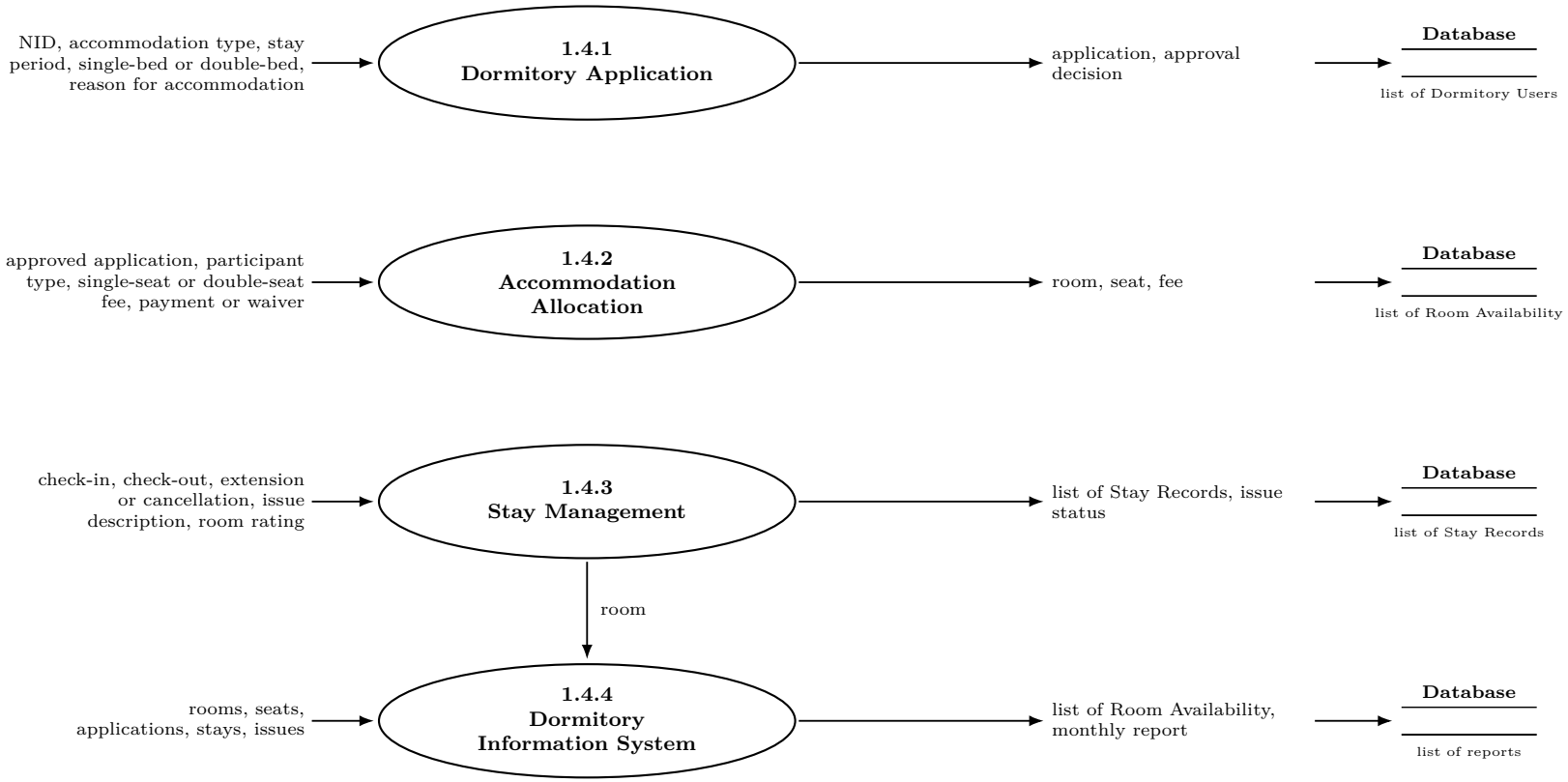
Level 1.3.4 — Function List

Sl.	Function	Parameters	Output
1	`discoverDigitalResources()`	`actorType, searchText, title, author, category, catalogueInfo`	`digitalResourceList, availability, publicAccessStatus`
2	`accessDigitalResource()`	`actorID, actorType, resourceID, accessMode`	`resourceContent, accessStatus, downloadStatus`
3	`getUsageAnalytics()`	`dateRange, resourceID, statisticType`	`usageStatistics`
4	`manageDigitalCollection()`	`staffID, collectionAction, file, title, author, ISBN, category, publicationInfo, downloadPermission`	`resourceID, collectionUpdateStatus`

Level 1.3.4: Digital Library

Level 1.4:

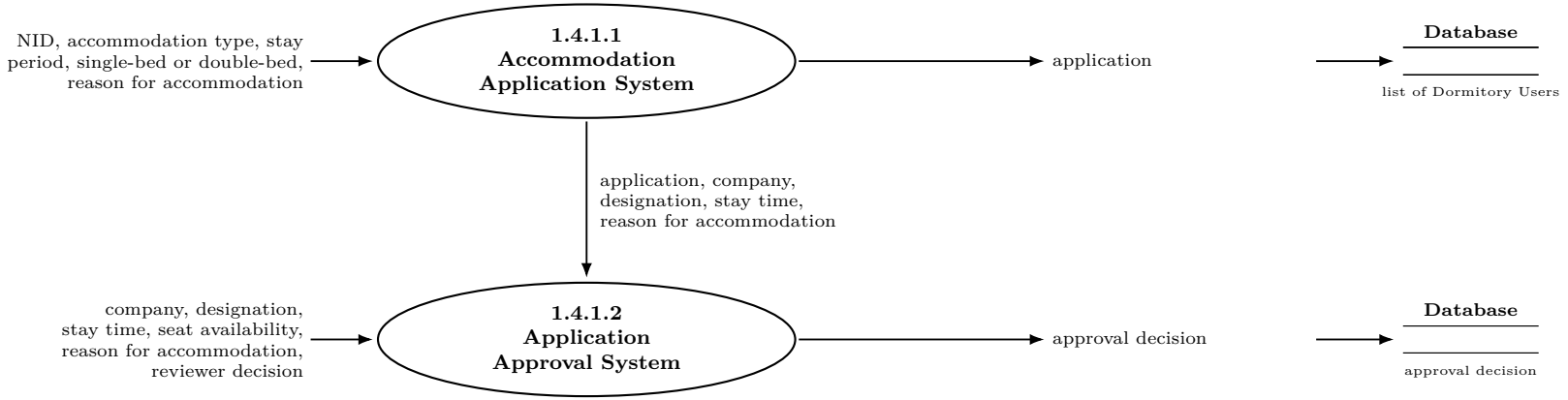
Name: Dormitory Management



Level 1.4: Dormitory Management

Level 1.4.1:

Name: Dormitory Application



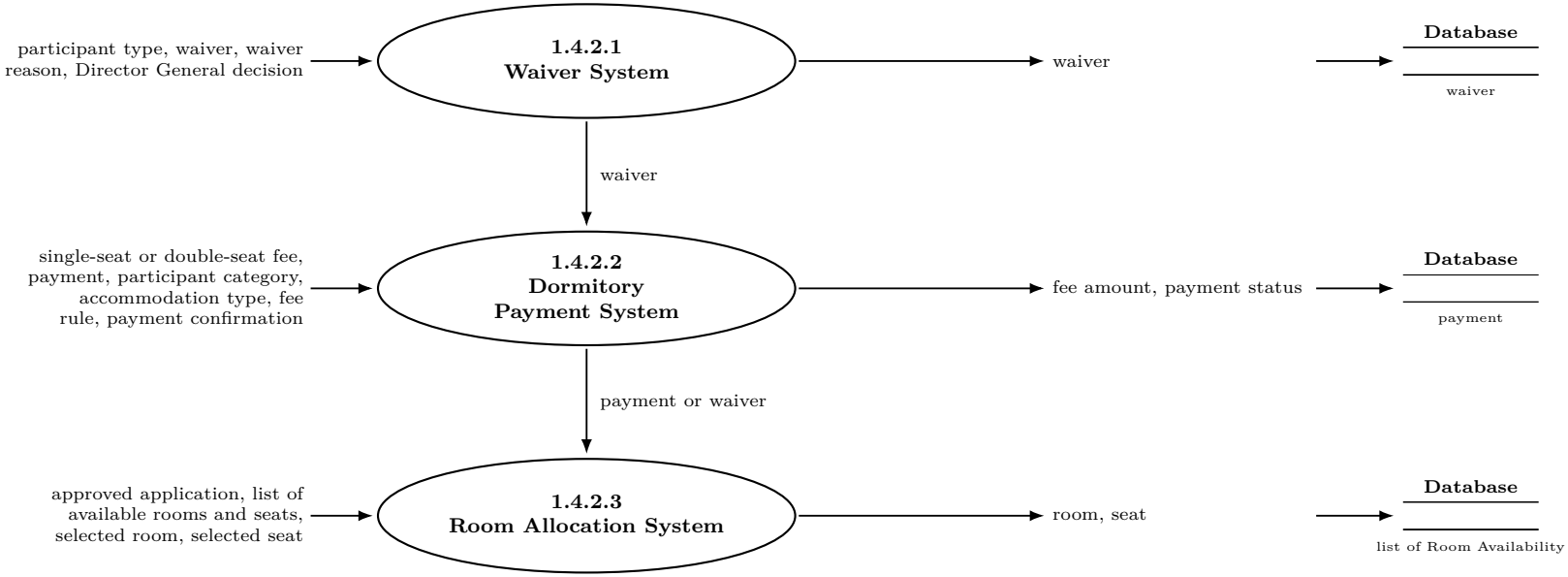
Level 1.4.1 — Function List

Sl.	Function	Parameters	Output
1	`submitAccommodationApplication()`	`accountNo, NID, accommodationType, seatPreference, stayStartDate, stayEndDate, accommodationReason`	`applicationID, submissionStatus`
2	`reviewApplication()`	`applicationID, completenessData, seatAvailability`	`boolean`
3	`recordApprovalDecision()`	`reviewerID, applicationID, approvalStage, decision, comment`	`approvalStatus, nextApprovalStage, finalDecision`
4	`sendApprovalLink()`	`applicationID, reviewerEmail, approvalStage`	`approvalNotificationStatus`

Level 1.4.1: Dormitory Application

Level 1.4.2:

Name: Accommodation Allocation



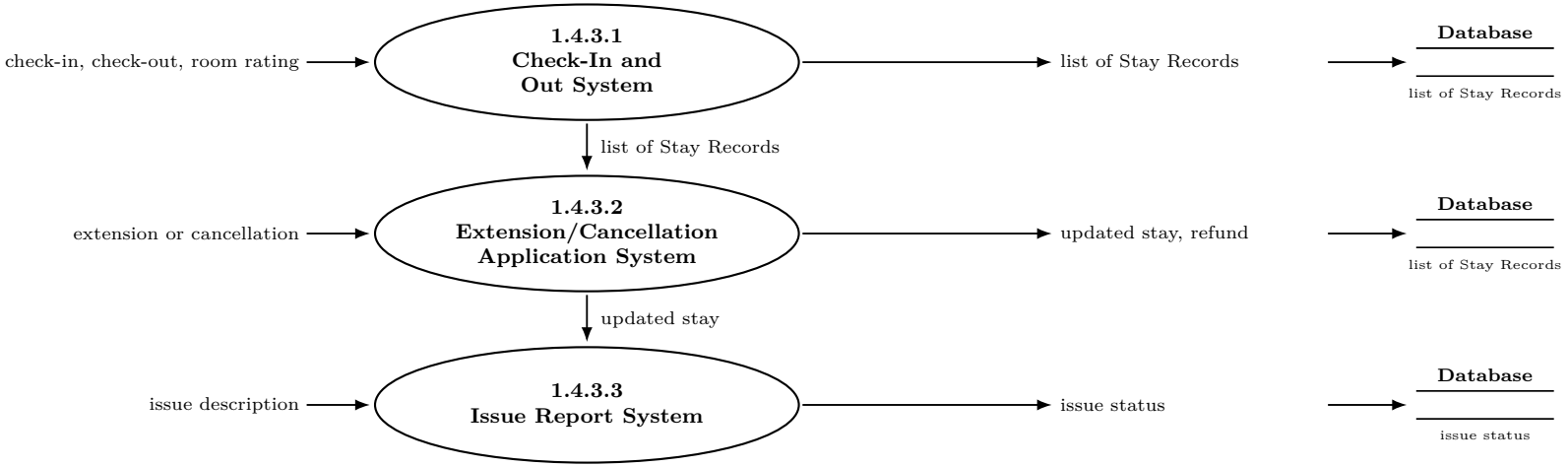
Level 1.4.2 — Function List

Sl.	Function	Parameters	Output
1	`requestFeeWaiver()`	`applicationID, participantCategory, company, designation, governmentEmployeeFlag, waiverReason`	`waiverRequestID, waiverEligibility, waiverStatus`
2	`approveFeeWaiver()`	`waiverRequestID, waiverDecision`	`waiverStatus`
3	`getDormitoryFee()`	`applicationID, participantCategory, accommodationType, feeRule, waiverStatus`	`feeAmount`
4	`processDormitoryPayment()`	`applicationID, paymentData, accountsConfirmation`	`paymentID, paymentStatus`
5	`checkAccommodationAvailability()`	`applicationID, accommodationType`	`availableRoomSeatList, pendingStatus`
6	`allocateRoomSeat()`	`applicationID, selectedRoomID, selectedSeatID, paymentStatus, waiverStatus`	`allocationID, allocationStatus, occupancyUpdate`
7	`getAllocationDetails()`	`applicationID`	`roomNo, seatNo, stayDetails`

Level 1.4.2: Accommodation Allocation

Level 1.4.3:

Name: Stay Management



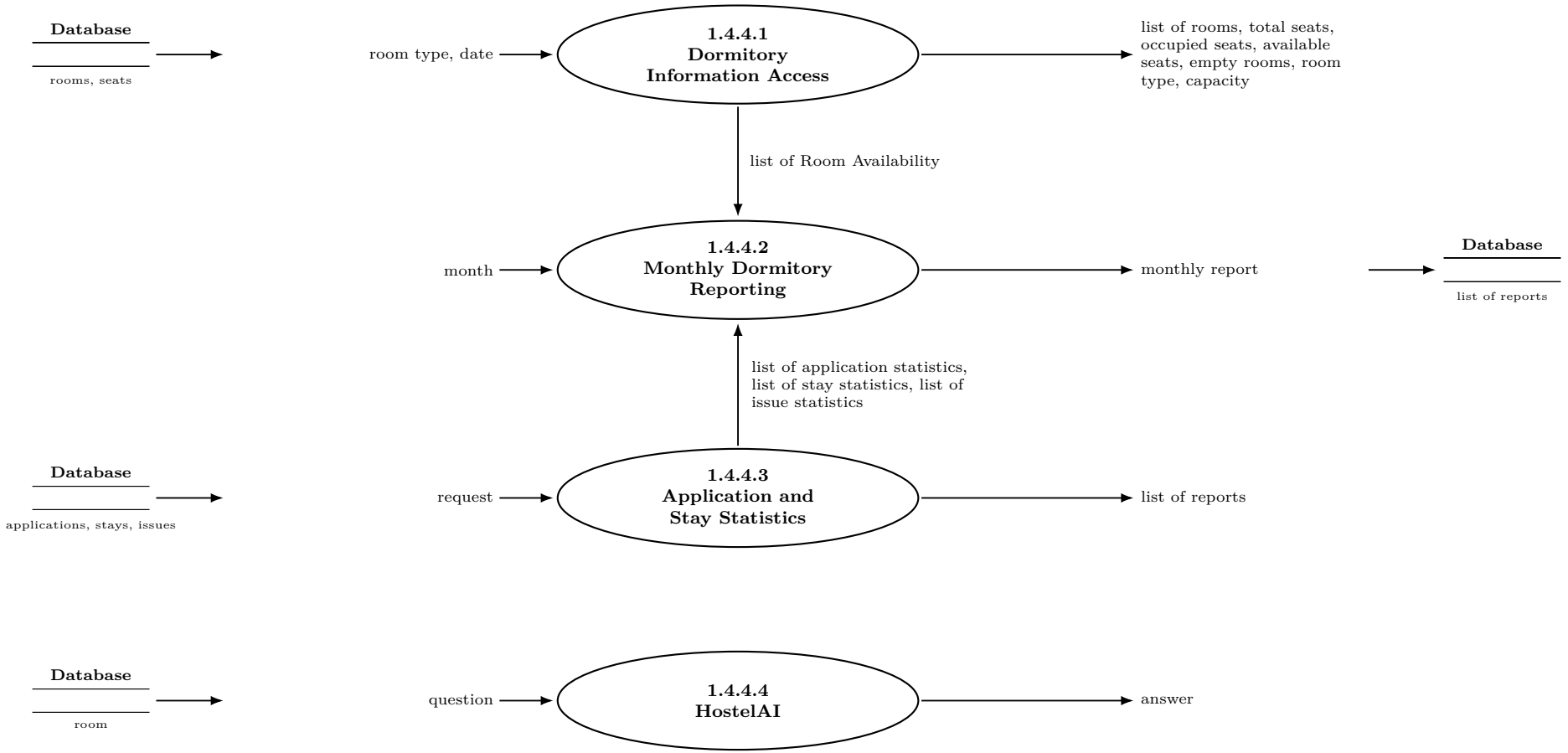
Level 1.4.3 — Function List

Sl.	Function	Parameters	Output
1	`recordCheckIn()`	`allocationID, staffID, checkInDateTime`	`stayID, checkInStatus, stayStatus`
2	`recordCheckOut()`	`stayID, staffID, checkOutDateTime, earlyDepartureFlag`	`checkOutStatus, stayStatus, releasedRoomSeatStatus`
3	`submitDormitoryRating()`	`stayID, rating, feedback`	`ratingStatus`
4	`submitStayAdjustmentRequest()`	`stayID, requestType, requestedDate, newEndDate, reason`	`requestID, requestStatus`
5	`resolveStayAdjustment()`	`requestID, managerDecision`	`requestStatus, updatedStayStatus`
6	`processRefund()`	`stayID, refundAmount, refundReason, paymentData`	`refundID, refundStatus`
7	`reportIssue()`	`stayID, participantID, issueType, description`	`issueID, issueStatus`
8	`assignIssueAction()`	`issueID, assignedStaff, action`	`issueStatus`
9	`updateIssueStatus()`	`issueID, action, newStatus`	`issueStatus`
10	`getIssueProgress()`	`participantID, issueID`	`issueProgress`

Level 1.4.3: Stay Management

Level 1.4.4:

Name: Dormitory Information System



Level 1.4.4 — Function List

Sl.	Function	Parameters	Output
1	`getDormitoryOccupancyInformation()`	`roomType, date`	`roomTotals, seatTotals, occupancy, availability, emptyRooms, capacity`
2	`generateMonthlyDormitoryReport()`	`month, year, reportFormat`	`monthlyDormitoryReport, reportGenerationStatus`
3	`getApplicationStayStatistics()`	`dateRange, statisticType, applicationStatusFilter, issueStatusFilter`	`application, approval, pending, allocation, stay, extension, cancellation, issueStatistics`
4	`askDormitoryInformationAI()`	`participantID, roomNo, questionOrAction, layoutQuery`	`hostelInformation, layoutGuidance, supportedActionResult`

Level 1.4.4: Dormitory Information System